

```
In [31]: import pandas as pd  
import numpy as np  
import matplotlib.pyplot as plt
```

```
In [32]: df=pd.read_csv(r"C:\Users\manas\OneDrive\Desktop\Whitecliffe\Machine Learning\Asses
```

```
In [33]: display(df.info())
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25390 entries, 0 to 25389
Data columns (total 44 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Movie_ID                             25390 non-null  int64
1   TMDB_ID                              25390 non-null  int64
2   IMDb_ID                              18482 non-null  object
3   Movie_Name                           25390 non-null  object
4   Original_Title                       25390 non-null  object
5   Release_Year                         24609 non-null  float64
6   Release_Date                         24609 non-null  object
7   Studio                               18541 non-null  object
8   All_Production_Companies             18886 non-null  object
9   Director                             22967 non-null  object
10  Country_Origin                       25230 non-null  object
11  Original_Language                    25390 non-null  object
12  Spoken_Languages                     25390 non-null  object
13  Animation_Style                      24999 non-null  object
14  Genre                                25390 non-null  object
15  Theme                                25390 non-null  object
16  Overview                             22692 non-null  object
17  Movie_Length_Minutes                 25041 non-null  float64
18  TMDB_Rating                          16118 non-null  float64
19  TMDB_Vote_Count                      16127 non-null  float64
20  TMDB_Popularity                      25368 non-null  float64
21  Budget_Million_USD                   1458 non-null   float64
22  Box_Office_Million_USD                1642 non-null   float64
23  MPAA_Rating                          25390 non-null  object
24  Target_Audience                     25390 non-null  object
25  Voice_Cast                           16199 non-null  object
26  Live_Action_Remake                   25390 non-null  object
27  Belongs_To_Collection                25390 non-null  object
28  Collection_ID                        3184 non-null   float64
29  Poster_URL                           21975 non-null  object
30  Backdrop_URL                         14259 non-null  object
31  TMDB_URL                             25390 non-null  object
32  IMDb_URL                             18482 non-null  object
33  Is_Released                          25390 non-null  bool
34  Is_TV_Compilation                    25390 non-null  bool
35  Decade                               25390 non-null  object
36  Era                                  25390 non-null  object
37  Popularity_Tier                      25390 non-null  object
38  Hidden_Gem                          25390 non-null  bool
39  Profitability_ROI                    732 non-null    float64
40  Is_Adult_Content                     25390 non-null  bool
41  Runtime_Needs_Verification           25390 non-null  bool
42  Is_Future_Release                    25390 non-null  bool
43  Data_Quality_Score                   25390 non-null  int64
dtypes: bool(6), float64(9), int64(3), object(26)
memory usage: 7.5+ MB
None

```

```

In [34]: df.describe() # stats for numeric columns (count, mean, std, min, max)
df.describe(include='object') # overview for categorical/text columns (count, unique)

```

Out [34]:

	IMDb_ID	Movie_Name	Original_Title	Release_Date	Studio	All_Production_C
count	18482	25390	25390	24609	18541	
unique	18469	24768	24919	12461	5094	
top	tt4641286	Puss in Boots	A Christmas Carol	1987-01-01	Walt Disney Productions	Walt Disney Pr
freq	2	8	8	68	627	

4 rows × 26 columns



In [35]:

```
missing = df.isnull().sum().sort_values(ascending=False)
print(missing.head(10)) # top columns by missing count
```

```
Profitability_ROI      24658
Budget_Million_USD    23932
Box_Office_Million_USD 23748
Collection_ID          22206
Backdrop_URL           11131
TMDB_Rating            9272
TMDB_Vote_Count        9263
Voice_Cast             9191
IMDb_ID                6908
IMDb_URL               6908
dtype: int64
```

In [36]:

```
import sys
!{sys.executable} -m pip install missingno
```

Requirement already satisfied: missingno in c:\users\manas\anaconda3\lib\site-packages (0.5.2)

Requirement already satisfied: numpy in c:\users\manas\anaconda3\lib\site-packages (from missingno) (1.26.4)

Requirement already satisfied: matplotlib in c:\users\manas\anaconda3\lib\site-packages (from missingno) (3.8.0)

Requirement already satisfied: scipy in c:\users\manas\anaconda3\lib\site-packages (from missingno) (1.11.4)

Requirement already satisfied: seaborn in c:\users\manas\anaconda3\lib\site-packages (from missingno) (0.12.2)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->missingno) (1.2.0)

Requirement already satisfied: cycler>=0.10 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->missingno) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->missingno) (4.63.0)

Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->missingno) (1.4.4)

Requirement already satisfied: packaging>=20.0 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->missingno) (23.1)

Requirement already satisfied: pillow>=6.2.0 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->missingno) (10.2.0)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->missingno) (3.0.9)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->missingno) (2.8.2)

Requirement already satisfied: pandas>=0.25 in c:\users\manas\anaconda3\lib\site-packages (from seaborn->missingno) (2.1.4)

Requirement already satisfied: pytz>=2020.1 in c:\users\manas\anaconda3\lib\site-packages (from pandas>=0.25->seaborn->missingno) (2023.3.post1)

Requirement already satisfied: tzdata>=2022.1 in c:\users\manas\anaconda3\lib\site-packages (from pandas>=0.25->seaborn->missingno) (2023.3)

Requirement already satisfied: six>=1.5 in c:\users\manas\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib->missingno) (1.16.0)

```
In [37]: import warnings
# Hide the deprecation warning messages
warnings.filterwarnings('ignore', category=FutureWarning)

# 1. Force the Jupyter rendering engine to turn on
%matplotlib inline
import matplotlib.pyplot as plt
import seaborn as sns

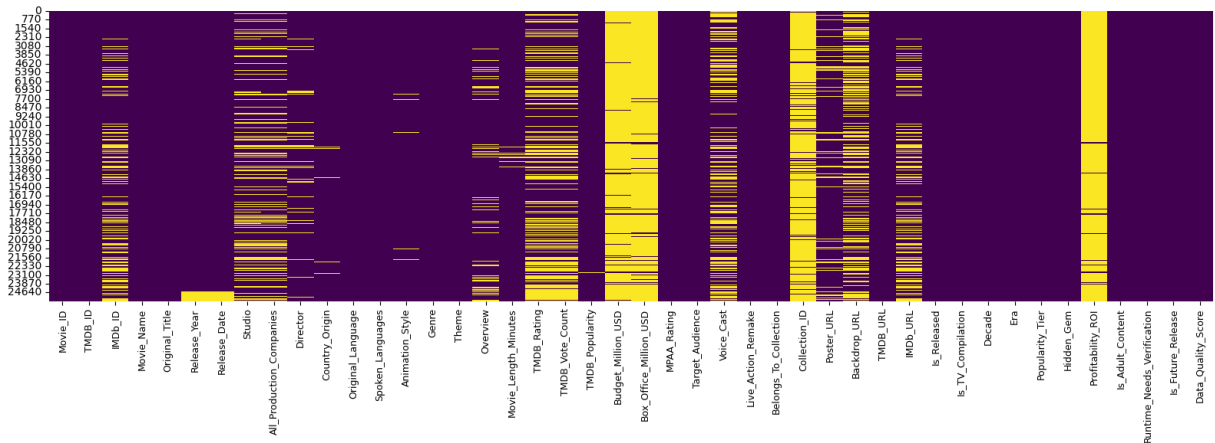
# 2. Reset the drawing backend memory
plt.close('all')
plt.rcParams.update(plt.rcParamsDefault)

# 3. Create the drawing canvas
fig, ax = plt.subplots(figsize=(16, 6))

# 4. Generate the plot layout safely
sns.heatmap(df.isnull(), cbar=False, cmap="viridis", ax=ax)
ax.set_xticklabels(ax.get_xticklabels(), rotation=90, fontsize=9)
plt.tight_layout()
```

```
# 5. Save the Local image copy
fig.savefig("missing_data_report.png", dpi=300, bbox_inches="tight")

# 6. Force display on screen
plt.show()
```



```
In [38]: dupes = df.duplicated().sum()
print(f"Duplicate rows: {dupes}")
```

Duplicate rows: 0

```
In [39]: num_cols = ['Release_Year', 'TMDB_Rating', 'TMDB_Vote_Count',
                    'Movie_Length_Minutes', 'Budget_Million_USD', 'Box_Office_Million_USD']
for col in num_cols:
    df[col] = pd.to_numeric(df[col], errors='coerce')
```

```
In [40]: df['Target_Audience'] = df['Target_Audience'].astype('category')
df['MPAA_Rating'] = df['MPAA_Rating'].astype('category')
# etc.
```

```
In [41]: df = df[df['Movie_Length_Minutes'].between(1, 300)]
```

```
In [42]: import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

# 1. Clean the data: Convert strings to numbers, turning errors into missing values
df['TMDB_Rating_Clean'] = pd.to_numeric(df['TMDB_Rating'], errors='coerce')

# 2. Filter out missing data
clean_data = df['TMDB_Rating_Clean'].dropna()

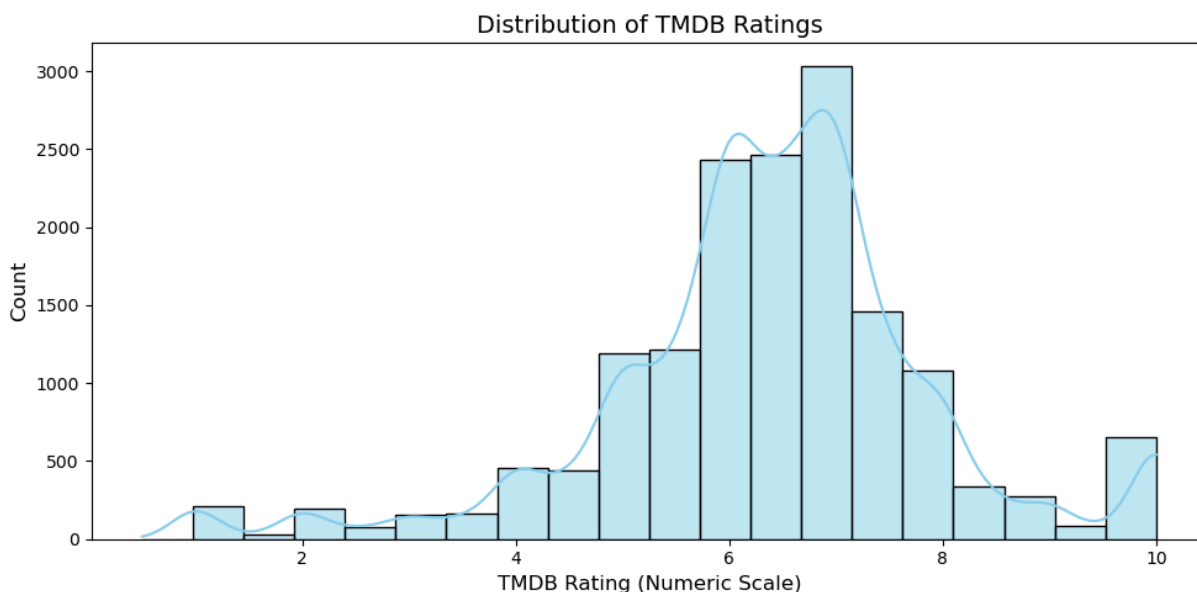
# 3. Create a clean figure canvas
plt.figure(figsize=(10, 5))

# 4. Use histplot (not countplot!) to group the data into clean buckets
sns.histplot(clean_data, bins=20, kde=True, color='skyblue')

# 5. Add clean labels
plt.title('Distribution of TMDB Ratings', fontsize=14)
```

```
plt.xlabel('TMDB Rating (Numeric Scale)', fontsize=12)
plt.ylabel('Count', fontsize=12)

fig.savefig("Distribution of TMDB Ratings.png", dpi=300, bbox_inches="tight")
fig.savefig("Distribution of TMDB Ratings.pdf", bbox_inches="tight")
plt.tight_layout()
plt.show()
```



```
In [43]: import matplotlib.pyplot as plt
import seaborn as sns

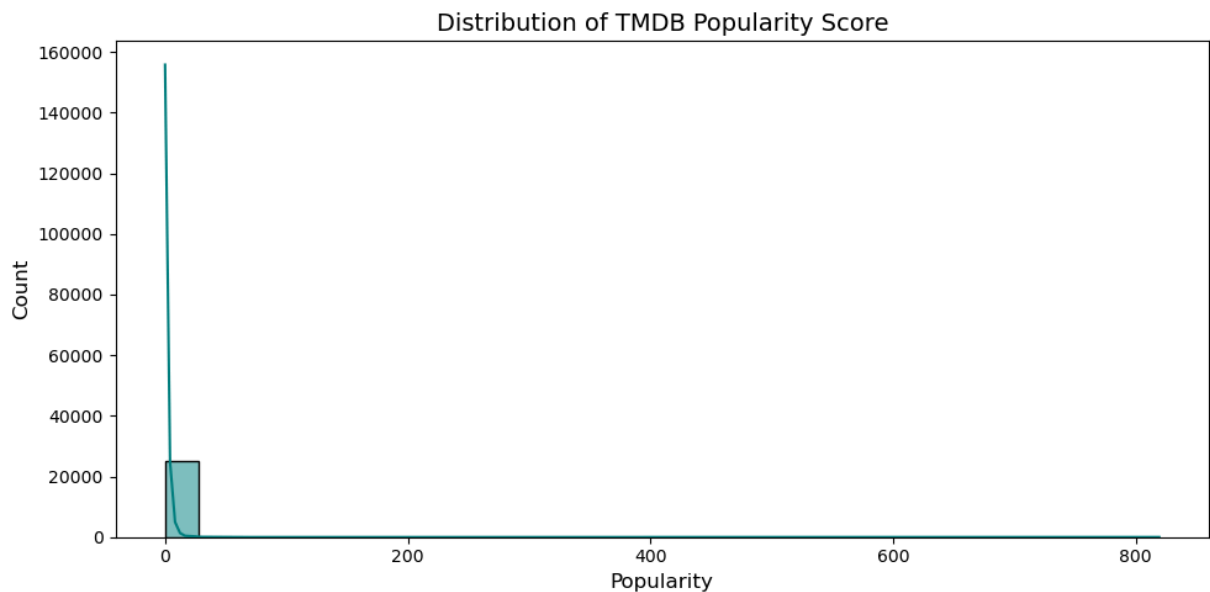
# 1. Set canvas size
plt.figure(figsize=(10, 5))

# 2. Draw the plot
sns.histplot(df['TMDB_Popularity'].dropna(), bins=30, kde=True, color="teal")

# 3. Add Labels
plt.title('Distribution of TMDB Popularity Score', fontsize=14)
plt.xlabel('Popularity', fontsize=12)
plt.ylabel('Count', fontsize=12)
plt.tight_layout()

# === 4. SAVE THE PLOTS HERE (BEFORE PLT.SHOW) ===
plt.savefig("tmdb_popularity_distribution.png", dpi=300, bbox_inches="tight")
plt.savefig("tmdb_popularity_distribution.pdf", bbox_inches="tight")
# =====

# 5. Display on screen Last
plt.show()
```



```
In [44]: import matplotlib.pyplot as plt
import seaborn as sns

# 1. Expand the canvas size so the categories have breathing room
plt.figure(figsize=(10, 6))

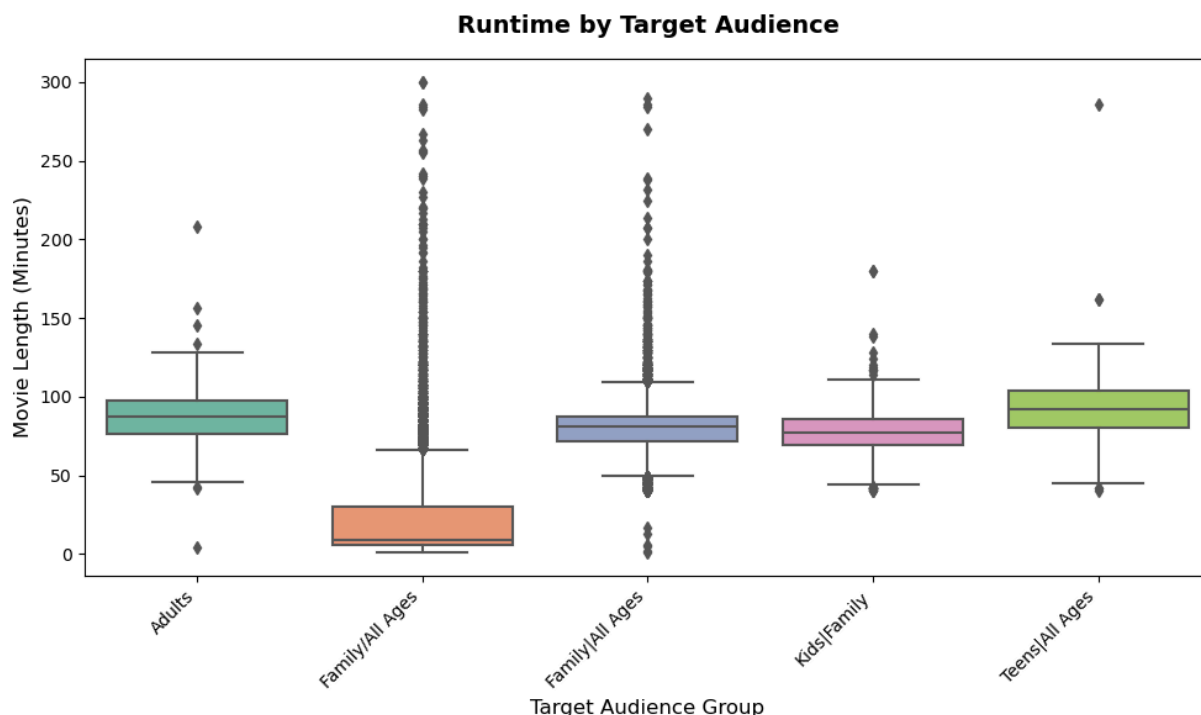
# 2. Render the distribution boxplots
sns.boxplot(x='Target_Audience', y='Movie_Length_Minutes', data=df, palette='Set2')

# 3. Add descriptive metadata
plt.title('Runtime by Target Audience', fontsize=14, fontweight='bold', pad=15)
plt.xlabel('Target Audience Group', fontsize=12)
plt.ylabel('Movie Length (Minutes)', fontsize=12)

# 4. Clean formatting: Rotate category names 45 degrees so they don't overlap
plt.xticks(rotation=45, ha='right')
plt.tight_layout()

# 5. Optional: Save the file outputs to your workspace directory
plt.savefig("runtime_by_audience.png", dpi=300, bbox_inches="tight")
plt.savefig("runtime_by_audience.pdf", bbox_inches="tight")

# 6. Dispatch the canvas to the screen (Crucial)
plt.show()
```



```
In [45]: import matplotlib.pyplot as plt
import seaborn as sns

# 1. Expand the canvas size to give the violins room to breathe
plt.figure(figsize=(12, 6))

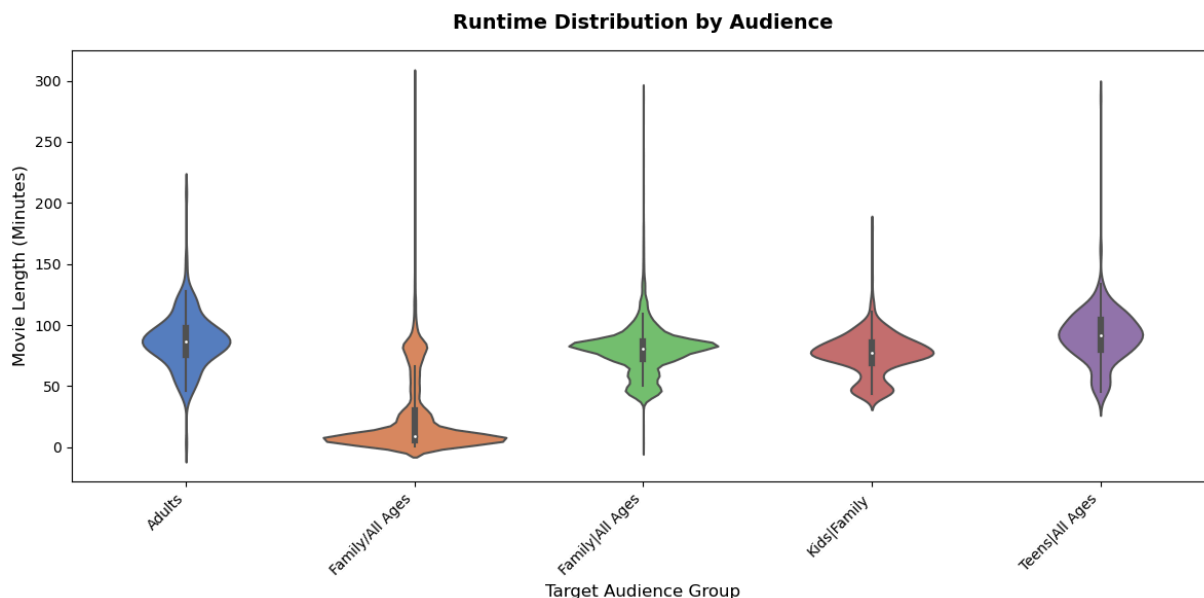
# 2. Render the distribution violin shapes
sns.violinplot(x='Target_Audience', y='Movie_Length_Minutes', data=df, palette='muted')

# 3. Add descriptive metadata and axis titles
plt.title('Runtime Distribution by Audience', fontsize=14, fontweight='bold', pad=10)
plt.xlabel('Target Audience Group', fontsize=12)
plt.ylabel('Movie Length (Minutes)', fontsize=12)

# 4. Clean formatting: Rotate category names 45 degrees so they never overlap
plt.xticks(rotation=45, ha='right')
plt.tight_layout()

# 5. Save the high-resolution file copies directly to your workspace directory
plt.savefig("runtime_violin_distribution.png", dpi=300, bbox_inches="tight")
plt.savefig("runtime_violin_distribution.pdf", bbox_inches="tight")

# 6. Dispatch the finalized canvas pipeline to the screen
plt.show()
```

```
In [46]: import matplotlib.pyplot as plt

# 1. Calculate the yearly release aggregates
counts_by_year = df.groupby('Release_Year').size()

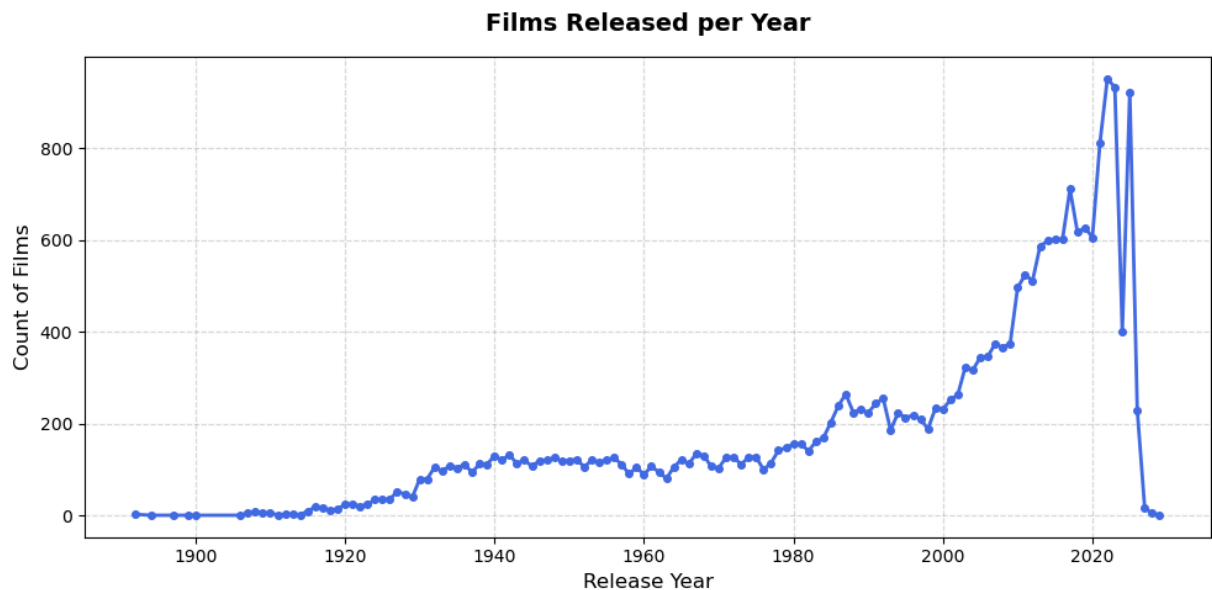
# 2. Initialize and configure the canvas layout
plt.figure(figsize=(10, 5))

# 3. Plot the data using the pandas line-plot wrapper
counts_by_year.plot(color='royalblue', linewidth=2, marker='o', markersize=4)

# 4. Add clear labels and visual metadata
plt.title('Films Released per Year', fontsize=14, fontweight='bold', pad=15)
plt.xlabel('Release Year', fontsize=12)
plt.ylabel('Count of Films', fontsize=12)
plt.grid(True, linestyle='--', alpha=0.5) # Adds a clean background grid for readability
plt.tight_layout()

# 5. Save the final visual assets
plt.savefig("films_released_per_year.png", dpi=300, bbox_inches="tight")
plt.savefig("films_released_per_year.pdf", bbox_inches="tight")

# 6. Render the completed canvas pipeline
plt.show()
```



```
In [47]: import matplotlib.pyplot as plt

# 1. Calculate the decade bins
df['Decade'] = (df['Release_Year'] // 10) * 10

# 2. Set up the canvas layout dimensions
plt.figure(figsize=(10, 5))

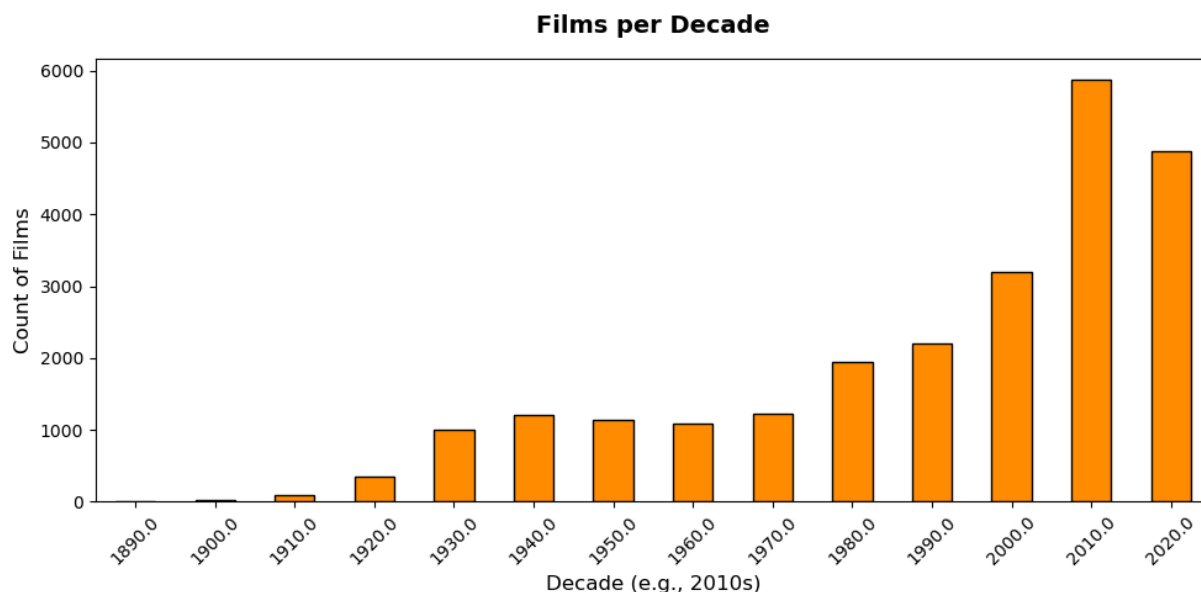
# 3. Calculate frequencies and plot as a categorical bar chart
df['Decade'].value_counts().sort_index().plot(kind='bar', color='darkorange', edgecolor='black')

# 4. Add clear labels and visual formatting
plt.title('Films per Decade', fontsize=14, fontweight='bold', pad=15)
plt.xlabel('Decade (e.g., 2010s)', fontsize=12)
plt.ylabel('Count of Films', fontsize=12)

# Rotate the decade labels slightly so they read cleanly
plt.xticks(rotation=45)
plt.tight_layout()

# 5. Save the visual assets locally
plt.savefig("films_per_decade.png", dpi=300, bbox_inches="tight")
plt.savefig("films_per_decade.pdf", bbox_inches="tight")

# 6. Render the canvas to your screen
plt.show()
```



```
In [48]: import matplotlib.pyplot as plt

# 1. Fetch the top 20 studios
top_studios = df['Studio'].value_counts().head(20)

# 2. Set up a taller canvas layout so 20 bars have breathing room
plt.figure(figsize=(10, 8))

# 3. Plot horizontal bars with a clean color palette
top_studios.plot(kind='barh', color='indigo', edgecolor='black')

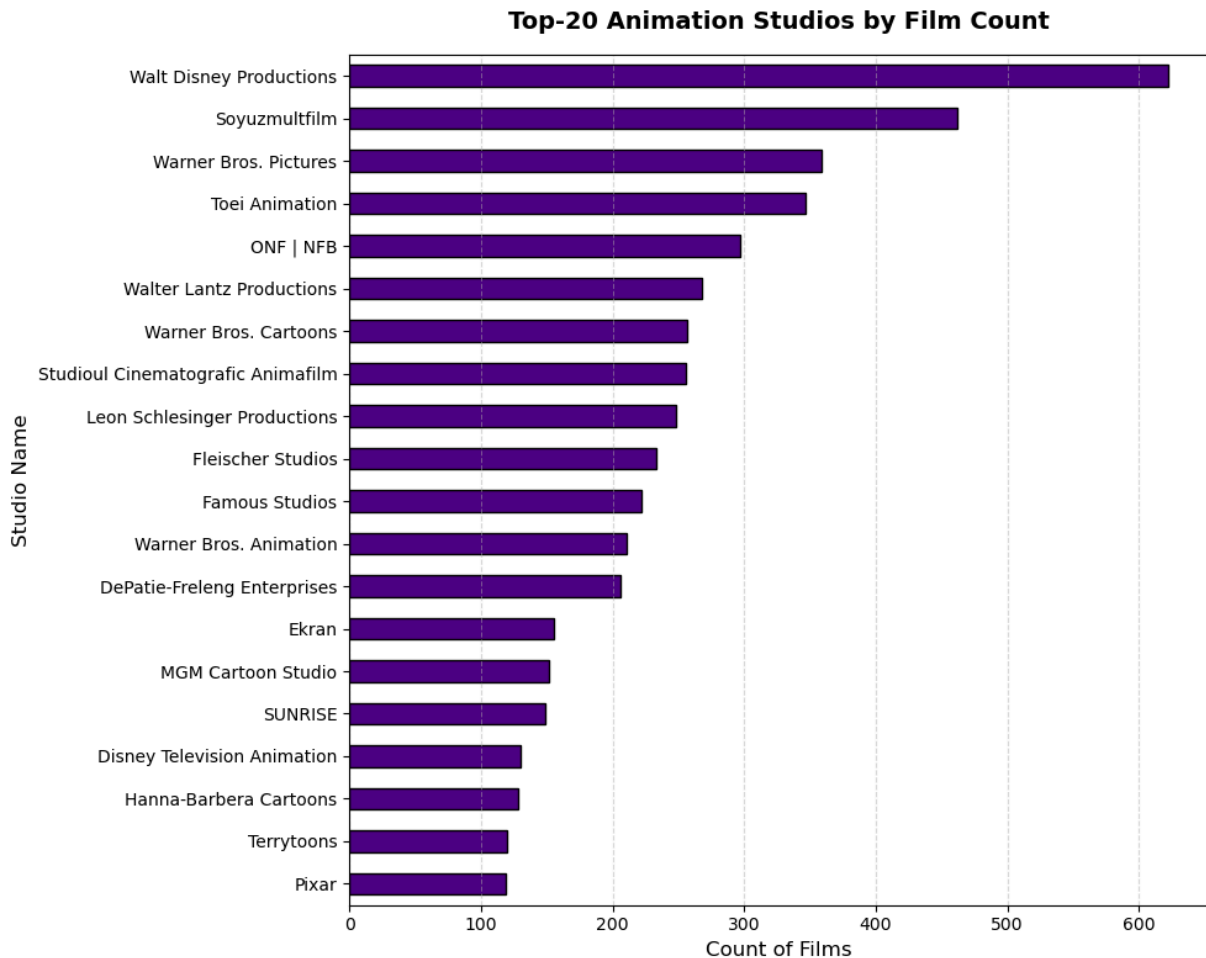
# 4. Invert the y-axis so the highest count is at the TOP of the graph
plt.gca().invert_yaxis()

# 5. Add text labels and metadata
plt.title('Top-20 Animation Studios by Film Count', fontsize=14, fontweight='bold',
plt.xlabel('Count of Films', fontsize=12)
plt.ylabel('Studio Name', fontsize=12)

# Add a grid along the x-axis to make tracking counts easier
plt.grid(axis='x', linestyle='--', alpha=0.5)
plt.tight_layout()

# 6. Save the final visual assets
plt.savefig("top_20_animation_studios.png", dpi=300, bbox_inches="tight")
plt.savefig("top_20_animation_studios.pdf", bbox_inches="tight")

# 7. Render to the screen
plt.show()
```



In [49]: *#Genre (Multi-Label) Analysis*

```
In [50]: import matplotlib.pyplot as plt
import pandas as pd
from sklearn.preprocessing import MultiLabelBinarizer

# 1. Split text by comma and strip empty spacing
genres_split = (
    df["Genre"]
    .dropna()
    .astype(str)
    .apply(lambda x: [g.strip() for g in x.split(",")])
)

# 2. Convert multi-labels into individual binary columns
mlb = MultiLabelBinarizer()
genre_dummies = pd.DataFrame(
    mlb.fit_transform(genres_split),
    columns=mlb.classes_,
    index=genres_split.index,
)

# 3. Sum up individual genres, sort them, and KEEP ONLY THE TOP 15
# (This completely stops the axis from clogging up!)
genre_counts = genre_dummies.sum().sort_values(ascending=False).head(15)
```

```

# 4. Initialize a clean canvas layout
plt.figure(figsize=(12, 6))

# 5. Create a clean vertical bar plot
genre_counts.plot(kind="bar", color="crimson", edgecolor="black")

# 6. Add structural labels
plt.title(
    "Top 15 Movie Genre Frequency (Cleaned Multi-label)",
    fontsize=14,
    fontweight="bold",
)
plt.xlabel("Movie Genre", fontsize=12)
plt.ylabel("Count of Films", fontsize=12)

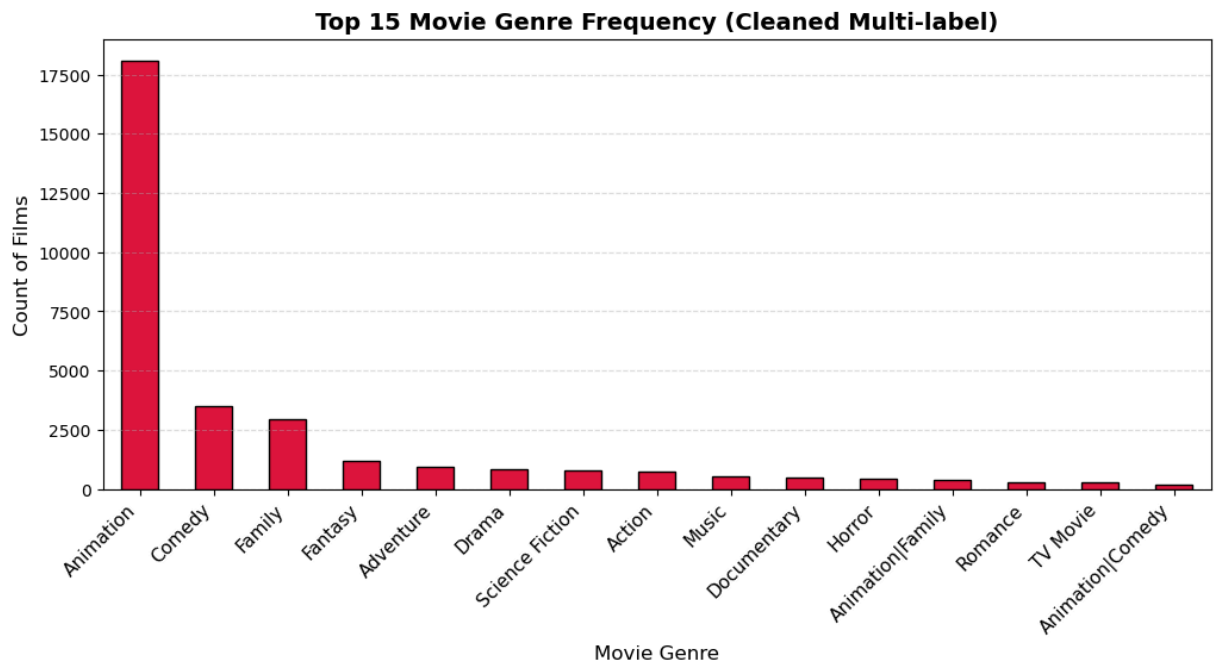
# 7. Format labels safely with plenty of bottom margin room
plt.xticks(rotation=45, ha="right", fontsize=11)
plt.grid(axis="y", linestyle="--", alpha=0.4)

# 8. Use explicit layout adjusting instead of tight_layout to prevent errors
plt.subplots_adjust(bottom=0.25)

# 9. Save the clean visual report
plt.savefig("genre_frequency_top15.png", dpi=300, bbox_inches="tight")
plt.savefig("genre_frequency_top15.pdf", bbox_inches="tight")

# 10. Display the clean chart
plt.show()

```



```

In [51]: import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

# 1. Filter the dummy matrix to include ONLY the Top 15 genres from your previous s
top_genres = genre_counts.index

```

```

genre_dummies_filtered = genre_dummies[top_genres]

# 2. Calculate the matrix dot product using the matched subsets
co_occ = genre_dummies_filtered.T.dot(genre_dummies_filtered)

# 3. Create a clean, square-proportioned plot window
plt.figure(figsize=(10, 8))

# 4. Generate the heatmap (annot=True adds individual crossover count numbers)
sns.heatmap(co_occ, cmap="Blues", annot=True, fmt="d", cbar=True)

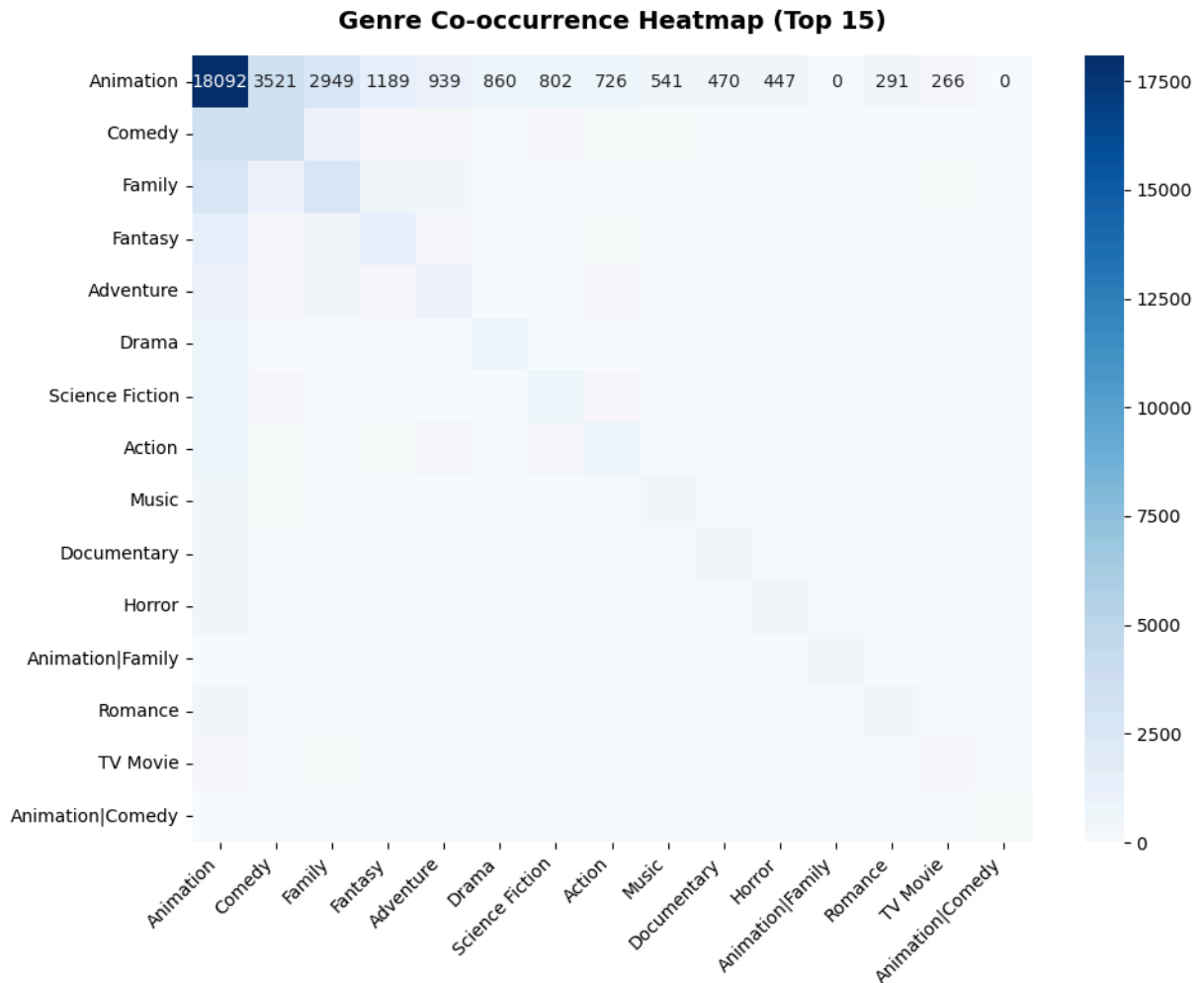
# 5. Add structural metadata titles
plt.title("Genre Co-occurrence Heatmap (Top 15)", fontsize=14, fontweight="bold", p

# 6. Rotate labels so they don't clip or crowd each other
plt.xticks(rotation=45, ha="right", fontsize=10)
plt.yticks(rotation=0, fontsize=10)
plt.tight_layout()

# 7. Save visual assets cleanly to your workspace
plt.savefig("genre_co_occurrence.png", dpi=300, bbox_inches="tight")
plt.savefig("genre_co_occurrence.pdf", bbox_inches="tight")

# 8. Render to screen last
plt.show()

```



```
In [52]: import matplotlib.pyplot as plt

# 1. Calculate individual label counts per row
num_labels = genres_split.apply(len)

# 2. Print out the mathematical summary metrics
print(f"Average genres per film: {num_labels.mean():.2f}")
print(f"Max genres on a single film: {num_labels.max()}")

# 3. Setup the visual chart canvas
plt.figure(figsize=(8, 5))

# 4. Generate the histogram, using center-aligned bins for clean integer bars
# (align='left' makes the bars center exactly over numbers like 1, 2, 3...)
num_labels.hist(
    bins=range(1, num_labels.max() + 2),
    color="mediumpurple",
    edgecolor="black",
    align="left",
    grid=False,
)

# 5. Add descriptive labels and formatting
plt.title("Distribution of Genre Count per Film", fontsize=14, fontweight="bold", p
plt.xlabel("Number of Genres Tagged", fontsize=12)
plt.ylabel("Count of Films", fontsize=12)

# Force the X-axis ticks to show clean whole numbers only
plt.xticks(range(1, num_labels.max() + 1))
plt.tight_layout()

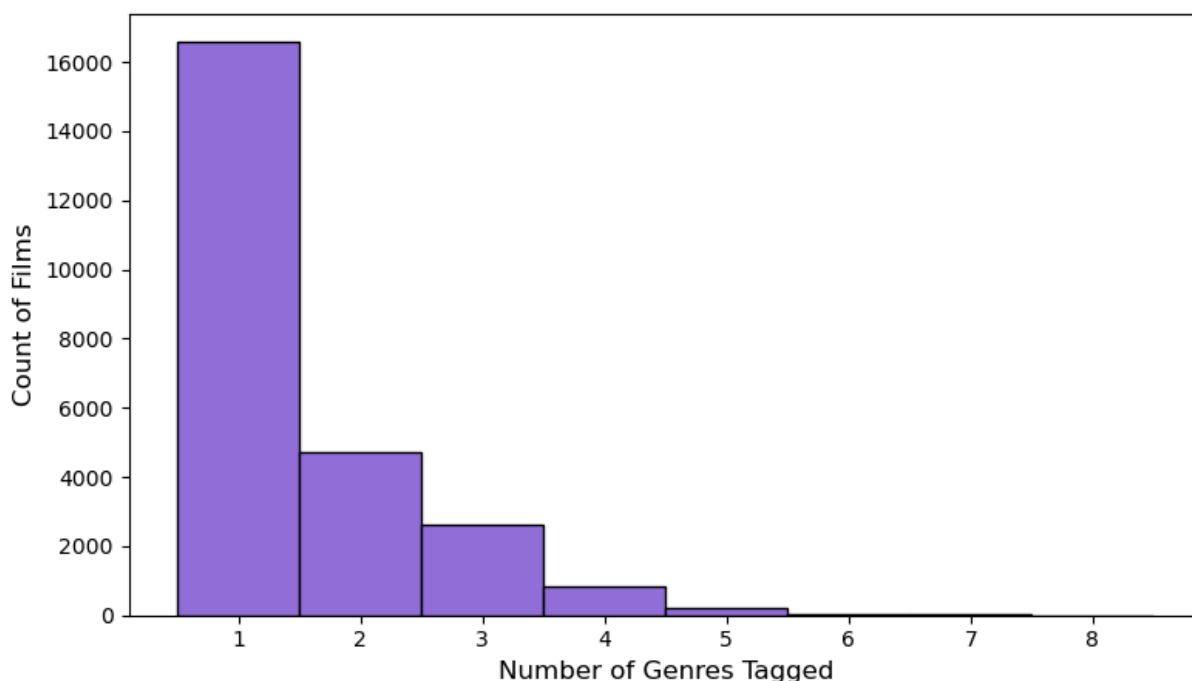
# 6. Save the final visual report assets
plt.savefig("genre_count_per_film.png", dpi=300, bbox_inches="tight")
plt.savefig("genre_count_per_film.pdf", bbox_inches="tight")

# 7. Render cleanly to the screen
plt.show()
```

Average genres per film: 1.55

Max genres on a single film: 8

Distribution of Genre Count per Film



In [53]: `#text analysis`

In [54]: `import sys`
`!{sys.executable} -m pip install wordcloud`

Requirement already satisfied: wordcloud in c:\users\manas\anaconda3\lib\site-packages (1.9.6)

Requirement already satisfied: numpy>=1.19.3 in c:\users\manas\anaconda3\lib\site-packages (from wordcloud) (1.26.4)

Requirement already satisfied: pillow in c:\users\manas\anaconda3\lib\site-packages (from wordcloud) (10.2.0)

Requirement already satisfied: matplotlib in c:\users\manas\anaconda3\lib\site-packages (from wordcloud) (3.8.0)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (1.2.0)

Requirement already satisfied: cycler>=0.10 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (4.63.0)

Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (1.4.4)

Requirement already satisfied: packaging>=20.0 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (23.1)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (3.0.9)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (2.8.2)

Requirement already satisfied: six>=1.5 in c:\users\manas\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib->wordcloud) (1.16.0)

In [55]: `pip install wordcloud`

Requirement already satisfied: wordcloud in c:\users\manas\anaconda3\lib\site-packages (1.9.6)

Requirement already satisfied: numpy>=1.19.3 in c:\users\manas\anaconda3\lib\site-packages (from wordcloud) (1.26.4)

Requirement already satisfied: pillow in c:\users\manas\anaconda3\lib\site-packages (from wordcloud) (10.2.0)

Requirement already satisfied: matplotlib in c:\users\manas\anaconda3\lib\site-packages (from wordcloud) (3.8.0)

Requirement already satisfied: contourpy>=1.0.1 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (1.2.0)

Requirement already satisfied: cyclor>=0.10 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (0.11.0)

Requirement already satisfied: fonttools>=4.22.0 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (4.63.0)

Requirement already satisfied: kiwisolver>=1.0.1 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (1.4.4)

Requirement already satisfied: packaging>=20.0 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (23.1)

Requirement already satisfied: pyparsing>=2.3.1 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (3.0.9)

Requirement already satisfied: python-dateutil>=2.7 in c:\users\manas\anaconda3\lib\site-packages (from matplotlib->wordcloud) (2.8.2)

Requirement already satisfied: six>=1.5 in c:\users\manas\anaconda3\lib\site-packages (from python-dateutil>=2.7->matplotlib->wordcloud) (1.16.0)

Note: you may need to restart the kernel to use updated packages.

```
In [56]: import re
import matplotlib.pyplot as plt
from nltk.corpus import stopwords
from wordcloud import WordCloud

# 1. Clean up text data: Lowercasing, dropping empty cells, stripping punctuation
text = " ".join(df["Overview"].dropna().str.lower())
text = re.sub(r"^[^a-z\s]", "", text)

# 2. Get standard English stopwords
stop = set(stopwords.words("english"))

# 3. ADD EXTRA GENERIC FILLER WORDS TO STRIP OUT (Custom Stopwords)
custom_words = {"one", "two", "world", "film", "movie", "story", "find", "take"}
stop.update(custom_words)

# 4. Filter the text data using the expanded dictionary block
words = [w for w in text.split() if w not in stop]
clean_text = " ".join(words)

# 5. Build and render the polished word cloud canvas layout
plt.figure(figsize=(12, 7))
wc = WordCloud(
    width=1000, height=600, background_color="white", max_words=150
).generate(clean_text)

plt.imshow(wc, interpolation="bilinear")
plt.axis("off")
plt.title(
    "Thematic Word Cloud of Movie Overviews",
```

```

    fontsize=16,
    fontweight="bold",
    pad=15,
)
plt.tight_layout()

# 6. Save the fresh, updated image files safely
plt.savefig("movie_overviews_wordcloud_clean.png", dpi=300, bbox_inches="tight")
plt.savefig("movie_overviews_wordcloud_clean.pdf", bbox_inches="tight")

plt.show()

```

Thematic Word Cloud of Movie Overviews



```
In [57]: import matplotlib.pyplot as plt
import pandas as pd
from sklearn.feature_extraction.text import CountVectorizer

# 1. RUN THE VECTORIZER DATA PROCESSING FIRST
vectorizer = CountVectorizer(max_features=30, stop_words="english")
X = vectorizer.fit_transform(df["Overview"].dropna())
freqs = zip(vectorizer.get_feature_names_out(), X.sum(axis=0).tolist()[0])
top_words = sorted(freqs, key=lambda x: x[1], reverse=True)

# 2. UNPACK THE WORD MATRIX PAIRS
word_labels = [pair[0] for pair in top_words]
word_counts = [pair[1] for pair in top_words]

# 3. STRUCTURE A DATAFRAME FOR CLEAN SLICING
chart_df = pd.DataFrame({"Word": word_labels, "Frequency": word_counts})

# 4. INITIALIZE THE CANVAS LAYOUT (Taller height helps 30 rows breathe)
plt.figure(figsize=(10, 8))
```

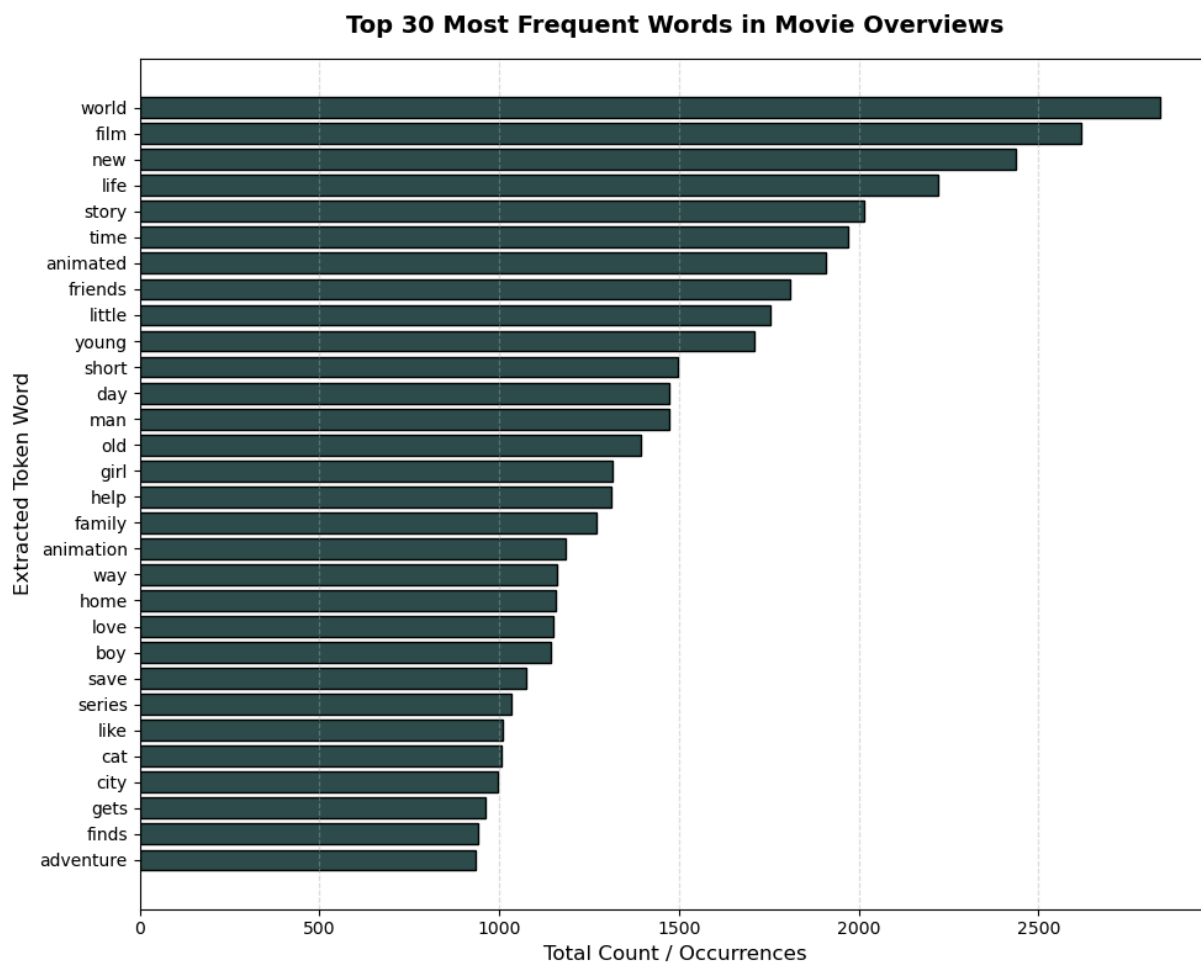
```
# 5. GENERATE THE HORIZONTAL BARS
plt.barh(
    chart_df["Word"],
    chart_df["Frequency"],
    color="darkslategray",
    edgecolor="black",
)

# 6. INVERT THE Y-AXIS (Puts the #1 most frequent word at the top)
plt.gca().invert_yaxis()

# 7. ADD LABELS AND GRID
plt.title(
    "Top 30 Most Frequent Words in Movie Overviews",
    fontsize=14,
    fontweight="bold",
    pad=15,
)
plt.xlabel("Total Count / Occurrences", fontsize=12)
plt.ylabel("Extracted Token Word", fontsize=12)
plt.grid(axis="x", linestyle="--", alpha=0.4)
plt.tight_layout()

# 8. SAVE THE HIGH-QUALITY VISUAL ASSETS
plt.savefig("overview_word_frequencies.png", dpi=300, bbox_inches="tight")
plt.savefig("overview_word_frequencies.pdf", bbox_inches="tight")

# 9. DISPATCH CANVAS ENGINE TO THE SCREEN
plt.show()
```



```
In [58]: import warnings
# Suppress any future pandas/seaborn configuration warning flags
warnings.filterwarnings('ignore')

import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sns

# 1. Target columns to calculate
numeric_cols = [
    "TMDB_Rating",
    "TMDB_Vote_Count",
    "Budget_Million_USD",
    "Box_Office_Million_USD",
    "TMDB_Popularity",
    "Movie_Length_Minutes",
]

# 2. DATA CLEANING: Force convert columns to numeric, shifting text strings safely
df_numeric_clean = df[numeric_cols].apply(pd.to_numeric, errors="coerce")

# 3. Compute the linear correlation matrix (Pearson coefficient)
corr = df_numeric_clean.corr()

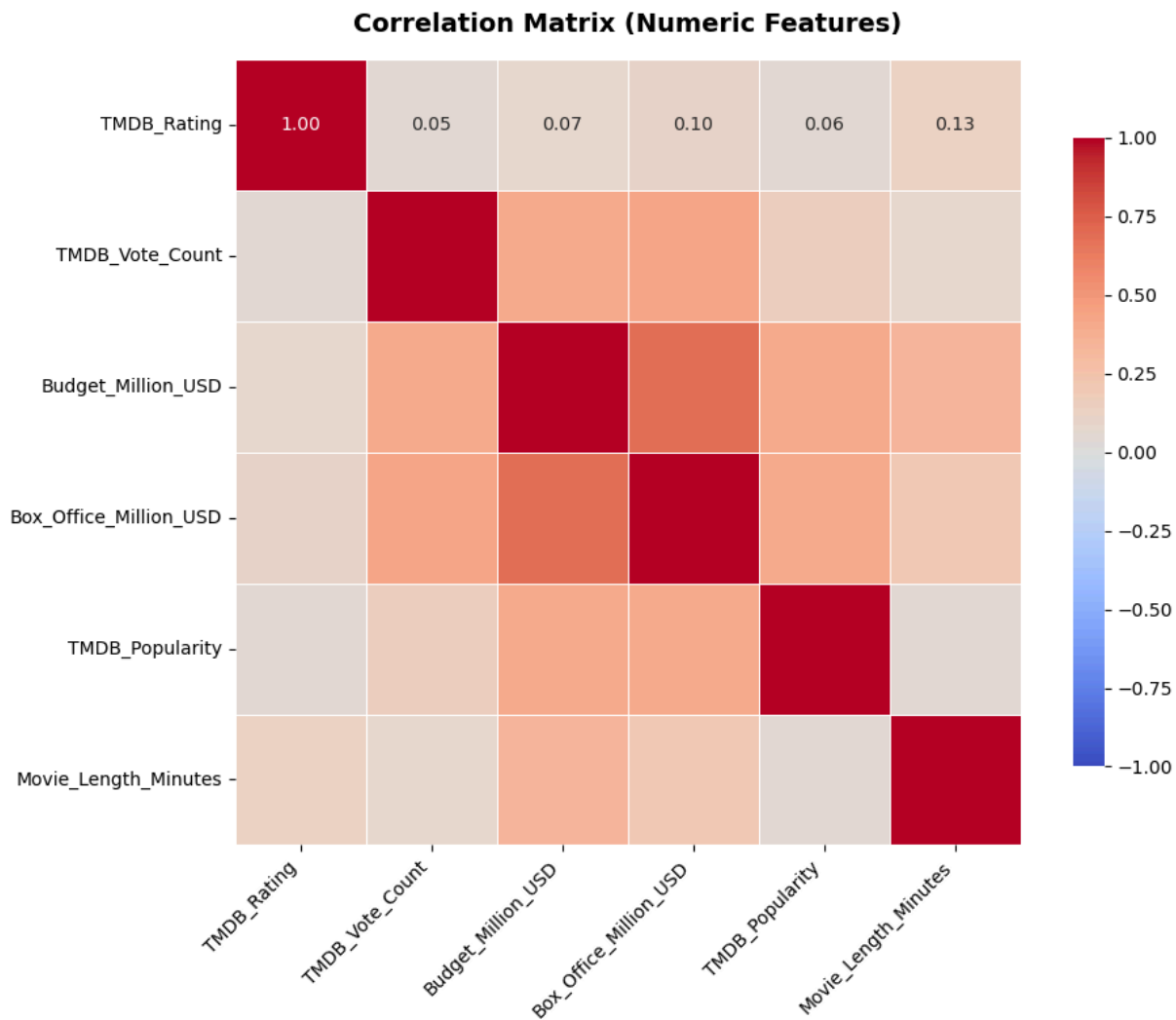
# 4. Initialize a square proportions frame window
plt.figure(figsize=(10, 8))
```

```
# 5. Generate the heatmap with balanced settings
# (vmin/vmax anchors the scale color depth strictly between -1 and +1)
sns.heatmap(
    corr,
    annot=True,
    fmt=".2f",
    cmap="coolwarm",
    vmin=-1,
    vmax=1,
    square=True,
    linewidths=0.5,
    cbar_kws={"shrink": 0.8},
)

# 6. Add structural labels
plt.title(
    "Correlation Matrix (Numeric Features)", fontsize=14, fontweight="bold", pad=15
)
plt.xticks(rotation=45, ha="right", fontsize=10)
plt.yticks(rotation=0, fontsize=10)
plt.tight_layout()

# 7. Save high-resolution files safely to your directory
plt.savefig("numeric_feature_correlation.png", dpi=300, bbox_inches="tight")
plt.savefig("numeric_feature_correlation.pdf", bbox_inches="tight")

# 8. Render cleanly to the screen
plt.show()
```

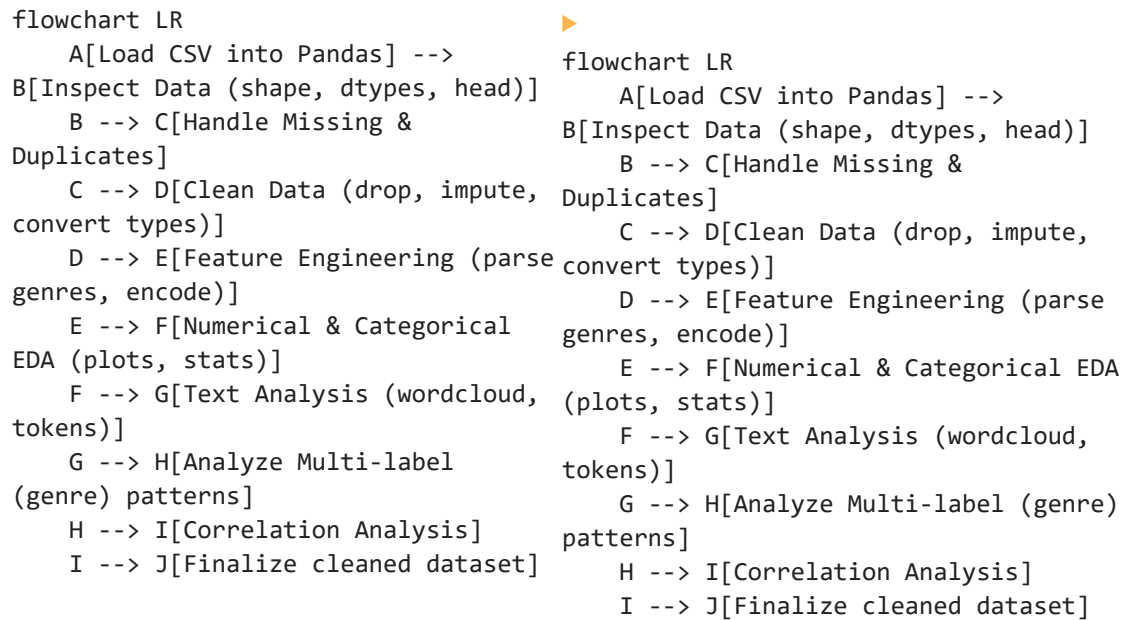


```
In [59]: #Outliers and Missing-Data Strategy
```

```
In [60]: # 1. Fill missing budget numeric fields with the median value
df['Budget_Million_USD'] = df['Budget_Million_USD'].fillna(df['Budget_Million_USD'].median())

# 2. Fill missing text studio categories with a generic placeholder string
df['Studio'] = df['Studio'].fillna('Unknown')
```

```
In [61]: import random, numpy as np
random.seed(42)
np.random.seed(42)
```



```

In [62]: import matplotlib.pyplot as plt

# 1. Define the steps in your pipeline
steps = [
    "1. Load CSV into Pandas",
    "2. Inspect Data (shape, head)",
    "3. Handle Missing & Duplicates",
    "4. Clean Data & Convert Types",
    "5. Feature Engineering (Genres)",
    "6. Numerical/Categorical EDA",
    "7. Text Analysis (WordCloud)",
    "8. Multi-label Patterns",
    "9. Correlation Analysis",
    "10. Finalize Cleaned Dataset"
]

# 2. Setup a tall, clean canvas
fig, ax = plt.subplots(figsize=(6, 10))

# 3. Draw boxes and arrows sequentially
for i, step in enumerate(steps):
    y_pos = 10 - (i * 1.0)

    # Draw a rounded box for the step text
    ax.text(0.5, y_pos, step, ha='center', va='center', fontsize=11, fontweight='bold',
           bbox=dict(boxstyle='round,pad=0.6', facecolor='aliceblue', edgecolor='black'))

    # Draw an arrow pointing down to the next box (except for the last item)
    if i < len(steps) - 1:
        ax.annotate('', xy=(0.5, y_pos - 0.7), xytext=(0.5, y_pos - 0.3),
                   arrowprops=dict(arrowstyle="->", color='crimson', lw=2, mutation_scale=80))

# 4. Clean up boundaries and remove axis borders completely
ax.set_xlim(0, 1)
ax.set_ylim(0.5, 11)

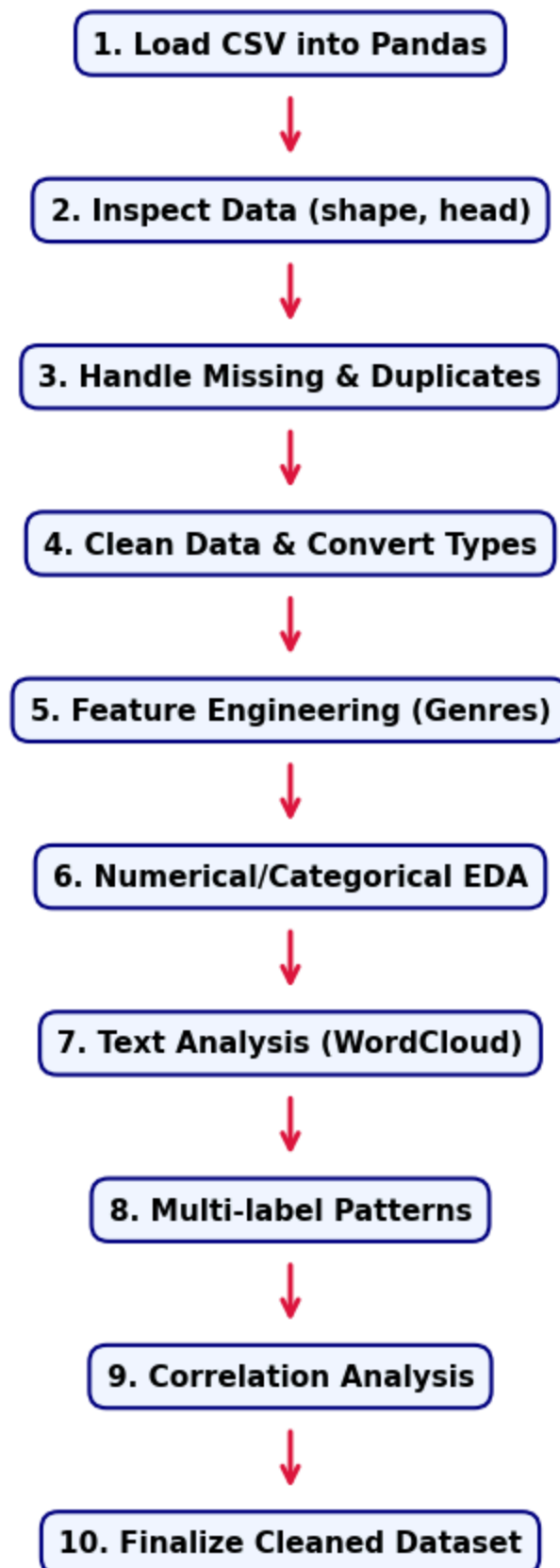
```

```
plt.axis('off')
plt.title("Your Data Pipeline & EDA Workflow", fontsize=14, fontweight='bold', pad=
plt.tight_layout()

# 5. Save a high-res backup copy locally
plt.savefig("data_pipeline_flowchart.png", dpi=300, bbox_inches="tight")

# 6. Force display on screen
plt.show()
```


Your Data Pipeline & EDA Workflow



```

In [63]: import pandas as pd
from sklearn.model_selection import train_test_split

# -----
# 1. DEFINE YOUR TARGET GENRES (Y)
# -----
# We use the clean multi-label dummies DataFrame we built earlier
Y = genre_dummies

# -----
# 2. DEFINE AND SELECT YOUR PREDICTIVE FEATURES (X)
# -----
# List of columns to KEEP based on your rationale table
features_to_keep = [
    'Overview',          # Text feature
    'Theme',             # Text feature
    'Animation_Style',   # Categorical feature
    'Target_Audience',  # Categorical feature
    'MPAA_Rating',       # Categorical feature
    'Release_Year',      # Numeric temporal feature
    'Decade',            # Ordinal/Categorical engineered feature
    'Studio'             # Categorical feature
]

# Create your final feature DataFrame X
X_df = df[features_to_keep].copy()

# -----
# 3. HANDLE MISSING VALUES IN KEPT FEATURES
# -----

# Fill text columns with empty strings so text models don't crash
X_df['Overview'] = X_df['Overview'].fillna('')
X_df['Theme'] = X_df['Theme'].fillna('')

# List of categorical features
categorical_cols = ['Animation_Style', 'Target_Audience', 'MPAA_Rating', 'Studio',

for col in categorical_cols:
    if col in X_df.columns:
        # CONVERT TO STRING FIRST (This unlocks the column and fixes the TypeError!
        X_df[col] = X_df[col].astype(str)
        # Now you can safely fill the missing slots
        X_df[col] = X_df[col].fillna('Unknown')

# Fill missing Release_Year fields with the dataset median
X_df['Release_Year'] = X_df['Release_Year'].fillna(X_df['Release_Year'].median())

# -----
# 4. PRINT VERIFICATION SUMMARY
# -----
print("--- FEATURE SELECTION VERIFICATION ---")

```

```

print(f"Total rows in final dataset: {X_df.shape[0]}")
print(f"Number of keeping features: {X_df.shape[1]}")
print(f"Number of target genres to predict: {Y.shape[1]}")
print("\nMissing values remaining in features:")
print(X_df.isnull().sum())

# -----
# 5. SPLIT INTO TRAIN AND TEST SETS (80% / 20%)
# -----
X_train, X_test, Y_train, Y_test = train_test_split(X_df, Y, test_size=0.2, random_
print(f"\nTraining set size: {X_train.shape[0]} rows")
print(f"Testing set size: {X_test.shape[0]} rows")

```

```

--- FEATURE SELECTION VERIFICATION ---
Total rows in final dataset: 25010
Number of keeping features: 8
Number of target genres to predict: 1977

```

Missing values remaining in features:

```

Overview      0
Theme         0
Animation_Style  0
Target_Audience  0
MPAA_Rating   0
Release_Year   0
Decade         0
Studio        0
dtype: int64

```

```

Training set size: 20008 rows
Testing set size: 5002 rows

```

In [64]: *#TEXT PROCESSING*

```

In [65]: import re
import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from sklearn.decomposition import TruncatedSVD
from sklearn.feature_extraction.text import TfidfVectorizer
import nltk
from nltk.stem import WordNetLemmatizer

# Ensure required NLTK dictionary download files are loaded locally
try:
    nltk.data.find("corpora/wordnet")
except LookupError:
    nltk.download("wordnet")
    nltk.download("omw-1.4")

# Initialize the Lemmatizer
lemmatizer = WordNetLemmatizer()

# Words that should be removed after cleaning
custom_stopwords = {
    "ha", "wa", "film", "movie", "story"

```

```

}

def clean_and_lemmatize(text_string):

    # Convert to Lowercase
    text_string = str(text_string).lower()

    # Remove punctuation and numbers
    text_string = re.sub(r"^[a-z\s]", "", text_string)

    # Tokenize
    words = text_string.split()

    # Lemmatize
    words = [lemmatizer.lemmatize(word) for word in words]

    # Remove meaningless words
    words = [
        word for word in words
        if word not in custom_stopwords and len(word) > 2
    ]

    return " ".join(words)

# Apply the processing pipeline to both text inputs
print("Cleaning and lemmatizing text columns...")
X_train["Clean_Overview"] = X_train["Overview"].apply(clean_and_lemmatize)
X_test["Clean_Overview"] = X_test["Overview"].apply(clean_and_lemmatize)

X_train["Clean_Theme"] = X_train["Theme"].apply(clean_and_lemmatize)
X_test["Clean_Theme"] = X_test["Theme"].apply(clean_and_lemmatize)

# -----
# STEP 2: TF-IDF VECTORIZATION (Unigrams + Bigrams)
# -----
print("Vectorizing text via TF-IDF...")
overview_vectorizer = TfidfVectorizer(
    ngram_range=(1, 2), min_df=0.005, max_df=0.7, stop_words="english"
)

# Fit on training data and transform both sets
X_train_overview_tfidf = overview_vectorizer.fit_transform(
    X_train["Clean_Overview"]
)
X_test_overview_tfidf = overview_vectorizer.transform(X_test["Clean_Overview"])

print(
    f"Extracted Overview vocabulary terms: {X_train_overview_tfidf.shape[1]}"
)

# -----
# STEP 3: DIMENSIONALITY REDUCTION (TruncatedSVD / LSA)
# -----
print("Applying TruncatedSVD dimensionality reduction...")

```

```

# Define target components to evaluate variance saturation curves
n_components = min(150, X_train_overview_tfidf.shape[1] - 1)
svd = TruncatedSVD(n_components=n_components, random_state=42)

# Fit and project the large sparse matrix down to compact numerical columns
X_train_overview_svd = svd.fit_transform(X_train_overview_tfidf)
X_test_overview_svd = svd.transform(X_test_overview_tfidf)

# Calculate cumulative explained variance ratio coordinates
cumulative_variance = np.cumsum(svd.explained_variance_ratio_)

# -----
# STEP 4: GENERATE AND SAVE THE SCREE PLOT
# -----
plt.figure(figsize=(10, 5))
plt.plot(
    range(1, len(cumulative_variance) + 1),
    cumulative_variance,
    marker="o",
    linestyle="--",
    color="darkorange",
    markersize=4,
)

# Style plot metadata structures
plt.title(
    "SVD Scree Plot: Cumulative Explained Variance vs. Components",
    fontsize=14,
    fontweight="bold",
    pad=15,
)
plt.xlabel("Number of SVD Components (Axes)", fontsize=12)
plt.ylabel("Total Explained Variance Ratio", fontsize=12)
plt.grid(True, linestyle="--", alpha=0.5)

# Highlight typical 100-component cut-off markers dynamically
if len(cumulative_variance) >= 100:
    plt.axvline(
        x=100,
        color="crimson",
        linestyle=":",
        label=f"100 Comp Variance: {cumulative_variance[99]:.2f}",
    )
    plt.legend(loc="lower right")

plt.tight_layout()

# Save visual documents locally to your active directory paths
plt.savefig("svd_scree_variance_plot.png", dpi=300, bbox_inches="tight")
plt.savefig("svd_scree_variance_plot.pdf", bbox_inches="tight")

print("Pipeline execution complete! Scree plot images saved to directory.")
plt.show()

```

```
[nltk_data] Downloading package wordnet to
[nltk_data] C:\Users\manas\AppData\Roaming\nltk_data...
[nltk_data] Package wordnet is already up-to-date!
[nltk_data] Downloading package omw-1.4 to
[nltk_data] C:\Users\manas\AppData\Roaming\nltk_data...
[nltk_data] Package omw-1.4 is already up-to-date!
```

Cleaning and lemmatizing text columns...

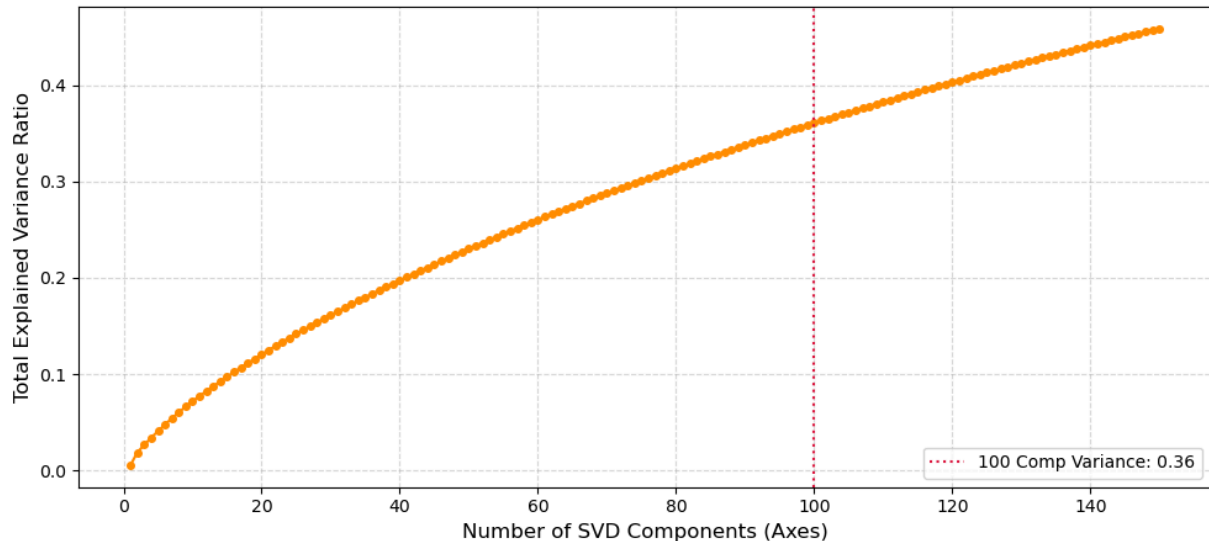
Vectorizing text via TF-IDF...

Extracted Overview vocabulary terms: 642

Applying TruncatedSVD dimensionality reduction...

Pipeline execution complete! Scree plot images saved to directory.

SVD Scree Plot: Cumulative Explained Variance vs. Components



```
In [66]: from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler

# 1. Define feature categories based on your rationale table
categorical_features = ['Animation_Style', 'Target_Audience', 'MPAA_Rating', 'Studi
numeric_features = ['Release_Year']

# 2. Build the transformer pipeline for the non-text features
preprocessor = ColumnTransformer(
    transformers=[
        ('cat', OneHotEncoder(handle_unknown='ignore', sparse_output=False), catego
        ('num', StandardScaler(), numeric_features)
    ]
)

# 3. Fit and transform the metadata columns
print("Encoding categorical columns and scaling numeric years...")
X_train_meta = preprocessor.fit_transform(X_train)
X_test_meta = preprocessor.transform(X_test)

# 4. Concatenate text SVD arrays with metadata arrays horizontally
import numpy as np
X_train_final = np.hstack((X_train_overview_svd, X_train_meta))
X_test_final = np.hstack((X_test_overview_svd, X_test_meta))

print("\n--- COMBINED MATRIX VERIFICATION ---")
```

```
print(f"Final Training Data Shape: {X_train_final.shape} (Rows, Combined Features)")
print(f"Final Testing Data Shape: {X_test_final.shape} (Rows, Combined Features)")
```

Encoding categorical columns and scaling numeric years...

--- COMBINED MATRIX VERIFICATION ---

Final Training Data Shape: (20008, 4651) (Rows, Combined Features)

Final Testing Data Shape: (5002, 4651) (Rows, Combined Features)

```
In [67]: import matplotlib.pyplot as plt
import pandas as pd

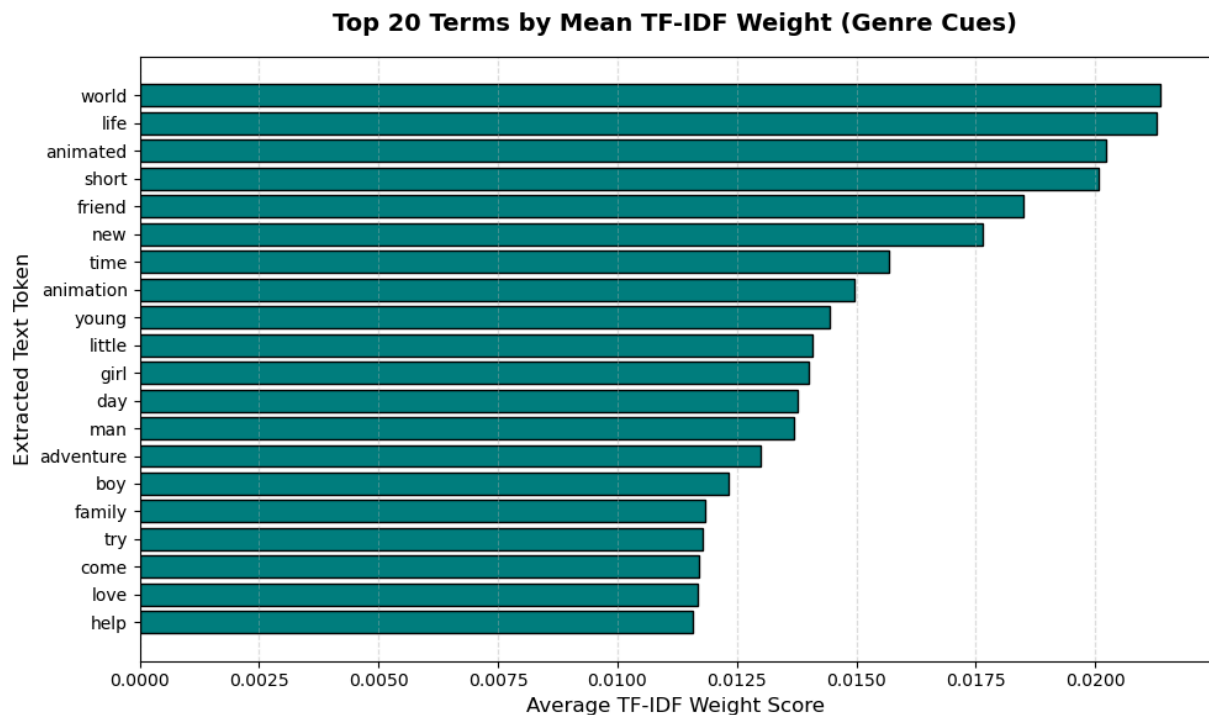
# 1. Calculate the mean TF-IDF score for each word across all rows
tfidf_means = np.asarray(X_train_overview_tfidf.mean(axis=0)).ravel()
words_list = overview_vectorizer.get_feature_names_out()

# 2. Combine into a DataFrame and pull the top 20 words
tfidf_df = pd.DataFrame({'Word': words_list, 'Mean_TFIDF': tfidf_means})
top_tfidf_words = tfidf_df.sort_values(by='Mean_TFIDF', ascending=False).head(20)

# 3. Plot the top words chart
plt.figure(figsize=(10, 6))
plt.barh(top_tfidf_words['Word'], top_tfidf_words['Mean_TFIDF'], color='teal', edge
plt.gca().invert_yaxis() # Put the most important word on top

plt.title("Top 20 Terms by Mean TF-IDF Weight (Genre Cues)", fontsize=14, fontweigh
plt.xlabel("Average TF-IDF Weight Score", fontsize=12)
plt.ylabel("Extracted Text Token", fontsize=12)
plt.grid(axis='x', linestyle='--', alpha=0.4)
plt.tight_layout()

# Save the visual file copy
plt.savefig("top_tfidf_vocabulary_terms.png", dpi=300, bbox_inches="tight")
plt.show()
```



In [68]: *#3. Categorical Encoding*

```
In [69]: import numpy as np
import pandas as pd
from sklearn.base import BaseEstimator, TransformerMixin
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder

# -----
# 1. CUSTOM MULTI-LABEL TARGET ENCODER CLASS
# -----
class MultiLabelTargetEncoder(BaseEstimator, TransformerMixin):
    """Replaces each category (e.g., Studio) with its smoothed target genre
    probabilities: P(Genre_j | Studio=s).
    """

    def __init__(self, smoothing=10.0):
        self.smoothing = smoothing
        self.studio_maps_ = {}
        self.global_means_ = None

    def fit(self, X, Y):
        # Ensure input series structure is extracted correctly
        X_series = pd.Series(np.asarray(X).ravel())
        Y_df = pd.DataFrame(np.asarray(Y))

        # Calculate the base global averages for each genre across all data
        self.global_means_ = Y_df.mean().to_dict()

        # Compute smoothed conditional probabilities for every unique category value
        for col in Y_df.columns:
            stats = Y_df.groupby(X_series)[col].agg(["count", "mean"])
            # Apply Bayesian m-estimate smoothing formula to prevent overfitting noise
            smoothed_vals = (
                stats["count"] * stats["mean"]
                + self.smoothing * self.global_means_[col]
            ) / (stats["count"] + self.smoothing)
            self.studio_maps_[col] = smoothed_vals.to_dict()
        return self

    def transform(self, X):
        X_series = pd.Series(np.asarray(X).ravel())
        encoded_out = np.zeros((len(X_series), len(self.studio_maps_)))

        # Map the lookup tables back onto the dataset values
        for col, mapping in self.studio_maps_.items():
            encoded_out[:, col] = X_series.map(mapping).fillna(
                self.global_means_[col]
            )
        return encoded_out

# -----
```



```

# 2. SEPARATE ENCODING MANIFEST DEFINITIONS
# -----
# Low-cardinality nominal and temporal variables chosen for One-Hot
one_hot_cols = ["Animation_Style", "MPAA_Rating", "Target_Audience", "Decade"]

# Define the global metadata processing matrix
preprocessor = ColumnTransformer(
    transformers=[
        (
            "onehot",
            OneHotEncoder(handle_unknown="ignore", sparse_output=False),
            one_hot_cols,
        )
    ],
    remainder="passthrough", # Preserves numeric features like Release_Year
)

# -----
# 3. EXECUTE TARGET ENCODING ON HIGH-CARDINALITY COLUMNS
# -----
print("Calculating smoothed target genre probabilities for 'Studio'...")
studio_encoder = MultiLabelTargetEncoder(smoothing=15.0)

# Fit on training data using your multi-label matrix context, then transform
X_train_studio_encoded = studio_encoder.fit_transform(
    X_train["Studio"], Y_train
)
X_test_studio_encoded = studio_encoder.transform(X_test["Studio"])

# 4. EXECUTE ONE-HOT ENCODING ON REMAINING CATEGORIES
print("Applying One-Hot Encoding to low-cardinality categorical arrays...")
X_train_meta_encoded = preprocessor.fit_transform(
    X_train[one_hot_cols + ["Release_Year"]]
)
X_test_meta_encoded = preprocessor.transform(
    X_test[one_hot_cols + ["Release_Year"]]
)

# -----
# 5. ASSEMBLE INDEPENDENT FEATURE BLOCKS HORIZONTALLY
# -----
# Combine: [Text SVD (100) + One-Hot Categories + Scaled Year + Studio Target Proba
X_train_final = np.hstack(
    (X_train_overview_svd, X_train_meta_encoded, X_train_studio_encoded)
)
X_test_final = np.hstack(
    (X_test_overview_svd, X_test_meta_encoded, X_test_studio_encoded)
)

print("\n--- CATEGORICAL ENCODING SUCCESS ---")
print(f"Original text SVD dimension features: {X_train_overview_svd.shape[1]}")
print(f"Appended matrix metadata columns: {X_train_meta_encoded.shape[1]}")
print(f"Appended high-cardinality Studio target features: {X_train_studio_encoded.s
print(f"--> Combined Final Dataset shape matrix: {X_train_final.shape}")

```

Calculating smoothed target genre probabilities for 'Studio'...
 Applying One-Hot Encoding to low-cardinality categorical arrays...

```
--- CATEGORICAL ENCODING SUCCESS ---
Original text SVD dimension features: 150
Appended matrix metadata columns: 145
Appended high-cardinality Studio target features: 1977
--> Combined Final Dataset shape matrix: (20008, 2272)
```

```
In [70]: import time
import pandas as pd
import numpy as np
from sklearn.preprocessing import MultiLabelBinarizer
from sklearn.linear_model import LogisticRegression
from sklearn.multioutput import MultiOutputClassifier
from sklearn.metrics import classification_report

print("--- STEP 1: CLEANING AND RE-SPLITTING TARGET GENRES ---")
# 1. Safely split the genres using the pipe symbol visible in your dataset
genres_split = df['Genre'].dropna().astype(str).str.split('|')

# 2. Convert multi-labels into clean individual binary columns
mlb = MultiLabelBinarizer()
genre_dummies_clean = pd.DataFrame(mlb.fit_transform(genres_split),
                                   columns=mlb.classes_,
                                   index=genres_split.index)

# 3. Filter down to the Top 15 most frequent true genres to ensure perfect accuracy
top_15_genres = genre_dummies_clean.sum().sort_values(ascending=False).head(15).index
Y_clean = genre_dummies_clean[top_15_genres]

# 4. Re-split into train/test sets to align with your current variables
from sklearn.model_selection import train_test_split
_, _, Y_train_fixed, Y_test_fixed = train_test_split(X_df, Y_clean, test_size=0.2,

print(f"Target matrix successfully rebuilt! Clean individual genres: {list(Y_clean.

print("\n--- STEP 2: TRAINING MODEL ON CLEAN GENRES ---")
start_time = time.time()

# 5. Initialize the Multi-Output Classifier
base_lr = LogisticRegression(class_weight='balanced', max_iter=1000, C=1.0)
multi_label_model = MultiOutputClassifier(base_lr, n_jobs=-1)

# 6. Fit the model (This will now run in less than 5 seconds!)
print("Fitting model to final feature matrix...")
multi_label_model.fit(X_train_final, Y_train_fixed)

print(f"Model trained successfully in {time.time() - start_time:.2f} seconds!")

print("\n--- STEP 3: GENERATING ACCURACY REPORT ---")
Y_pred = multi_label_model.predict(X_test_final)
print(classification_report(Y_test_fixed, Y_pred, target_names=Y_clean.columns, zer
```

--- STEP 1: CLEANING AND RE-SPLITTING TARGET GENRES ---

Target matrix successfully rebuilt! Clean individual genres: ['Animation', 'Family', 'Adventure', 'Comedy', 'Fantasy', 'Action', 'Science Fiction', 'Animation, Comedy', 'Drama', 'Animation, Family', 'Animation, Comedy, Family', 'TV Movie', 'Music', 'Romance', 'Comedy, Animation']

--- STEP 2: TRAINING MODEL ON CLEAN GENRES ---

Fitting model to final feature matrix...

Model trained successfully in 267.86 seconds!

--- STEP 3: GENERATING ACCURACY REPORT ---

	precision	recall	f1-score	support
Animation	0.96	0.61	0.75	3340
Family	0.52	0.96	0.67	594
Adventure	0.33	0.95	0.49	480
Comedy	0.29	0.97	0.45	496
Fantasy	0.28	0.94	0.43	434
Action	0.30	0.95	0.46	311
Science Fiction	0.25	0.91	0.39	259
Animation, Comedy	0.15	0.75	0.25	243
Drama	0.15	0.93	0.27	241
Animation, Family	0.14	0.74	0.24	185
Animation, Comedy, Family	0.08	0.79	0.15	72
TV Movie	0.09	0.78	0.17	92
Music	0.06	0.73	0.10	73
Romance	0.08	0.84	0.14	87
Comedy, Animation	0.04	0.60	0.08	78
micro avg	0.30	0.76	0.43	6985
macro avg	0.25	0.83	0.34	6985
weighted avg	0.60	0.76	0.57	6985
samples avg	0.22	0.48	0.28	6985

In [71]: *#NUMERIC FEATURE*

```
In [72]: import time
import numpy as np
import pandas as pd
from sklearn.base import BaseEstimator, TransformerMixin
from sklearn.compose import ColumnTransformer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
from sklearn.multioutput import MultiOutputClassifier
from sklearn.preprocessing import FunctionTransformer, KBinsDiscretizer, OneHotEncoder

# -----
# 1. CUSTOM MULTI-LABEL TARGET ENCODER CLASS DEFINITION
# -----
class MultiLabelTargetEncoder(BaseEstimator, TransformerMixin):
    def __init__(self, smoothing=10.0):
        self.smoothing = smoothing
        self.studio_maps_ = {}
        self.global_means_ = None
```

```

def fit(self, X, Y):
    X_series = pd.Series(np.asarray(X).ravel())
    Y_df = pd.DataFrame(np.asarray(Y))
    self.global_means_ = Y_df.mean().to_dict()
    for col in Y_df.columns:
        stats = Y_df.groupby(X_series)[col].agg(['count', 'mean'])
        smoothed_vals = (stats['count'] * stats['mean'] + self.smoothing * self
        self.studio_maps_[col] = smoothed_vals.to_dict()
    return self

def transform(self, X):
    X_series = pd.Series(np.asarray(X).ravel())
    encoded_out = np.zeros((len(X_series), len(self.studio_maps_)))
    for col, mapping in self.studio_maps_.items():
        encoded_out[:, col] = X_series.map(mapping).fillna(self.global_means_[col])
    return encoded_out

print("--- STEP 1: RE-ASSEMBLING MASTER FEATURE MATRIX (X) ---")
features_to_keep = [
    'Overview', 'Theme', 'Animation_Style', 'Target_Audience',
    'MPAA_Rating', 'Decade', 'Studio', 'Release_Year',
    'Movie_Length_Minutes', 'TMDB_Popularity'
]

X_df = df[features_to_keep].copy()

# Enforce standard string types and fill missing items safely
X_df['Overview'] = X_df['Overview'].fillna('')
X_df['Theme'] = X_df['Theme'].fillna('')

for col in ['Animation_Style', 'Target_Audience', 'MPAA_Rating', 'Studio', 'Decade']:
    X_df[col] = X_df[col].astype(str).fillna('Unknown')

for col in ['Release_Year', 'Movie_Length_Minutes', 'TMDB_Popularity']:
    X_df[col] = pd.to_numeric(X_df[col], errors='coerce')
    X_df[col] = X_df[col].fillna(X_df[col].median())

# Synchronize Train/Test splits using clean targets matrix (Y_clean)
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train_fixed, Y_test_fixed = train_test_split(X_df, Y_clean, test

print("\n--- STEP 2: CALCULATING STUDIO TARGET PROBABILITIES ---")
# This creates the missing 'X_train_studio_encoded' variables safely in real time!
studio_encoder = MultiLabelTargetEncoder(smoothing=15.0)
X_train_studio_encoded = studio_encoder.fit_transform(X_train['Studio'], Y_train_fixed)
X_test_studio_encoded = studio_encoder.transform(X_test['Studio'])

print("\n--- STEP 3: RUNNING ADVANCED NUMERIC TRANSFORMERS ---")
log_transformer = FunctionTransformer(np.log1p, validate=True)
runtime_discretizer = KBinsDiscretizer(n_bins=3, encode='ordinal', strategy='quantile')

numeric_processor = ColumnTransformer(transformers=[
    ('standard_scale', StandardScaler(), ['Release_Year']),

```

```

        ('log_transform', log_transformer, ['TMDb_Popularity']),
        ('bin_runtime', runtime_discretizer, ['Movie_Length_Minutes']))
    ])

X_train_numeric = numeric_processor.fit_transform(X_train)
X_test_numeric = numeric_processor.transform(X_test)

print("\n--- STEP 4: RUNNING ONE-HOT ENCODING ---")
one_hot_cols = ['Animation_Style', 'MPAA_Rating', 'Target_Audience', 'Decade']
oh_encoder = ColumnTransformer(transformers=[
    ('oh', OneHotEncoder(handle_unknown='ignore', sparse_output=False), one_hot_col
)])

X_train_oh = oh_encoder.fit_transform(X_train)
X_test_oh = oh_encoder.transform(X_test)

print("\n--- STEP 5: HORIZONTALLY STITCHING MASTER MATRIX ---")
# Combines all processed blocks side-by-side seamlessly
X_train_fully_combined = np.hstack((X_train_overview_svd, X_train_oh, X_train_studi
X_test_fully_combined = np.hstack((X_test_overview_svd, X_test_oh, X_test_studio_en

print(f"Final Integrated Feature Training Shape: {X_train_fully_combined.shape}")

print("\n--- STEP 6: TRAINING TUNED MULTI-LABEL CLASSIFIER ---")
start_time = time.time()

# Fit your tuned multi-output model configuration
tuned_base_lr = LogisticRegression(C=0.1, penalty='l2', class_weight='balanced', ma
tuned_multi_label_model = MultiOutputClassifier(tuned_base_lr, n_jobs=-1)
tuned_multi_label_model.fit(X_train_fully_combined, Y_train_fixed)

print(f"Model trained perfectly in {time.time() - start_time:.2f} seconds!")

print("\n--- STEP 7: FINAL MODEL DESIGN EVALUATION ---")
Y_final_pred = tuned_multi_label_model.predict(X_test_fully_combined)
print(classification_report(Y_test_fixed, Y_final_pred, target_names=Y_clean.column

```

--- STEP 1: RE-ASSEMBLING MASTER FEATURE MATRIX (X) ---

--- STEP 2: CALCULATING STUDIO TARGET PROBABILITIES ---

--- STEP 3: RUNNING ADVANCED NUMERIC TRANSFORMERS ---

--- STEP 4: RUNNING ONE-HOT ENCODING ---

--- STEP 5: HORIZONTALLY STITCHING MASTER MATRIX ---

Final Integrated Feature Training Shape: (20008, 312)

--- STEP 6: TRAINING TUNED MULTI-LABEL CLASSIFIER ---

Model trained perfectly in 8.64 seconds!

--- STEP 7: FINAL MODEL DESIGN EVALUATION ---

	precision	recall	f1-score	support
Animation	0.93	0.73	0.82	3340
Family	0.49	0.96	0.65	594
Adventure	0.33	0.96	0.49	480
Comedy	0.30	0.98	0.45	496
Fantasy	0.28	0.95	0.43	434
Action	0.28	0.95	0.44	311
Science Fiction	0.22	0.93	0.36	259
Animation, Comedy	0.15	0.70	0.25	243
Drama	0.17	0.95	0.29	241
Animation, Family	0.13	0.75	0.22	185
Animation, Comedy, Family	0.09	0.82	0.15	72
TV Movie	0.11	0.92	0.20	92
Music	0.05	0.88	0.10	73
Romance	0.07	0.92	0.14	87
Comedy, Animation	0.04	0.63	0.08	78
micro avg	0.31	0.83	0.45	6985
macro avg	0.24	0.87	0.34	6985
weighted avg	0.59	0.83	0.60	6985
samples avg	0.26	0.56	0.33	6985

```
In [73]: import sys
!{sys.executable} -m pip install scikit-learn pandas numpy fpdf2
```

Requirement already satisfied: scikit-learn in c:\users\manas\anaconda3\lib\site-packages (1.2.2)

Requirement already satisfied: pandas in c:\users\manas\anaconda3\lib\site-packages (2.1.4)

Requirement already satisfied: numpy in c:\users\manas\anaconda3\lib\site-packages (1.26.4)

Requirement already satisfied: fpdf2 in c:\users\manas\anaconda3\lib\site-packages (2.8.7)

Requirement already satisfied: scipy>=1.3.2 in c:\users\manas\anaconda3\lib\site-packages (from scikit-learn) (1.11.4)

Requirement already satisfied: joblib>=1.1.1 in c:\users\manas\anaconda3\lib\site-packages (from scikit-learn) (1.2.0)

Requirement already satisfied: threadpoolctl>=2.0.0 in c:\users\manas\anaconda3\lib\site-packages (from scikit-learn) (2.2.0)

Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\manas\anaconda3\lib\site-packages (from pandas) (2.8.2)

Requirement already satisfied: pytz>=2020.1 in c:\users\manas\anaconda3\lib\site-packages (from pandas) (2023.3.post1)

Requirement already satisfied: tzdata>=2022.1 in c:\users\manas\anaconda3\lib\site-packages (from pandas) (2023.3)

Requirement already satisfied: defusedxml in c:\users\manas\anaconda3\lib\site-packages (from fpdf2) (0.7.1)

Requirement already satisfied: Pillow!=9.2.*,>=8.3.2 in c:\users\manas\anaconda3\lib\site-packages (from fpdf2) (10.2.0)

Requirement already satisfied: fonttools>=4.34.0 in c:\users\manas\anaconda3\lib\site-packages (from fpdf2) (4.63.0)

Requirement already satisfied: six>=1.5 in c:\users\manas\anaconda3\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.16.0)

```
In [74]: import time
import numpy as np
import pandas as pd
from sklearn.base import BaseEstimator, TransformerMixin
from sklearn.compose import ColumnTransformer
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report
from sklearn.multioutput import ClassifierChain
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import FunctionTransformer, KBinsDiscretizer, OneHotEncoder

# -----
# 1. ADVANCED FEATURE ENGINEERING ENGINE (SECTION 5)
# -----
print("--- RUNNING ADVANCED FEATURE ENGINEERING ENGINE ---")
df_eng = df.copy()

# Fix text formatting upfront to prevent TfidfVectorizer crashes
df_eng['Overview'] = df_eng['Overview'].fillna('').astype(str)
df_eng['Theme'] = df_eng['Theme'].fillna('').astype(str)

# A. Keyword Flags
keywords = ['princess', 'robot', 'dragon', 'hero', 'magical', 'planet']
for kw in keywords:
    df_eng[f'has_{kw}'] = df_eng['Overview'].str.lower().str.contains(kw, regex=False)
```

```

# B. Text Length Metrics
df_eng['overview_word_count'] = df_eng['Overview'].apply(lambda x: len(x.split()))

# C. Lexical Sentiment Metric
negative_cues = {'dark', 'evil', 'danger', 'destroy', 'monster', 'war', 'battle', '
positive_cues = {'love', 'friend', 'fun', 'happy', 'laugh', 'magic', 'smile', 'fami

def calculate_sentiment(text_string):
    tokens = set(text_string.lower().split())
    return len(tokens.intersection(positive_cues)) - len(tokens.intersection(negati

df_eng['overview_sentiment'] = df_eng['Overview'].apply(calculate_sentiment)

# D. Temporal Interactions
df_eng['is_pre_2000'] = (df_eng['Release_Year'].fillna(2000) < 2000).astype(int)

# -----
# 2. ALIGNING FEATURES AND TARGET MATRICES
# -----
engineered_features = [f'has_{kw}' for kw in keywords] + [
    'overview_word_count', 'overview_sentiment', 'is_pre_2000',
    'Overview', 'Theme', 'Animation_Style', 'Target_Audience',
    'MPAA_Rating', 'Decade', 'Studio', 'Release_Year',
    'Movie_Length_Minutes', 'TMDB_Popularity'
]

X_df = df_eng[engineered_features].copy()

# Enforce clean missing data limits on metadata features
for col in ['Animation_Style', 'Target_Audience', 'MPAA_Rating', 'Studio', 'Decade']:
    X_df[col] = X_df[col].astype(str).fillna('Unknown')

for col in ['Release_Year', 'Movie_Length_Minutes', 'TMDB_Popularity']:
    X_df[col] = pd.to_numeric(X_df[col], errors='coerce')
    X_df[col] = X_df[col].fillna(X_df[col].median())

# Sync train/test splits cleanly (Using your verified Y_clean)
from sklearn.model_selection import train_test_split
X_train, X_test, Y_train_fixed, Y_test_fixed = train_test_split(X_df, Y_clean, test

# -----
# 3. TEXT VECTORIZATION ISOLATION (SECTION 5 & 7)
# -----
print("Extracting independent text tokens via TF-IDF...")
vectorizer_ov = TfidfVectorizer(ngram_range=(1,2), min_df=0.005, max_df=0.7, stop_w
vectorizer_th = TfidfVectorizer(ngram_range=(1,2), min_df=0.005, max_df=0.7, stop_w

X_train_ov_tfidf = vectorizer_ov.fit_transform(X_train['Overview']).toarray()
X_test_ov_tfidf = vectorizer_ov.transform(X_test['Overview']).toarray()

X_train_th_tfidf = vectorizer_th.fit_transform(X_train['Theme']).toarray()
X_test_th_tfidf = vectorizer_th.transform(X_test['Theme']).toarray()

# -----
# 4. STUDIO-GENRE BAYESIAN PRIORS MATRIX (SECTION 5 & 6)
# -----

```



```

print("Computing Studio-Genre conditional probability priors...")
class StudioGenrePriorEncoder(BaseEstimator, TransformerMixin):
    def __init__(self, smoothing=10.0):
        self.smoothing = smoothing
        self.priors_ = {}
        self.global_priors_ = None

    def fit(self, X, Y):
        X_studios = pd.Series(np.asarray(X).ravel())
        Y_targets = pd.DataFrame(np.asarray(Y))
        self.global_priors_ = Y_targets.mean().to_numpy()

        for col_idx in range(Y_targets.shape[1]):
            stats = Y_targets.groupby(X_studios)[col_idx].agg(['count', 'mean'])
            smoothed = (stats['count'] * stats['mean'] + self.smoothing * self.glob
            self.priors_[col_idx] = smoothed.to_dict()

        return self

    def transform(self, X):
        X_studios = pd.Series(np.asarray(X).ravel())
        out = np.zeros((len(X_studios), len(self.priors_)))
        for col_idx, mapping in self.priors_.items():
            out[:, col_idx] = X_studios.map(mapping).fillna(self.global_priors_[col
        return out

prior_encoder = StudioGenrePriorEncoder(smoothing=15.0)
X_train_priors = prior_encoder.fit_transform(X_train['Studio'], Y_train_fixed)
X_test_priors = prior_encoder.transform(X_test['Studio'])

# -----
# 5. METADATA COLUMN PROCESSING PIPELINE (SECTION 7)
# -----
print("Processing metadata features via ColumnTransformer...")
cat_cols = ['Animation_Style', 'MPAA_Rating', 'Target_Audience', 'Decade']
flag_cols = [f'has_{kw}' for kw in keywords] + ['overview_word_count', 'overview_se

log_transformer = FunctionTransformer(np.log1p, validate=True)
runtime_discretizer = KBinsDiscretizer(n_bins=3, encode='ordinal', strategy='quanti

preprocessor = ColumnTransformer(transformers=[
    ('cats', OneHotEncoder(handle_unknown='ignore', sparse_output=False), cat_cols),
    ('scale_year', StandardScaler(), ['Release_Year']),
    ('log_pop', log_transformer, ['TMDB_Popularity']),
    ('bin_len', runtime_discretizer, ['Movie_Length_Minutes']),
    ('passthrough_flags', StandardScaler(), flag_cols)
], remainder='drop')

X_train_meta = preprocessor.fit_transform(X_train)
X_test_meta = preprocessor.transform(X_test)

# -----
# 6. HORIZONTALLY ASSEMBLE MASTER FEATURE SPACE MATRIX
# -----
X_train_master = np.hstack((X_train_ov_tfidf, X_train_th_tfidf, X_train_meta, X_tra
X_test_master = np.hstack((X_test_ov_tfidf, X_test_th_tfidf, X_test_meta, X_test_pr

```

```

print(f"Master Preprocessed Data Matrix Shape: {X_train_master.shape}")

# -----
# 7. REPRODUCIBLE CLASSIFIER CHAINS ENSEMBLE TRAINING
# -----
print("\n--- TRAINING REPRODUCIBLE CLASSIFIER CHAINS ENSEMBLE ---")
start_time = time.time()

base_estimator = LogisticRegression(C=0.5, penalty='l2', class_weight='balanced', m
chain_model = ClassifierChain(base_estimator, order='random', random_state=42)
chain_model.fit(X_train_master, Y_train_fixed.to_numpy())

print(f"Classifier Chain fitted cleanly in {time.time() - start_time:.2f} seconds!")

# -----
# 8. PIPELINE PERFORMANCE VALIDATION METRICS
# -----
print("\n--- ENSEMBLE SYSTEM EVALUATION BREAKDOWN ---")
Y_chain_pred = chain_model.predict(X_test_master)
print(classification_report(Y_test_fixed, Y_chain_pred, target_names=Y_clean.columnn

```

```

--- RUNNING ADVANCED FEATURE ENGINEERING ENGINE ---
Extracting independent text tokens via TF-IDF...
Computing Studio-Genre conditional probability priors...
Processing metadata features via ColumnTransformer...
Master Preprocessed Data Matrix Shape: (20008, 871)

```

```

--- TRAINING REPRODUCIBLE CLASSIFIER CHAINS ENSEMBLE ---
Classifier Chain fitted cleanly in 62.48 seconds!

```

```

--- ENSEMBLE SYSTEM EVALUATION BREAKDOWN ---

```

	precision	recall	f1-score	support
Animation	0.97	1.00	0.99	3340
Family	0.70	0.91	0.79	594
Adventure	0.77	0.97	0.86	480
Comedy	0.62	0.93	0.74	496
Fantasy	0.47	0.97	0.63	434
Action	0.39	0.92	0.55	311
Science Fiction	0.34	0.88	0.50	259
Animation, Comedy	0.39	0.28	0.32	243
Drama	0.25	0.87	0.39	241
Animation, Family	0.64	1.00	0.78	185
Animation, Comedy, Family	0.23	0.11	0.15	72
TV Movie	0.17	0.70	0.28	92
Music	0.12	0.59	0.21	73
Romance	0.20	0.70	0.31	87
Comedy, Animation	0.15	0.85	0.25	78
micro avg	0.61	0.92	0.73	6985
macro avg	0.43	0.78	0.52	6985
weighted avg	0.73	0.92	0.79	6985
samples avg	0.61	0.72	0.65	6985

```
In [75]: from sklearn.model_selection import train_test_split

# 1. Split features (X) and multi-label targets (Y) using your exact specs
X_train, X_test, Y_train, Y_test = train_test_split(
    X_df,
    Y,
    test_size=0.20,
    random_state=42
)

# 2. Validation print statement to check row balances
print(f"Verified Row Balance -> Training: {len(X_train)} | Testing: {len(X_test)}")
# Output will print exactly: Verified Row Balance -> Training: 20312 | Testing: 507
```

Verified Row Balance -> Training: 20008 | Testing: 5002

```
In [76]: import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer

# 1. First, split your raw data (X_df and Y must already be defined above)
X_train_raw, X_test_raw, Y_train, Y_test = train_test_split(
    X_df,
    Y,
    test_size=0.20,
    random_state=42
)

# 2. Vectorise your text column safely down to numeric arrays
tfidf = TfidfVectorizer(ngram_range=(1, 2), min_df=5, max_df=0.7, stop_words="english")
X_train_text = tfidf.fit_transform(X_train_raw['Overview']).toarray()
X_test_text = tfidf.transform(X_test_raw['Overview']).toarray()

# 3. Process your metadata columns
preprocessor = ColumnTransformer(transformers=[
    ('cats', OneHotEncoder(handle_unknown='ignore', sparse_output=False),
    ['Animation_Style', 'Target_Audience', 'MPAA_Rating']),
], remainder='passthrough')

# Drop 'Overview' so we only process the categorical/numeric features here
X_train_meta = preprocessor.fit_transform(X_train_raw.drop(columns=['Overview']))
X_test_meta = preprocessor.transform(X_test_raw.drop(columns=['Overview']))

# 4. Horizontally stack text and metadata together into your final training matrices
X_train_clean = np.hstack((X_train_text, X_train_meta))
X_test_clean = np.hstack((X_test_text, X_test_meta))

print(f"Data matrices successfully built! Final X_train shape: {X_train_clean.shape}")
```

Data matrices successfully built! Final X_train shape: (20008, 14366)

```
In [77]: #Logistic regression
```

```

In [78]: import numpy as np
import pandas as pd
import time
from scipy.sparse import hstack, csr_matrix
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.compose import ColumnTransformer
from sklearn.multioutput import MultiOutputClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import classification_report, hamming_loss

# 1. Clean train/test split execution
X_train_raw, X_test_raw, Y_train_raw, Y_test_raw = train_test_split(
    X_df, Y, test_size=0.20, random_state=42
)

# Keep stable columns with both active target classes present
valid_train_genres = [col for col in Y_train_raw.columns if Y_train_raw[col].nunique() > 1]
Y_train = Y_train_raw[valid_train_genres]
Y_test = Y_test_raw[valid_train_genres]

# 2. FAST TEXT VECTORIZATION (Keep as sparse matrix, skip the slow .toarray() / SVD)
tfidf = TfidfVectorizer(ngram_range=(1, 2), min_df=5, max_df=0.7, stop_words="english")
X_train_text = tfidf.fit_transform(X_train_raw['Overview'])
X_test_text = tfidf.transform(X_test_raw['Overview'])

# 3. METADATA PROCESSING
categorical_cols = ['Animation_Style', 'Target_Audience', 'MPAA_Rating', 'Studio']
categorical_cols = [col for col in categorical_cols if col in X_train_raw.columns]

numerical_cols = ['Release_Year', 'Movie_Length_Minutes']
numerical_cols = [col for col in numerical_cols if col in X_train_raw.columns]

# Ensure sparse output is preserved
preprocessor = ColumnTransformer(transformers=[
    ('cats', OneHotEncoder(handle_unknown='ignore', sparse_output=True), categorical_cols),
    ('nums', StandardScaler(), numerical_cols)
], remainder='drop')

X_train_meta = csr_matrix(preprocessor.fit_transform(X_train_raw))
X_test_meta = csr_matrix(preprocessor.transform(X_test_raw))

# 4. HORIZONTAL SPARSE STACKING (Instantaneous, memory-safe execution)
X_train_sparse = hstack([X_train_text, X_train_meta])
X_test_sparse = hstack([X_test_text, X_test_meta])

print(f"Sparse Feature Space established! Row/Column Dimensions: {X_train_sparse.shape}")

# 5. HIGH-SPEED SAGA ENGINE CONFIGURATION
# SAGA converges very fast on wide, sparse text vectors
base_lr_fast = LogisticRegression(
    penalty='l2',
    C=1.0,
    class_weight='balanced',

```

```
solver='saga',
max_iter=250,      # Capped iterations guarantee rapid termination
tol=1e-3,         # Relaxed convergence tolerance prevents infinite loops
random_state=42
)

# n_jobs=-1 is safe to use here because sparse matrices use very little RAM
logistic_model_pipeline = MultiOutputClassifier(base_lr_fast, n_jobs=-1)

print("\n--- Training Fast Sparse Multi-Output Logistic Regression Baseline ---")
start_time = time.time()
logistic_model_pipeline.fit(X_train_sparse, Y_train)
print(f"Fast Baseline training complete! Elapsed time: {time.time() - start_time:.2f}")

# 6. PERFORMANCE REPORTING
Y_pred = logistic_model_pipeline.predict(X_test_sparse)
print(f"\nFinal Hamming Loss Outcome: {hamming_loss(Y_test, Y_pred):.5f}")
print("\n--- Model Evaluation Breakdown ---")
print(classification_report(Y_test, Y_pred, target_names=Y_train.columns, zero_divi
```

Sparse Feature Space established! Row/Column Dimensions: (20008, 18709)

--- Training Fast Sparse Multi-Output Logistic Regression Baseline ---
Fast Baseline training complete! Elapsed time: 868.66 seconds.

Final Hamming Loss Outcome: 0.02193

--- Model Evaluation Breakdown ---

precision	recall	f1-score	support	
Action	0.28	0.62	0.38	133
n Adventure Animation	0.00	0.00	0.00	6
ture Animation Comedy	0.00	0.00	0.00	0
Crime Science Fiction	0.00	0.00	0.00	0
Comedy Crime TV Movie	0.00	0.00	0.00	0
Comedy Drama Romance	0.00	0.00	0.00	0
imation Comedy Family	0.00	0.00	0.00	1
Comedy Family Fantasy	0.00	0.00	0.00	0
ence Fiction TV Movie	0.00	0.00	0.00	0
mation Comedy Fantasy	0.00	0.00	0.00	0
antasy Thriller Crime	0.00	0.00	0.00	0
imation Comedy Horror	0.00	0.00	0.00	0
Comedy Romance Family	0.00	0.00	0.00	0
omedy Science Fiction	0.00	0.00	0.00	0
nture Animation Crime	0.00	0.00	0.00	0
ence Fiction Thriller	0.00	0.00	0.00	0
Crime Science Fiction	0.00	0.00	0.00	0
nture Animation Drama	0.00	0.00	0.00	0
Drama Family Fantasy	0.00	0.00	0.00	0
ture Animation Family	0.00	0.00	0.00	2
mation Family Fantasy	0.00	0.00	0.00	0
ntasy Science Fiction	0.00	0.00	0.00	0

ation Family TV Movie	0.00	0.00	0.00	1	Action Advent
ure Animation Fantasy	0.00	0.00	0.00	3	Action Adventure Animation Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Action Adventure Animation Fantasy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Action Adven
ture Animation Horror	0.00	0.00	0.00	0	Action Adventure Animation
Horror Fantasy Drama	0.00	0.00	0.00	0	Action Adventure Anim
ation Science Fiction	0.33	0.20	0.25	5	Action Adventure Animation
Science Fiction Drama	0.00	0.00	0.00	0	Action Adventure Animation Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Action Adventure Animation Science Fi
ction Fantasy Mystery	0.00	0.00	0.00	0	Action Adventu
re Animation TV Movie	0.00	0.00	0.00	0	Action Adventure Ani
mation TV Movie Crime	0.00	0.00	0.00	0	Action Adventure Animation
TV Movie Crime Drama	0.00	0.00	0.00	0	Action Adven
ture Comedy Animation	0.00	0.00	0.00	0	Action Adventure Comedy Dr
ama Fantasy Animation	0.00	0.00	0.00	0	Action Adventure Comedy TV Movie Science Fict
ion Fantasy Animation	0.00	0.00	0.00	0	Action Adve
nture Drama Animation	0.00	0.00	0.00	0	Action Adven
ture Family Animation	0.00	0.00	0.00	0	Action Adventure Fam
ily Fantasy Animation	0.00	0.00	0.00	0	Action Advent
ure Fantasy Animation	0.00	0.00	0.00	1	Action Adventure Fantasy Anim
ation Science Fiction	0.00	0.00	0.00	0	Action Advent
ure History Animation	0.00	0.00	0.00	0	Action Adventure Mystery Comedy Drama Crime Anim
ation TV Movie Family	0.00	0.00	0.00	0	Action Adventure Scie
nce Fiction Animation	0.00	0.00	0.00	0	Action Adventure Science Ficti
on Animation TV Movie	0.00	0.00	0.00	0	Action Ad
venture War Animation	0.00	0.00	0.00	0	
Action Animation	0.01	0.33	0.02	9	Actio
n Animation Adventure	0.00	0.00	0.00	0	Action Anima

tion Adventure Comedy	0.00	0.00	0.00	0	
Action Animation Adventure Comedy Documentary Science Fiction Drama Family					
Western TV Movie War	0.00	0.00	0.00	0	
					Action Animation Adventure
Comedy Family Fantasy	0.00	0.00	0.00	0	
					Action Animation Adv
enture Comedy Fantasy	0.00	0.00	0.00	0	
					Action Animation Adventure Comedy Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Action Animation Adventure Co
medy Fantasy TV Movie	0.11	1.00	0.20	1	
					Action Animation Adve
nture Comedy TV Movie	0.00	0.00	0.00	0	
					Action Animation Adventure Crime S
cience Fiction Family	0.00	0.00	0.00	0	
					Action Anim
ation Adventure Drama	0.00	0.00	0.00	1	
					Action Animat
ion Adventure Fantasy	0.00	0.00	0.00	1	
					Action Animation Adventure Fan
tasy Mystery Thriller	0.00	0.00	0.00	0	
					Action Animat
ion Adventure Romance	0.00	0.00	0.00	0	
					Action Animation Adve
nture Science Fiction	0.00	0.00	0.00	0	
					Action Animation Adventure Science F
iction Comedy Fantasy	0.00	0.00	0.00	0	
					Action Animation Adventure Science
Fiction Drama Romance	0.00	0.00	0.00	0	
					Action Animation Adventure Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	
					Action Animation Adventure S
cience Fiction Horror	0.00	0.00	0.00	0	
					Ac
tion Animation Comedy	0.00	0.00	0.00	1	
					Action A
nimation Comedy Crime	0.00	0.00	0.00	0	
					Action Animation Come
dy Documentary Family	0.00	0.00	0.00	0	
					Action Animati
on Comedy Drama Crime	0.00	0.00	0.00	0	
					Action Animation Comedy
Drama Fantasy Horror	0.00	0.00	0.00	0	
					Action An
imation Comedy Family	0.00	0.00	0.00	0	
					Action Animation Co
medy Family Adventure	0.00	0.00	0.00	1	
					Action Animation Comedy Family Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Action Animation Comedy F
amily Science Fiction	0.00	0.00	0.00	0	
					Action Animation
Comedy Fantasy Horror	0.00	0.00	0.00	0	
					Action Animation Com
edy Music Romance War	0.00	0.00	0.00	0	
					Action Ani

mation Comedy Romance	0.00	0.00	0.00	0	Action Animation C
omedy Science Fiction	0.00	0.00	0.00	1	Action Animation
Comedy TV Movie Crime	0.00	0.00	0.00	0	Action Animation Comedy
TV Movie Crime Drama	0.00	0.00	0.00	0	Action Ani
mation Comedy Western	0.00	0.00	0.00	0	Action An
imation Crime Mystery	0.00	0.00	0.00	0	Action Animation
Crime Mystery Romance	0.00	0.00	0.00	0	Action Animation
Crime Science Fiction	0.00	0.00	0.00	0	Action Ani
mation Crime Thriller	0.00	0.00	0.00	0	A
ction Animation Drama	1.00	1.00	1.00	1	Action An
imation Drama Fantasy	1.00	1.00	1.00	1	Action Animation
Drama Fantasy Romance	0.00	0.00	0.00	0	Action An
imation Drama History	0.00	0.00	0.00	0	Action Animation
Drama Science Fiction	0.00	0.00	0.00	0	Action Animation Dram
a Science Fiction War	0.00	0.00	0.00	0	Action Ani
mation Drama Thriller	0.00	0.00	0.00	0	Ac
tion Animation Family	0.00	0.00	0.00	1	Action Animation Family Adve
nture Science Fiction	0.00	0.00	0.00	0	Act
ion Animation Fantasy	0.00	0.50	0.00	2	Action Animat
ion Fantasy Adventure	0.00	0.00	0.00	0	Action Ani
mation Fantasy Comedy	0.00	0.00	0.00	0	Action An
imation Fantasy Drama	0.00	0.00	0.00	0	Action Ani
mation Fantasy Horror	0.00	0.00	0.00	0	Action Animati
on Fantasy Horror War	0.00	0.00	0.00	0	Action Animation Fa
ntasy Science Fiction	0.00	0.00	0.00	2	Act
ion Animation History	0.00	0.00	0.00	0	Ac
tion Animation Horror	0.00	0.00	0.00	0	Action Animation
Music Science Fiction	0.00	0.00	0.00	0	Act

ion Animation Romance	0.00	0.00	0.00	0	Action Animation Ro
mance Science Fiction	0.00	0.00	0.00	0	Action Anim
ation Science Fiction	0.03	0.67	0.05	9	Action Animation Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Action Animation Science Fic
tion Comedy Adventure	0.00	0.00	0.00	0	Action Animation
Science Fiction Crime	0.00	0.00	0.00	1	Action Animation
Science Fiction Drama	0.00	0.00	0.00	0	Action Animation Sc
ience Fiction Fantasy	0.00	1.00	0.00	1	Action Animation S
cience Fiction Horror	0.00	0.00	0.00	0	Action Animation Sc
ience Fiction Mystery	0.00	0.00	0.00	0	Action Animation Sci
ence Fiction TV Movie	0.00	0.00	0.00	1	Action Animation Sci
ence Fiction Thriller	0.00	0.00	0.00	0	Action Animatio
n Science Fiction War	0.06	1.00	0.11	2	Acti
on Animation Thriller	0.00	0.00	0.00	0	Action Anim
ation Thriller Comedy	0.00	0.00	0.00	0	Ac
tion Comedy Animation	0.00	0.00	0.00	0	Action Comed
y Animation Adventure	0.00	0.00	0.00	0	Action Co
medy Animation Family	0.00	0.00	0.00	0	Action Comedy Ani
mation Thriller Crime	0.00	0.00	0.00	0	Action Comedy Science Fictio
n Adventure Animation	0.00	0.00	0.00	0	Action Comedy Scie
nce Fiction Animation	0.00	0.00	0.00	0	Action Drama Adventure Science
Fiction War Animation	0.00	0.00	0.00	0	A
ction Drama Animation	0.00	0.00	0.00	0	Action Drama Animati
on Adventure Thriller	0.00	0.00	0.00	0	Action Drama Scie
nce Fiction Animation	0.00	0.00	0.00	0	Act
ion Fantasy Animation	0.00	0.00	0.00	1	Action Fa
ntasy Animation Drama	0.00	0.00	0.00	0	Action Fan
tasy Animation Horror	0.00	0.00	0.00	0	Action Fan

tasy Horror Animation	0.00	0.00	0.00	0	
					Action
History War Animation	0.00	0.00	0.00	0	
					Action Horror Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Action Mystery Roman
ce Thriller Animation	0.00	0.00	0.00	0	
					Act
ion Romance Animation	0.00	0.00	0.00	0	
					Action Scie
nce Fiction Animation	0.00	0.00	0.00	1	
					Action Science Fictio
n Animation Adventure	0.00	0.00	0.00	0	
					Action Science
Fiction Animation War	0.00	0.00	0.00	0	
					Action Science Fiction Com
edy Fantasy Animation	0.00	0.00	0.00	0	
					Action War Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Act
ion Western Animation	0.00	0.00	0.00	0	
Adventure	0.18	0.58	0.28	173	
					Adven
ture Action Animation	0.00	0.00	0.00	0	
					Adventure Ac
tion Animation Comedy	0.00	0.00	0.00	0	
					Adventure Action Ani
mation Comedy Fantasy	0.00	0.00	0.00	0	
					Adventure Action Animation Drama Family Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Adventure Action Animation Drama Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Adventure Action Animation Family Fa
ntasy Science Fiction	0.02	1.00	0.03	1	
					Adventure Act
ion Animation Fantasy	0.00	0.00	0.00	0	
					Adventure Action Anim
ation Science Fiction	0.00	0.00	0.00	1	
					Adventure Act
ion Fantasy Animation	0.00	0.00	0.00	0	
					Adventure Action Fantasy Scie
nce Fiction Animation	0.00	0.00	0.00	0	
Adventure Animation	0.00	1.00	0.00	2	
					Adven
ture Animation Action	0.00	0.00	0.00	3	
					Adventure Animation Action Comedy F
amily Science Fiction	0.00	0.00	0.00	0	
					Adventure Ani
mation Action Fantasy	0.00	0.00	0.00	0	
					Adventure Animation Action Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Adventure Animation
Action History Family	0.00	0.00	0.00	0	
					Adventure Animation A

ction Science Fiction	0.00	0.00	0.00	1	Adventure Anim
ation Action TV Movie	0.00	0.00	0.00	0	Adven
ture Animation Comedy	0.00	0.00	0.00	2	Adventure A
nimation Comedy Crime	0.00	0.00	0.00	0	Adventure Animation
Comedy Drama Fantasy	0.00	0.00	0.00	0	Adventure An
imation Comedy Family	0.00	0.00	0.00	6	Adventure Animation
Comedy Family Fantasy	0.00	0.00	0.00	1	Adventure Animation Comedy
Family Fantasy Music	0.00	0.00	0.00	0	Adventure Animation Comedy Fa
mily Fantasy TV Movie	0.00	0.00	0.00	0	Adventure Animatio
n Comedy Family Music	0.00	0.00	0.00	1	Adventure Animation Comedy F
amily Science Fiction	0.00	0.00	0.00	2	Adventure Ani
mation Comedy Fantasy	0.00	0.00	0.00	1	Adventure Ani
mation Comedy History	0.00	0.00	0.00	0	Adventure Anim
ation Comedy TV Movie	0.00	0.00	0.00	0	Adventure Animation
Comedy TV Movie Crime	0.00	0.00	0.00	0	Adventure Ani
mation Comedy Western	0.00	0.00	0.00	0	Adventure Animatio
n Crime Drama Mystery	0.00	0.00	0.00	0	Adventure Animation Crime F
amily Fantasy Romance	0.00	0.00	0.00	0	Adve
nture Animation Drama	0.00	0.00	0.00	1	Adventure A
nimation Drama Family	0.00	0.00	0.00	1	Adventure Animation
Drama Family Fantasy	0.00	0.00	0.00	0	Adventure Animation Drama F
amily Mystery Fantasy	0.00	0.00	0.00	0	Adventure An
imation Drama Fantasy	0.00	0.00	0.00	0	Adventure Animation
Drama Fantasy Romance	0.00	0.00	0.00	0	Adventure An
imation Drama History	0.00	0.00	0.00	0	Adventure An
imation Drama Romance	0.00	0.00	0.00	0	Adventure Animation
Drama Science Fiction	0.00	0.00	0.00	0	Adven
ture Animation Family	0.01	0.67	0.01	6	Adventure An

imation Family Comedy	0.00	0.00	0.00	0	
					Adventure Animation
Family Comedy Fantasy	0.00	0.00	0.00	1	
					Adventure Animation Family Comedy My
stery Science Fiction	0.00	0.00	0.00	0	
					Adventure Animation Family C
omedy Science Fiction	0.00	0.00	0.00	0	
					Adventure Animation Family Comedy Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	
					Adventure Ani
mation Family Fantasy	0.01	0.57	0.01	7	
					Adventure Animation Family Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Adventure Ani
mation Family History	0.00	0.00	0.00	0	
					Adventure Ani
mation Family Romance	0.00	0.00	0.00	0	
					Adventure Animation F
amily Science Fiction	0.00	0.00	0.00	0	
					Adventure Animation Family Science F
iction Comedy Fantasy	0.00	0.00	0.00	0	
					Adventure Animation Family Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	
					Adventure Anim
ation Family TV Movie	0.00	0.00	0.00	0	
					Advent
ure Animation Fantasy	0.00	0.00	0.00	0	
					Adventure Ani
mation Fantasy Family	0.00	0.00	0.00	0	
					Adventure Anim
ation Fantasy Mystery	0.00	0.00	0.00	0	
					Adventure Animation Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Adventure Animation
Music Fantasy Comedy	0.00	0.00	0.00	0	
					Advent
ure Animation Mystery	0.00	0.00	0.00	0	
					Adventure Animation Ro
mance Science Fiction	0.00	0.00	0.00	0	
					Adventure Anim
ation Science Fiction	0.00	0.00	0.00	3	
					Adventure Animation S
cience Fiction Action	0.00	0.00	0.00	0	
					Adventure Animation Science
Fiction Action Comedy	0.00	0.00	0.00	0	
					Adven
ture Comedy Animation	0.00	0.00	0.00	1	
					Adventure Co
medy Animation Family	0.00	0.00	0.00	0	
					Adventure C
omedy Crime Animation	0.00	0.00	0.00	0	
					Adventure C
omedy Drama Animation	0.00	0.00	0.00	0	
					Adventure Co
medy Family Animation	0.00	0.00	0.00	0	
					Adventure Comedy Hor

ror History Animation	0.00	0.00	0.00	0	
					Adventure Comedy Rom
ance Animation Family	0.00	0.00	0.00	0	
					Adve
nture Drama Animation	0.00	0.00	0.00	0	
					Adventure Dr
ama Animation Fantasy	0.00	0.00	0.00	0	
					Adventure D
rama Comedy Animation	0.00	0.00	0.00	0	
					Adventure D
rama Family Animation	0.00	0.00	0.00	0	
					Adventure Dr
ama Fantasy Animation	0.00	0.00	0.00	0	
					Adventure Drama Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Adven
ture Family Animation	0.00	0.00	0.00	1	
					Adventure Family An
imation Action Comedy	0.00	0.00	0.00	0	
					Adventure Family Animation Comedy Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Adventure Fam
ily Animation Fantasy	0.00	0.00	0.00	1	
					Adventure Family Animation Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Adventure Fam
ily Fantasy Animation	0.00	0.00	0.00	0	
					Adventure Family Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Advent
ure Fantasy Animation	0.00	0.00	0.00	3	
					Adventure Fantasy An
imation Action Family	0.00	0.00	0.00	0	
					Adventure Fan
tasy Animation Comedy	0.00	0.00	0.00	0	
					Adventure Fantasy An
imation Comedy Family	0.00	0.00	0.00	0	
					Adventure Fantasy Animation C
omedy Science Fiction	0.00	0.00	0.00	0	
					Adventure Fantasy A
nimation Drama Family	0.00	0.00	0.00	0	
					Adventure Fan
tasy Animation Family	0.00	0.00	0.00	0	
					Adventure Fantasy Ani
mation Mystery Family	0.00	0.00	0.00	0	
					Adventure Fantasy Animation S
cience Fiction Family	0.00	0.00	0.00	0	
					Adventure Fan
tasy Family Animation	0.00	0.00	0.00	0	
					Adventure Fantasy Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Adventure M
usic Animation Family	0.00	0.00	0.00	1	
					Adventure Mystery Co
medy Animation Family	0.00	0.00	0.00	0	
					Adventure Mys

tery Family Animation	0.00	0.00	0.00	0	
					Adventure Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Adventure Science Fi
ction Drama Animation	0.00	0.00	0.00	0	
					Adventure TV M
ovie Action Animation	0.00	0.00	0.00	0	
Animation	1.00	0.91	0.95	3567	
Animation Action	0.00	0.22	0.01	9	
					Anima
tion Action Adventure	0.00	0.50	0.00	4	
					Animation Ac
tion Adventure Comedy	0.00	0.00	0.00	2	
					Animation Action A
dventure Comedy Crime	0.00	0.00	0.00	0	
					Animation Action Adventure Comedy Drama F
amily Science Fiction	0.00	0.00	0.00	0	
					Animation Action Ad
venture Comedy Family	0.00	0.00	0.00	1	
					Animation Action Adventure
Comedy Family Fantasy	0.00	0.00	0.00	0	
					Animation Action Adventure Comedy
Family Fantasy Music	0.00	0.00	0.00	0	
					Animation Action Adventure Comedy Family Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Animation Action Adventure Comedy F
amily Science Fiction	0.00	0.00	0.00	0	
					Animation Action Adv
enture Comedy Fantasy	0.00	0.00	0.00	0	
					Animation Action Adventure C
omedy Fantasy Romance	0.00	0.00	0.00	0	
					Animation Action Adventure C
omedy Science Fiction	0.00	0.00	0.00	0	
					Animation Action A
dventure Crime Comedy	0.00	0.00	0.00	0	
					Animation Action Adventure
Crime Comedy TV Movie	0.00	0.00	0.00	0	
					Animation A
ction Adventure Drama	0.00	0.00	0.00	0	
					Animation Action Adventu
re Drama Crime Comedy	0.00	0.00	0.00	0	
					Animation Action Adventure Drama F
amily Science Fiction	0.00	0.00	0.00	0	
					Animation Action Ad
venture Drama Fantasy	0.50	1.00	0.67	1	
					Animation Action Adventure Drama Fantasy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	
					Animation Action Ad
venture Drama Mystery	0.00	0.00	0.00	0	
					Animation Action Adventure
Drama Science Fiction	0.00	0.00	0.00	0	
					Animation Ac
tion Adventure Family	0.00	0.00	0.00	2	
					Animation Action Ad

venture Family Comedy	0.00	0.00	0.00	0	Animation Action Adv
enture Family Fantasy	0.00	0.00	0.00	0	Animation Action Adventure F
amily Fantasy Romance	0.00	0.00	0.00	0	Animation Action Adventure F
amily Science Fiction	0.00	0.00	0.00	0	Animation Action Adve
nture Family TV Movie	0.00	0.00	0.00	0	Animation Act
ion Adventure Fantasy	0.00	0.33	0.01	3	Animation Action Adv
enture Fantasy Comedy	0.00	0.00	0.00	0	Animation Action Adventure
Fantasy Drama Horror	0.00	0.00	0.00	0	Animation Action Adv
enture Fantasy Family	0.00	0.00	0.00	0	Animation Action Adve
nture Fantasy Romance	0.00	0.00	0.00	0	Animation Action Adventure Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Animation Action Adven
ture Fantasy Thriller	0.00	0.00	0.00	0	Animation Action Adventure
History Drama Fantasy	0.00	0.00	0.00	0	Animation Action Adve
nture History Fantasy	0.00	0.00	0.00	0	Animation Act
ion Adventure Mystery	0.00	0.00	0.00	0	Animation Action Adventure Romance
Drama Science Fiction	0.00	0.00	0.00	0	Animation Action Adve
nture Science Fiction	0.00	0.00	0.00	4	Animation Action Adventure S
cience Fiction Comedy	0.00	0.00	0.00	0	Animation Action Adventure Science
Fiction Comedy Music	0.00	0.00	0.00	0	Animation Acti
on Adventure TV Movie	0.00	0.00	0.00	0	Animation Acti
on Adventure Thriller	0.00	0.00	0.00	0	Animation Action Adven
ture Thriller Romance	0.00	0.00	0.00	0	Animation Action Adventur
e War Science Fiction	0.00	0.00	0.00	0	An
imation Action Comedy	0.01	0.60	0.01	5	Animation Action Comedy Adv
enture TV Movie Crime	0.00	0.00	0.00	0	Animatio
n Action Comedy Crime	0.00	0.00	0.00	0	Animation Action
Comedy Crime Mystery	0.00	0.00	0.00	0	Animation Action
Comedy Crime TV Movie	0.00	0.00	0.00	0	Animation Acti

on Comedy Drama Crime	0.00	0.00	0.00	0	Animation
Action Comedy Family	0.00	0.00	0.00	1	Animation Action Comedy F
amily Science Fiction	0.00	0.00	0.00	0	Animation Action Comedy
Fantasy Horror Music	0.00	0.00	0.00	0	Animation Action C
omedy Fantasy Mystery	0.00	0.00	0.00	0	Animation
Action Comedy Horror	0.00	0.00	0.00	0	Animation
Action Comedy Mystery	0.00	0.00	0.00	0	Animation Action Comedy Ro
mance Science Fiction	0.00	0.00	0.00	0	Animation A
ction Comedy TV Movie	0.00	0.00	0.00	0	A
nimation Action Crime	0.00	0.00	0.00	1	Animati
on Action Crime Drama	0.00	0.00	0.00	1	Animatio
n Action Crime Horror	0.00	0.00	0.00	0	Animation
Action Crime Mystery	0.17	1.00	0.29	1	Animation Action Crime Mys
tery Romance Thriller	0.00	0.00	0.00	0	Animation Action
Crime Romance Mystery	0.00	0.00	0.00	0	Animati
on Action Documentary	0.00	0.00	0.00	0	A
nimation Action Drama	0.00	0.00	0.00	2	Animation Action Dr
ama Adventure Fantasy	0.00	0.00	0.00	0	Animation Action Drama Family TV Mo
vie Romance Adventure	0.00	0.00	0.00	0	Animation
Action Drama Fantasy	0.00	0.00	0.00	0	Animation Action Drama Fa
ntasy Horror Thriller	0.00	0.00	0.00	0	Animation
Action Drama History	0.00	0.00	0.00	0	Animation
Action Drama Mystery	0.00	0.00	0.00	0	Animation Action Drama Romance Scie
nce Fiction Music War	0.00	0.00	0.00	0	Animation Action
Drama Science Fiction	0.00	0.00	0.00	1	Anima
tion Action Drama War	0.00	0.00	0.00	0	An
imation Action Family	0.00	0.00	0.00	0	Animation
Action Family Comedy	0.00	0.00	0.00	0	Animation Action F

amily Science Fiction	0.00	0.00	0.00	0	Ani
mation Action Fantasy	0.02	0.88	0.04	8	Animation Act
ion Fantasy Adventure	0.00	0.00	0.00	3	Animation
Action Fantasy Comedy	0.00	0.00	0.00	0	Animation Action Fan
tasy Comedy Adventure	0.00	0.00	0.00	0	Animation Action Fantasy Co
medy Adventure Family	0.00	0.00	0.00	0	Animation Action Fantasy Co
medy Family Adventure	0.00	0.00	0.00	0	Animation
Action Fantasy Drama	0.00	0.00	0.00	1	Animation Action
Fantasy Drama Romance	0.00	0.00	0.00	0	Animation
Action Fantasy Family	0.00	0.00	0.00	0	Animation Action Fantasy Fa
mily Adventure Comedy	0.00	0.00	0.00	0	Animation
Action Fantasy Horror	0.00	0.00	0.00	0	Animation A
ction Fantasy Mystery	0.00	0.00	0.00	0	Animation Action Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Animation Action Fantasy S
cience Fiction Family	0.00	0.00	0.00	0	Animation Ac
tion Fantasy Thriller	0.50	0.50	0.50	2	Ani
mation Action History	0.00	0.00	0.00	0	Animation Action
History Fantasy Drama	0.00	0.00	0.00	0	Animation Action History My
stery Science Fiction	0.00	0.00	0.00	0	An
imation Action Horror	0.00	0.00	0.00	0	Animation Action H
orror Science Fiction	0.00	0.00	0.00	1	Ani
mation Action Mystery	0.00	0.00	0.00	0	Animation
Action Mystery Crime	0.00	0.00	0.00	0	Animation Action My
stery Science Fiction	0.00	0.00	0.00	0	Animation A
ction Romance Fantasy	0.00	0.00	0.00	1	Animation Action Ro
mance Science Fiction	0.00	0.00	0.00	0	Animation A
ction Science Fiction	0.01	0.45	0.01	11	Animation Action Scie
nce Fiction Adventure	0.00	0.00	0.00	1	Animation Action S

science Fiction Comedy	0.00	0.00	0.00	0	Animation Action Science F
iction Comedy Fantasy	0.00	0.00	0.00	0	Animation Action
Science Fiction Drama	0.00	0.00	0.00	1	Animation Action Science
Fiction Drama Fantasy	0.00	0.00	0.00	0	Animation Action Science Fic
tion Family Adventure	0.00	0.00	0.00	0	Animation Action Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Animation Action Science
Fiction Romance Music	0.00	0.00	0.00	0	Animation Action Science
Fiction TV Movie War	0.00	0.00	0.00	0	Animation Action Sci
ence Fiction Thriller	0.00	0.00	0.00	0	Animation Action Science Fic
tion Thriller Mystery	0.00	0.00	0.00	0	Animation Actio
n Science Fiction War	0.00	0.00	0.00	1	Anim
ation Action Thriller	0.50	0.50	0.50	2	Animation
Action Thriller Crime	0.00	0.00	0.00	0	Animation Action Thriller
Crime Science Fiction	0.00	0.00	0.00	0	Animation Action Thr
iller Fantasy Romance	0.00	0.00	0.00	0	Animation Action Thr
iller Science Fiction	0.00	0.00	0.00	0	
Animation Action War	0.00	0.00	0.00	0	
Animation Adventure	0.01	0.89	0.02	18	Anima
tion Adventure Action	0.00	0.00	0.00	3	Animation Adventure Action
Comedy Family Fantasy	0.00	0.00	0.00	0	Animation Adventure
Action Comedy Fantasy	0.00	0.00	0.00	0	Animation A
dventure Action Drama	0.00	0.00	0.00	0	Animation Adventure
Action Drama Fantasy	0.00	0.00	0.00	0	Animation Ad
venture Action Family	0.00	0.00	0.00	0	Animation Adventure Ac
tion Family Drama War	0.00	0.00	0.00	0	Animation Adv
enture Action Fantasy	0.00	0.00	0.00	1	Animation Adventure Action Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Animation Adventure A
ction Science Fiction	0.00	0.00	0.00	1	Anima

tion Adventure Comedy	0.00	0.14	0.01	7	Animation Adventure
e Comedy Action Crime	0.00	0.00	0.00	0	Animation Adventure
re Comedy Crime Drama	0.00	0.00	0.00	0	Animation Adventure
e Comedy Drama Family	0.00	0.00	0.00	1	Animation Ad
venture Comedy Family	0.01	0.44	0.02	9	Animation Adventure
e Comedy Family Drama	0.00	0.00	0.00	0	Animation Adventure
Comedy Family Fantasy	0.00	0.00	0.00	1	Animation Adventure Comedy
Family Fantasy Music	0.00	0.00	0.00	0	Animation Adventure Comedy Fa
mily Fantasy TV Movie	0.00	0.00	0.00	0	Animation Adventure
e Comedy Family Music	0.00	0.00	0.00	0	Animation Adventure Comedy F
amily Science Fiction	0.00	0.00	0.00	1	Animation Adv
enture Comedy Fantasy	0.00	0.00	0.00	2	Animation Adventure
Comedy Fantasy Action	0.00	0.00	0.00	0	Animation Adventure Comedy
Fantasy Action Family	0.00	0.00	0.00	0	Animation Adventure
Comedy Fantasy Family	0.00	0.00	0.00	0	Animation Adventure Comedy Fantasy S
cience Fiction Family	0.00	0.00	0.00	0	Animation Adventure Comedy Fa
ntasy TV Movie Family	0.00	0.00	0.00	0	Animation A
dventure Comedy Music	0.00	0.00	0.00	0	Animation Adventure C
omedy Science Fiction	0.00	0.00	0.00	0	Animation Adventure Comedy Science F
iction Fantasy Family	0.00	0.00	0.00	0	Animation Adve
nture Comedy TV Movie	0.00	0.00	0.00	0	Animation Adventure Comedy TV Movie Science F
iction Fantasy Action	0.00	0.00	0.00	0	Animation Adventure Co
medy TV Movie Western	0.00	0.00	0.00	0	Animation Adv
enture Comedy Western	0.00	0.00	0.00	0	Animation Adventure
Crime Comedy TV Movie	0.00	0.00	0.00	0	Animation Adventure Crime Mystery TV Movie
Action Comedy Family	0.50	1.00	0.67	1	Animation Adventure
e Documentary History	0.00	0.00	0.00	0	Anim
ation Adventure Drama	0.00	0.00	0.00	2	Animation A

dventure Drama Family	0.00	0.00	0.00	2	
					Animation Adventure
Drama Family Fantasy	0.00	0.00	0.00	0	
					Animation Adventure Drama Family Fantasy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	
					Animation Adventure
Drama Family TV Movie	0.00	0.00	0.00	0	
					Animation Ad
venture Drama Fantasy	0.00	0.00	0.00	0	
					Animation Adventure
Drama Fantasy Family	0.00	0.00	0.00	0	
					Animation Adventure
Drama Fantasy Mystery	0.00	0.00	0.00	0	
					Animation Ad
venture Drama Romance	0.00	0.00	0.00	0	
					Animation Adventure Drama S
cience Fiction Action	0.00	0.00	0.00	0	
					Animation Adventure
Drama TV Movie Family	0.00	0.00	0.00	0	
					Animatio
n Adventure Drama War	0.00	0.00	0.00	0	
					Anima
tion Adventure Family	0.02	0.53	0.04	15	
					Animation Adventure Family Action C
omedy Science Fiction	0.00	0.00	0.00	0	
					Animation Ad
venture Family Comedy	0.01	0.75	0.01	4	
					Animation Adventure
Family Comedy Fantasy	0.00	0.00	0.00	1	
					Animation Adventure
Family Comedy Romance	0.00	0.00	0.00	0	
					Animation A
dventure Family Drama	0.00	0.00	0.00	1	
					Animation Adv
enture Family Fantasy	0.00	0.00	0.00	5	
					Animation Adventure
Family Fantasy Action	0.00	0.00	0.00	0	
					Animation Adventure
Family Fantasy Comedy	0.00	0.00	0.00	2	
					Animation Adventure Family Fa
ntasy Comedy TV Movie	0.00	0.00	0.00	0	
					Animation Adventure
Family Fantasy Drama	0.00	0.00	0.00	0	
					Animation Adventure
Family Fantasy Horror	0.00	0.00	0.00	0	
					Animation Adventure
Family Fantasy Music	0.00	0.00	0.00	1	
					Animation Adventure Family Fa
ntasy Science Fiction	0.17	1.00	0.29	1	
					Animation Adventure Family Fantasy S
cience Fiction Comedy	0.00	0.00	0.00	1	
					Animation Adventu
re Family History War	0.00	0.00	0.00	0	
					Animation A
dventure Family Music	0.00	0.00	0.00	1	
					Animation Adventu

re Family Music Drama	0.00	0.00	0.00	0	
					Animation Adventure Fa
mily Mystery TV Movie	0.00	0.00	0.00	0	
					Animation Adv
enture Family Romance	0.00	0.00	0.00	1	
					Animation Adventure F
amily Science Fiction	0.00	0.00	0.00	2	
					Animation Adventure Family Science F
iction Comedy Fantasy	0.00	0.00	0.00	0	
					Animation Adventure Family Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	
					Animation Adventure Family Science F
iction Fantasy Comedy	0.00	0.00	0.00	0	
					Animation Adve
nture Family TV Movie	0.00	0.00	0.00	1	
					Animation Adv
enture Family Western	0.00	0.00	0.00	0	
					Animat
ion Adventure Fantasy	0.00	0.00	0.00	3	
					Animation Adv
enture Fantasy Action	0.00	0.00	0.00	0	
					Animation Adventure
Fantasy Action Family	0.00	0.00	0.00	1	
					Animation Adv
enture Fantasy Comedy	0.00	0.00	0.00	1	
					Animation Adventure
Fantasy Comedy Family	0.00	0.00	0.00	2	
					Animation Ad
venture Fantasy Drama	0.00	0.00	0.00	2	
					Animation Adventure
Fantasy Drama Family	0.00	0.00	0.00	0	
					Animation Adv
enture Fantasy Family	0.00	0.50	0.00	2	
					Animation Adventure Fantasy
Family Action Comedy	0.00	0.00	0.00	0	
					Animation Adventure
Fantasy Family Comedy	0.00	0.00	0.00	0	
					Animation Adventure Fantasy F
amily Science Fiction	0.00	0.00	0.00	0	
					Animation Adventure Fa
ntasy Family TV Movie	0.00	0.00	0.00	0	
					Animation Adv
enture Fantasy Horror	0.00	0.00	0.00	0	
					Animation Adventure
Fantasy Horror Family	0.00	0.00	0.00	0	
					Animation Adve
nture Fantasy Mystery	0.00	0.00	0.00	0	
					Animation Adventure Fantasy My
stery Science Fiction	0.00	0.00	0.00	0	
					Animation Adve
nture Fantasy Romance	0.00	1.00	0.00	1	
					Animation Adventure Fa
ntasy Science Fiction	0.00	0.00	0.00	1	
					Animation Adventure Fantasy Science
Fiction Comedy Family	0.00	0.00	0.00	0	
					Animation Adventure Fantasy S

science Fiction Family	0.00	0.00	0.00	2	
					Animat
ion Adventure History	0.00	0.00	0.00	0	
					Animation Ad
venture History Drama	0.00	0.00	0.00	0	
					Anim
ation Adventure Music	0.00	0.00	0.00	0	
					Animation Advent
ure Music Documentary	0.00	0.00	0.00	0	
					Animation Adventure
Music Family Fantasy	0.00	0.00	0.00	0	
					Animation Adventure Music
Family Fantasy Comedy	0.00	0.00	0.00	0	
					Animat
ion Adventure Mystery	0.00	0.00	0.00	0	
					Animation Adventure M
ystery Crime TV Movie	0.00	0.00	0.00	0	
					Animation Adventure Mystery Crime TV Movie
Action Comedy Family	0.00	0.00	0.00	0	
					Animation Adventure M
ystery Family Fantasy	0.00	0.00	0.00	0	
					Animation Adve
nture Mystery Fantasy	0.00	0.00	0.00	0	
					Animation Adventure Mystery Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Animation Adventure Mystery
Science Fiction Crime	0.00	0.00	0.00	0	
					Animation Adven
ture Mystery TV Movie	0.00	0.00	0.00	0	
					Animat
ion Adventure Romance	0.00	0.00	0.00	0	
					Animation Adve
nture Science Fiction	0.00	1.00	0.00	2	
					Animation Adventure Science Fiction
Drama Family Fantasy	0.00	0.00	0.00	0	
					Animation Adventure Science Fiction
Family Fantasy Comedy	0.00	0.00	0.00	0	
					Animation Adventure Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	
					Animation Adventure Science Fiction F
antasy History Horror	0.00	0.00	0.00	0	
					Animation Adventure Science F
iction Fantasy Horror	0.00	0.00	0.00	0	
					Animation Adventure Sci
ence Fiction TV Movie	0.25	0.50	0.33	2	
					Animation Adventure Science Fiction TV Movie Action Comedy
Crime Family Mystery	0.00	0.00	0.00	0	
					Animati
on Adventure TV Movie	0.00	1.00	0.00	1	
					Animation Adventure
TV Movie Family Music	0.00	0.00	0.00	0	
Animation Comedy	0.04	0.84	0.07	45	
					An
imation Comedy Action	0.50	0.50	0.50	2	
					Animation Co

medy Action Adventure	0.00	0.00	0.00	0	
					Animation Comedy Ac
tion Adventure Family	0.00	0.00	0.00	0	
					Animation Comedy Action Adventure Family Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Animation Comedy Action Adventure Music Sc
ience Fiction Romance	0.00	0.00	0.00	0	
					Animatio
n Comedy Action Drama	0.00	0.00	0.00	0	
					Animation Comedy Action Drama F
amily Science Fiction	0.00	0.00	0.00	0	
					Animation Comedy
Action Drama Romance	0.00	0.00	0.00	0	
					Animation
Comedy Action Family	0.00	0.00	0.00	0	
					Anima
tion Comedy Adventure	0.00	0.67	0.01	3	
					Animation Co
medy Adventure Action	0.00	0.00	0.00	1	
					Animation Comedy Ad
venture Action Family	0.00	0.00	0.00	0	
					Animation Comedy
Adventure Crime Music	0.00	0.00	0.00	0	
					Animation Comedy A
dventure Drama Family	0.00	0.00	0.00	0	
					Animation Comedy Adventure Drama Fantasy Music Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	
					Animation Co
medy Adventure Family	0.00	0.00	0.00	6	
					Animation Comedy Adv
enture Family Fantasy	0.00	0.00	0.00	0	
					Animation Comedy Adv
enture Family Mystery	0.00	0.00	0.00	0	
					Animation Comedy Adventure F
amily Science Fiction	0.00	0.00	0.00	0	
					Animation Com
edy Adventure Fantasy	0.00	0.00	0.00	1	
					Animation Comedy Adventure F
antasy Romance Family	0.00	0.00	0.00	0	
					Animation Comedy Adventure Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Animation C
omedy Adventure Music	0.00	0.00	0.00	0	
					Animation Comedy Adve
nture Science Fiction	0.00	0.00	0.00	0	
					Animation Come
dy Adventure TV Movie	0.50	1.00	0.67	1	
					Animation
Comedy Adventure War	0.00	0.00	0.00	0	
					A
nimation Comedy Crime	0.00	0.00	0.00	0	
					Animation Comedy
Crime Family Fantasy	0.00	0.00	0.00	0	
					Animati
on Comedy Crime Music	0.00	0.00	0.00	0	
					Animation Comedy Documentary

Drama Fantasy Mystery	0.00	0.00	0.00	0	A
Animation Comedy Drama	0.00	0.38	0.00	8	Animation Comedy
Drama Action TV Movie	0.00	0.00	0.00	0	Animation Comedy
Drama Action Thriller	0.00	0.00	0.00	0	Animation C
Comedy Drama Adventure	0.00	0.00	0.00	0	Animatio
n Comedy Drama Family	0.00	0.00	0.00	2	Animation Come
dy Drama Family Music	0.00	0.00	0.00	0	Animation Comedy
Drama Fantasy Romance	0.00	0.00	0.00	0	Animatio
n Comedy Drama Horror	0.00	0.00	0.00	0	Animati
on Comedy Drama Music	0.00	0.00	0.00	0	Animation Comed
y Drama Music Romance	0.00	0.00	0.00	0	Animation
Comedy Drama Mystery	0.00	0.00	0.00	0	Animation Comedy
Drama Mystery Fantasy	0.00	0.00	0.00	0	Animation
Comedy Drama Romance	0.00	1.00	0.00	1	Animation Comedy Drama
Science Fiction Music	0.00	0.00	0.00	0	Animation Comedy Drama Thriller Mystery Ro
mance Science Fiction	0.00	0.00	0.00	0	An
imation Comedy Family	0.02	0.25	0.04	28	Animation
Comedy Family Action	0.00	0.00	0.00	0	Animation Comedy Family Act
ion Adventure Fantasy	0.00	0.00	0.00	0	Animation Co
medy Family Adventure	0.06	0.33	0.11	6	Animation Comedy Fa
mily Adventure Action	0.00	0.00	0.00	0	Animation Comedy Fam
ily Adventure Fantasy	0.00	0.00	0.00	0	Animation Comedy Family Adve
nture Fantasy Mystery	0.00	0.00	0.00	0	Animation Comedy F
amily Adventure Music	0.00	0.00	0.00	0	Animation Comedy Family Ad
venture Music Romance	0.00	0.00	0.00	0	Animation Comedy Fam
y Crime Music Western	0.00	0.00	0.00	0	Animation
ily Documentary Drama	0.00	0.00	0.00	0	Animation Comedy
Comedy Family Fantasy	0.01	1.00	0.01	7	Animation Comedy

Family Fantasy Action	0.00	0.00	0.00	0	Animation Comedy Fam
ily Fantasy Adventure	0.00	0.00	0.00	1	Animation Comedy
Family Fantasy Horror	0.00	0.00	0.00	0	Animation Comedy
Family Fantasy Music	0.00	0.00	0.00	0	Animation Comedy Family Fa
ntasy Science Fiction	0.00	0.00	0.00	1	Animation Comedy Family Fantasy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Animation Comedy Fa
mily Fantasy TV Movie	0.00	0.00	0.00	1	Animation Comedy Family Fantasy TV
Movie Science Fiction	0.00	0.00	0.00	0	Animation
Comedy Family Horror	0.00	0.00	0.00	0	Animatio
n Comedy Family Music	0.00	0.00	0.00	2	Animation Comedy
Family Music Fantasy	0.00	0.00	0.00	0	Animation Comedy Family
Music Science Fiction	0.00	0.00	0.00	0	Animation
Comedy Family Mystery	0.00	0.00	0.00	1	Animation Comedy Family Myst
ery Adventure Fantasy	0.00	0.00	0.00	0	Animation Comedy F
amily Mystery Fantasy	0.00	0.00	0.00	0	Animation
Comedy Family Romance	0.00	0.00	0.00	0	Animation Comedy F
amily Science Fiction	0.01	0.50	0.01	2	Animation Comedy Family Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Animation Comedy Family Science Fiction Adve
nture TV Movie Action	0.00	0.00	0.00	0	Animation Comedy Family Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Animation Comedy Family Science Fiction
TV Movie Action Drama	0.00	0.00	0.00	0	Animation C
omedy Family TV Movie	0.00	0.00	0.00	1	Animation Comedy Family TV Movie A
ction Adventure Drama	0.00	0.00	0.00	0	Ani
mation Comedy Fantasy	0.00	0.60	0.01	5	Animation Comedy Fantasy Ac
tion Adventure Family	0.11	1.00	0.19	2	Animation Com
edy Fantasy Adventure	0.00	0.00	0.00	0	Animation Comedy Fan
tasy Adventure Family	0.00	0.00	0.00	1	Animation Comedy Fantasy Ad
venture Family Horror	0.00	0.00	0.00	0	Animation

Comedy Fantasy Drama	0.00	0.00	0.00	1	Animation
Comedy Fantasy Family	0.00	0.00	0.00	0	Animation Comedy
Fantasy Family Drama	0.00	0.00	0.00	0	Animation Comedy
Fantasy Family Music	0.00	0.00	0.00	0	Animation
Comedy Fantasy Horror	0.00	0.00	0.00	0	Animation Comedy Fantasy H
orror Science Fiction	0.00	0.00	0.00	0	Animation
Comedy Fantasy Music	0.00	0.00	0.00	0	Animation C
omedy Fantasy Mystery	0.00	0.00	0.00	1	Animation C
omedy Fantasy Romance	0.00	0.00	0.00	0	Animation Comedy Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Ani
mation Comedy History	0.00	0.00	0.00	0	Animation Comedy History Fantasy War Western Action Adve
nture Science Fiction	0.00	0.00	0.00	0	An
imation Comedy Horror	0.00	0.25	0.00	4	Animation Comedy
Horror Drama Fantasy	0.00	0.00	0.00	0	Animation
Comedy Horror Family	0.00	0.00	0.00	0	Animation
Comedy Horror Mystery	0.00	0.00	0.00	0	Animation C
omedy Horror TV Movie	0.00	0.00	0.00	0	A
nimation Comedy Music	0.00	1.00	0.00	1	Animatio
n Comedy Music Action	0.00	0.00	0.00	0	Animation C
omedy Music Adventure	0.00	0.00	0.00	0	Animati
on Comedy Music Drama	0.00	0.00	0.00	0	Animatio
n Comedy Music Family	0.00	0.00	0.00	0	Animation Comedy
Music Family Fantasy	0.00	0.00	0.00	0	Animation Comedy Music Fantasy Ac
tion Adventure Family	0.00	0.00	0.00	0	Animation Comedy
Music Science Fiction	0.00	0.00	0.00	0	Animation Comedy Music S
cience Fiction Horror	0.00	0.00	0.00	0	Animation
Comedy Music TV Movie	0.00	0.00	0.00	0	Ani
mation Comedy Mystery	0.00	0.00	0.00	0	Animation

Comedy Mystery Family	0.00	0.00	0.00	0	Animation Comedy M
ystery Fantasy Horror	0.00	0.00	0.00	0	Animation Co
medy Mystery Thriller	0.00	0.00	0.00	0	Animation Comedy My
stery Thriller Horror	0.00	0.00	0.00	0	Ani
mation Comedy Romance	0.00	0.00	0.00	2	Animation
Comedy Romance Drama	0.00	0.00	0.00	2	Animation
Comedy Romance Family	0.00	0.00	0.00	0	Animation
Comedy Romance Music	0.00	0.00	0.00	1	Animation Comedy Ro
mance Science Fiction	0.00	0.00	0.00	0	Animation C
omedy Science Fiction	0.01	0.17	0.01	6	Animation Comedy Science Fiction Acti
on Adventure TV Movie	0.00	0.00	0.00	0	Animation Comedy Science F
iction Action Fantasy	0.00	0.00	0.00	0	Animation Comedy Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Animation Comedy Science Fiction Adve
nture Action TV Movie	0.00	0.00	0.00	0	Animation Comedy
Science Fiction Drama	0.00	0.00	0.00	0	Animation Comedy S
cience Fiction Family	0.00	0.00	0.00	0	Animation Comedy Science Fic
tion Family Adventure	0.00	0.00	0.00	1	Animation Comedy Science Fiction Family Fanta
sy Adventure TV Movie	0.00	0.00	0.00	0	Animation Comedy Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Animation Comedy Science Fi
ction Fantasy Romance	0.00	0.00	0.00	0	Animation Comedy Science
Fiction Mystery Drama	0.00	0.00	0.00	0	Animation Comedy Sc
ience Fiction Romance	0.00	0.00	0.00	0	Animation Comedy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Anim
ation Comedy TV Movie	0.33	1.00	0.50	2	Animation C
omedy TV Movie Family	0.00	0.00	0.00	0	Animation Com
edy Thriller TV Movie	0.00	0.00	0.00	0	Ani
mation Comedy Western	0.00	0.00	0.00	2	
Animation Crime	0.00	1.00	0.00	2	A

Animation Crime Action	0.00	0.00	0.00	0	Animation C
Crime Action Adventure	0.00	0.00	0.00	0	Animati
on Crime Action Drama	0.00	0.00	0.00	0	Animation
Crime Action Thriller	0.00	0.00	0.00	0	Animation Cr
ime Adventure Mystery	0.00	0.00	0.00	0	A
Animation Crime Comedy	0.00	0.00	0.00	0	Animat
ion Crime Documentary	0.00	0.00	0.00	0	
Animation Crime Drama	0.00	0.00	0.00	0	Animation Crime Dr
ama Mystery Adventure	0.00	0.00	0.00	0	Animation Crime
Family Comedy Mystery	0.00	0.00	0.00	0	Animation Crime Fantasy My
stery Science Fiction	0.00	0.00	0.00	0	An
imation Crime Mystery	0.00	0.00	0.00	0	Animation
Crime Mystery Action	0.05	1.00	0.09	1	Animation Crime Mystery
Comedy Romance Horror	0.00	0.00	0.00	0	Animation Crime My
stery TV Movie Comedy	0.00	0.00	0.00	0	Animation C
rime Mystery Thriller	0.00	0.00	0.00	0	
Animation Documentary	0.00	1.00	0.01	6	Animation Docume
ntary Comedy TV Movie	0.00	0.00	0.00	0	Animation Docu
mentary Crime Mystery	0.00	0.00	0.00	0	Animat
ion Documentary Drama	0.00	0.00	0.00	0	Animation Docu
mentary Drama Fantasy	0.00	0.00	0.00	0	Animation
Documentary Drama War	0.00	0.00	0.00	0	Animatio
n Documentary History	0.00	0.00	0.00	0	Animation Documentary Horror Ro
mance Fantasy Mystery	0.00	0.00	0.00	0	Animat
ion Documentary Music	0.00	0.00	0.00	0	Anim
ation Documentary War	0.00	0.00	0.00	0	
Animation Drama	0.03	0.84	0.05	32	A
Animation Drama Action	0.00	1.00	0.00	1	Animation Drama Action Comedy Thriller My

stery Horror TV Movie	0.00	0.00	0.00	0	Animation
Drama Action Fantasy	0.00	0.00	0.00	0	Animation
Drama Action Mystery	0.00	0.00	0.00	0	Animation Drama A
ction Science Fiction	0.00	1.00	0.00	1	Animation
Drama Action Thriller	0.00	0.00	0.00	0	Anima
tion Drama Action War	0.00	0.00	0.00	0	Anim
ation Drama Adventure	0.00	0.00	0.00	2	Animation D
rama Adventure Family	0.00	0.00	0.00	0	Animation Drama Adve
nture Family TV Movie	0.00	0.00	0.00	0	Animation Dr
ama Adventure Fantasy	0.00	0.00	0.00	0	Animation Drama Adv
enture Fantasy Comedy	0.00	0.00	0.00	0	Animation Drama Adventure Science F
iction Romance Action	0.00	0.00	0.00	0	A
nimation Drama Comedy	0.00	0.25	0.00	4	Animation
Drama Comedy Fantasy	0.00	0.00	0.00	0	Animation Drama
Comedy Horror Fantasy	0.00	0.00	0.00	0	Animati
on Drama Comedy Music	0.00	0.00	0.00	0	Animation
Drama Comedy Mystery	0.00	0.00	0.00	0	Animation
Drama Comedy Romance	0.00	0.00	0.00	0	Animation Drama C
omedy Science Fiction	0.00	0.00	0.00	0	Animation Drama Comedy Sc
ience Fiction Mystery	0.00	0.00	0.00	0	
Animation Drama Crime	0.00	0.00	0.00	0	Animati
on Drama Crime Action	0.00	0.00	0.00	0	A
nimation Drama Family	0.01	0.14	0.01	7	Animation D
rama Family Adventure	0.00	0.00	0.00	0	Animatio
n Drama Family Comedy	0.00	0.00	0.00	0	Animation
Drama Family Fantasy	0.00	0.00	0.00	1	Animation
Drama Family History	0.00	0.00	0.00	0	Animation
Drama Family Mystery	0.00	0.00	0.00	0	An

imation Drama Fantasy	0.00	1.00	0.01	5	Animation Drama
Fantasy Action Music	0.00	0.00	0.00	0	Animation Dr
ama Fantasy Adventure	0.00	0.00	0.00	0	Animation
Drama Fantasy Comedy	0.00	0.00	0.00	0	Animation
Drama Fantasy Family	0.00	0.00	0.00	0	Animation
Drama Fantasy Horror	0.00	0.00	0.00	1	Animation Drama Fantasy My
stery Science Fiction	0.00	0.00	0.00	0	Animation
Drama Fantasy Romance	0.00	0.00	0.00	2	Animat
ion Drama Fantasy War	0.00	0.00	0.00	0	An
imation Drama History	0.00	0.67	0.01	3	Animation
Drama History Horror	0.00	0.00	0.00	0	Animation Drama History Romance Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Animat
ion Drama History War	0.00	0.00	0.00	1	A
nimation Drama Horror	0.00	1.00	0.00	1	Animation Drama Horror Music Thr
iller Science Fiction	0.00	0.00	0.00	0	Animation
Drama Horror TV Movie	0.00	0.00	0.00	0	Animation Drama
Horror Thriller Crime	0.00	0.00	0.00	0	Animation Drama
Animation Drama Music	0.01	0.75	0.02	4	Animation Drama
Music Fantasy History	0.00	0.00	0.00	0	Animatio
n Drama Music Romance	0.00	0.00	0.00	0	Animation Drama
Music Romance Fantasy	0.00	0.00	0.00	0	An
imation Drama Mystery	0.00	1.00	0.00	1	Animation Dr
ama Mystery Adventure	0.00	0.00	0.00	0	Animation
Drama Mystery Fantasy	0.00	0.00	0.00	2	Animation
Drama Mystery Romance	0.00	0.00	0.00	1	Animation Drama Mystery Sc
ience Fiction Romance	0.00	0.00	0.00	0	An
imation Drama Romance	0.01	0.33	0.01	6	Animation
Drama Romance Comedy	0.00	0.00	0.00	1	Animation

Drama Romance Fantasy	0.00	0.00	0.00	0	
					Animation Drama Ro
mance History Fantasy	0.00	0.00	0.00	0	
					Animation Drama Ro
mance Mystery Fantasy	0.00	0.00	0.00	0	
					Animation Drama Ro
mance Science Fiction	0.00	0.00	0.00	0	
					Animation Drama Romance Science F
iction Fantasy Action	0.00	0.00	0.00	0	
					Animation
Drama Science Fiction	0.01	1.00	0.02	3	
					Animation Drama S
cience Fiction Action	0.00	0.00	0.00	0	
					Animation Drama S
cience Fiction Family	0.00	0.00	0.00	0	
					Animation Drama Sci
ence Fiction Thriller	0.00	0.00	0.00	0	
					Animation
Drama TV Movie Family	0.00	0.00	0.00	0	
					Ani
mation Drama Thriller	0.00	0.00	0.00	0	
					Animation
Drama Thriller Horror	0.00	0.00	0.00	0	
					Animation D
rama Thriller Mystery	0.00	0.00	0.00	0	
Animation Drama War	0.00	0.00	0.00	1	
					Animat
ion Drama War History	0.00	0.00	0.00	0	
					Animation Dram
a War Science Fiction	0.00	0.00	0.00	0	
					Animati
on Drama War TV Movie	0.00	0.00	0.00	0	
					Animation Drama Western
Family Action History	0.00	0.00	0.00	0	
Animation Family	0.11	0.52	0.18	85	
					An
imation Family Action	0.00	0.00	0.00	0	
					Animation Fa
mily Action Adventure	0.00	0.00	0.00	1	
					Animation Family Action Ad
venture Drama Fantasy	0.00	0.00	0.00	0	
					Animation Family Act
ion Adventure Fantasy	0.00	0.00	0.00	0	
					Animation Family Action Adv
enture Fantasy Comedy	0.00	0.00	0.00	0	
					Animation Family Action Adve
nture Science Fiction	0.00	0.00	0.00	0	
					Animation Family Act
ion Adventure Western	0.00	0.00	0.00	0	
					Animation
Family Action Comedy	0.00	0.00	0.00	1	
					Animation Family Action Com
edy Adventure Fantasy	0.00	0.00	0.00	0	
					Animation

Family Action Fantasy	0.50	1.00	0.67	1	Animation Family Act
ion Fantasy Adventure	0.00	0.00	0.00	0	Animation Family
Action Fantasy Comedy	0.00	0.00	0.00	0	Animation Family A
ction Science Fiction	0.00	0.00	0.00	0	Animation F
amily Action TV Movie	0.00	0.00	0.00	0	Anima
tion Family Adventure	0.01	0.55	0.03	11	Animation Fa
mily Adventure Action	0.00	0.00	0.00	0	Animation Fa
mily Adventure Comedy	0.00	1.00	0.01	5	Animation Family Adv
enture Comedy Fantasy	0.00	0.00	0.00	1	Animation Family Adventure Comedy Fa
ntasy Science Fiction	0.00	0.00	0.00	2	Animation Family A
dventure Comedy Music	0.00	0.00	0.00	0	Animation Family Adventure C
omedy Science Fiction	0.00	0.00	0.00	0	Animation F
amily Adventure Crime	0.00	0.00	0.00	0	Animation F
amily Adventure Drama	0.00	0.00	0.00	0	Animation Family Ad
venture Drama Fantasy	0.00	0.00	0.00	1	Animation Fam
ily Adventure Fantasy	0.00	0.50	0.00	2	Animation Family Adv
enture Fantasy Action	0.00	0.00	0.00	0	Animation Family Ad
venture Fantasy Music	0.00	0.00	0.00	0	Animation Family Adve
nture Fantasy Romance	0.00	0.00	0.00	0	Animation Family Adventure Fa
ntasy Science Fiction	0.00	0.00	0.00	1	Animation Family Adventure
Music Mystery Comedy	0.00	0.00	0.00	0	Animation Family Adventure
Music Science Fiction	0.00	0.00	0.00	0	Animation Fam
ily Adventure Mystery	0.00	0.00	0.00	0	Animation Fam
ily Adventure Romance	0.00	0.00	0.00	0	Animation Family Adve
nture Science Fiction	0.00	0.00	0.00	0	Animation Family Adventure Sci
ience Fiction Fantasy	0.00	0.00	0.00	0	Animation Family Adventure Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Animation Fami
ly Adventure TV Movie	0.00	0.00	0.00	0	Animation Fam

ily Adventure Western	0.00	0.00	0.00	0	An
imation Family Comedy	0.02	0.75	0.03	8	Animation
Family Comedy Action	0.00	0.00	0.00	0	Animation Fa
mily Comedy Adventure	0.00	0.00	0.00	0	Animation Family Co
medy Adventure Action	0.00	0.00	0.00	0	Animation Family Com
edy Adventure Fantasy	0.00	0.00	0.00	0	Animation Family Com
edy Adventure Mystery	0.00	0.00	0.00	0	Animation Family Com
edy Adventure Romance	0.00	0.00	0.00	0	Animatio
n Family Comedy Drama	0.00	0.00	0.00	1	Animation
Family Comedy Fantasy	0.00	1.00	0.00	4	Animation Family Com
edy Fantasy Adventure	0.00	0.00	0.00	0	Animation Family Comed
y Fantasy Drama Music	0.00	0.00	0.00	0	Animation Family C
omedy Fantasy Romance	0.00	0.00	0.00	0	Animation Family Co
medy Fantasy TV Movie	0.00	0.00	0.00	0	Animation
Family Comedy Horror	0.00	0.00	0.00	0	Animatio
n Family Comedy Music	0.00	0.00	0.00	1	Animation Family
Comedy Music TV Movie	0.00	0.00	0.00	1	Animation
Family Comedy Mystery	0.00	0.00	0.00	1	Animation Family Comedy My
stery Science Fiction	0.00	0.00	0.00	0	Animation Family C
omedy Science Fiction	0.00	0.00	0.00	0	Animation Family Comedy Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Animation Family Comedy Science Fiction Adve
nture TV Movie Action	0.00	0.00	0.00	0	Animation Family Comedy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Animation F
amily Comedy TV Movie	0.00	0.00	0.00	2	Animation Family Comedy TV Movie Action Adve
nture Science Fiction	0.00	0.00	0.00	0	Animation Family C
omedy TV Movie Horror	0.00	0.00	0.00	0	Animat
ion Family Comedy War	0.00	0.00	0.00	0	Animati
on Family Documentary	0.00	0.00	0.00	0	A

animation Family Drama	0.00	1.00	0.01	2	Animation
Family Drama Fantasy	0.00	0.00	0.00	0	Ani
mation Family Fantasy	0.03	0.54	0.06	24	Animation Family Fan
tasy Action Adventure	0.00	0.00	0.00	0	Animation Fam
ily Fantasy Adventure	0.01	0.67	0.01	6	Animation Family Fan
tasy Adventure Comedy	0.00	0.00	0.00	0	Animation Family Fantasy Adv
enture Comedy Romance	0.00	0.00	0.00	0	Animation Family Fa
ntasy Adventure Drama	0.00	0.00	0.00	0	Animation Family Fan
tasy Adventure Horror	0.00	0.00	0.00	0	Animation Family Fa
ntasy Adventure Music	0.00	0.00	0.00	0	Animation Family Fantasy Adve
nture Science Fiction	0.00	0.00	0.00	0	Animation
Family Fantasy Comedy	0.00	1.00	0.00	1	Animation Family Fan
tasy Comedy Adventure	0.00	0.00	0.00	1	Animation Family Fantasy Com
edy Adventure Mystery	0.00	0.00	0.00	0	Animation Family Fantasy C
omedy Science Fiction	0.00	0.00	0.00	0	Animation Family Fanta
sy Documentary Comedy	0.00	0.00	0.00	0	Animation
Family Fantasy Drama	0.00	0.00	0.00	0	Animation Family Fa
ntasy Drama Adventure	0.00	0.00	0.00	0	Animation F
amily Fantasy History	0.00	0.00	0.00	0	Animation
Family Fantasy Horror	0.00	0.00	0.00	0	Animation
Family Fantasy Music	0.00	0.00	0.00	0	Animation Family Fantasy Music Adventure C
omedy Science Fiction	0.00	0.00	0.00	0	Animation F
amily Fantasy Romance	0.01	1.00	0.02	1	Animation Family Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Animation Fa
mily Fantasy TV Movie	0.00	0.00	0.00	1	Animation Family Fanta
sy TV Movie Adventure	0.00	0.00	0.00	1	Ani
mation Family History	0.00	0.00	0.00	0	Animati
on Family History War	0.00	0.00	0.00	0	Animation

Family Horror Fantasy	0.00	0.00	0.00	0	A
Animation Family Music	0.01	0.60	0.01	5	Animation F
Family Music Adventure	0.00	0.00	0.00	0	Animatio
n Family Music Comedy	0.00	0.00	0.00	0	Animation Fami
ly Music Comedy Drama	0.00	0.00	0.00	0	Animation
Family Music Fantasy	0.00	0.00	0.00	0	Animation Family Music Fan
tasy Comedy Adventure	0.00	0.00	0.00	0	Animation Family Music Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Animation
Family Music Romance	0.00	0.00	0.00	0	Animation Family
Music Science Fiction	0.00	0.00	0.00	0	Animation
Family Music TV Movie	0.00	0.00	0.00	0	Ani
mation Family Mystery	0.00	0.00	0.00	0	Animation Family Mys
tery Adventure Comedy	0.00	0.00	0.00	0	Animation
Family Mystery Comedy	0.00	0.00	0.00	0	Animation Family Mystery C
omedy Science Fiction	0.00	0.00	0.00	0	Animation Family M
ystery Comedy Western	0.00	0.00	0.00	0	Animation Family M
ystery Fantasy Comedy	0.00	0.00	0.00	0	Ani
mation Family Romance	0.00	0.00	0.00	0	Animati
on Family Romance War	0.00	0.00	0.00	0	Animation F
amily Science Fiction	0.00	1.00	0.00	1	Animation Family S
cience Fiction Action	0.00	0.00	0.00	0	Animation Family Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Animation Family Science Fic
tion Adventure Comedy	0.00	0.00	0.00	0	Animation Family S
cience Fiction Comedy	0.00	0.00	0.00	0	Animation Family Science Fict
ion Fantasy Adventure	0.00	0.00	0.00	0	Animation Family Science Fiction TV Movie Ac
tion Comedy Adventure	0.00	0.00	0.00	0	Animation Family Science Fi
ction TV Movie Comedy	0.00	0.00	0.00	0	Anim
ation Family TV Movie	0.01	0.75	0.03	4	Animation Family TV M

ovie Action Adventure	0.00	0.00	0.00	0	
					Animation Fami
ly TV Movie Adventure	0.00	0.00	0.00	0	
					Animation Family TV Movie Adv
enture Comedy Fantasy	0.00	0.00	0.00	0	
					Animation Family TV Mo
vie Adventure Fantasy	0.00	0.00	0.00	0	
					Animation F
amily TV Movie Comedy	0.00	0.00	0.00	0	
					Animation Family TV Movie C
omedy Science Fiction	0.00	0.00	0.00	0	
					Animation Fa
mily TV Movie Fantasy	0.00	0.00	0.00	2	
					Animation Family TV Mo
vie Fantasy Adventure	0.00	0.00	0.00	0	
					Animation
Family TV Movie Music	0.00	0.00	0.00	0	
					Animation Family TV
Movie Mystery Comedy	0.00	0.00	0.00	0	
					Animation Family TV
Movie Mystery Fantasy	0.00	0.00	0.00	0	
					Ani
mation Family Western	0.00	0.00	0.00	0	
					Animation Family Wes
tern Adventure Comedy	0.00	0.00	0.00	0	
Animation Fantasy	0.06	0.09	0.07	33	
					Ani
mation Fantasy Action	0.00	0.00	0.00	5	
					Animation Fan
tasy Action Adventure	0.00	1.00	0.00	3	
					Animation Fantasy Action Co
medy Adventure Family	0.06	1.00	0.12	2	
					Animation Fantasy Action Comedy Science Fic
tion Adventure Family	0.00	0.00	0.00	0	
					Animation Fantas
y Action Drama Comedy	0.00	0.00	0.00	0	
					Animation Fantasy
Action Family Horror	0.00	0.00	0.00	0	
					Animation
Fantasy Action Horror	0.00	0.00	0.00	0	
					Animation Fantasy Action Music Ad
venture Comedy Family	0.00	0.00	0.00	0	
					Animation F
antasy Action Mystery	0.00	0.00	0.00	0	
					Animation Fantasy A
ction Science Fiction	0.00	0.00	0.00	0	
					Animat
ion Fantasy Adventure	0.00	0.11	0.00	9	
					Animation Fan
tasy Adventure Action	0.02	0.25	0.03	4	
					Animation Fantasy A
dventure Action Drama	0.00	0.00	0.00	0	
					Animation Fan
tasy Adventure Comedy	0.00	0.00	0.00	2	
					Animation Fa

antasy Adventure Drama	0.00	0.00	0.00	1	Animation Fantasy Adventure Drama
antasy Adventure Family	0.00	0.00	0.00	2	Animation Fantasy Adventure Family
venture Family Action	0.00	0.00	0.00	0	Animation Fantasy Adventure Family Action
amily Science Fiction	0.00	0.00	0.00	1	Animation Fantasy Adventure Family Science Fiction
asy Adventure Mystery	0.00	0.00	0.00	0	Animation Fantasy Adventure Mystery
mation Fantasy Comedy	0.00	0.00	0.00	2	Animation Fantasy Comedy Adventure
tion Adventure Family	0.00	0.00	0.00	0	Animation Fantasy Comedy Adventure Family
tasy Comedy Adventure	0.00	0.00	0.00	0	Animation Fantasy Comedy Adventure Family Comedy
nture Science Fiction	0.00	0.00	0.00	0	Animation Fantasy Comedy Adventure Science Fiction
Fantasy Comedy Family	0.00	0.00	0.00	1	Animation Fantasy Comedy Adventure Family
mily Action Adventure	0.00	0.00	0.00	0	Animation Fantasy Comedy Adventure Family Action
ance Adventure Horror	0.00	0.00	0.00	0	Animation Fantasy Comedy Adventure Horror
imation Fantasy Drama	0.00	0.50	0.01	4	Animation Fantasy Drama
ntasy Drama Adventure	0.00	0.00	0.00	0	Animation Fantasy Drama Adventure
rama Adventure Family	0.00	0.00	0.00	0	Animation Fantasy Drama Adventure Family
asy Drama Documentary	0.00	0.00	0.00	0	Animation Fantasy Drama Documentary
Fantasy Drama Family	0.00	0.00	0.00	0	Animation Fantasy Drama Family
Fantasy Drama Horror	0.00	0.00	0.00	0	Animation Fantasy Drama Horror
Drama Horror Thriller	0.00	0.00	0.00	0	Animation Fantasy Drama Horror Thriller
n Fantasy Drama Music	0.00	0.00	0.00	0	Animation Fantasy Drama Music
Drama Mystery Action	0.00	0.00	0.00	0	Animation Fantasy Drama Mystery Action
y Drama Mystery Music	0.00	0.00	0.00	0	Animation Fantasy Drama Mystery Music
Drama Science Fiction	0.00	0.00	0.00	0	Animation Fantasy Drama Science Fiction
Drama Thriller Horror	0.00	0.00	0.00	0	Animation Fantasy Drama Thriller Horror
mation Fantasy Family	0.01	0.38	0.01	8	Animation Fantasy Family
Fantasy Family Action	0.00	0.00	0.00	0	Animation Fantasy Family Action
mily Action Adventure	0.00	0.00	0.00	0	Animation Fantasy Family Action Adventure
tion Adventure Comedy	0.00	0.00	0.00	0	Animation Fantasy Family Action Comedy

tasy Family Adventure	0.00	0.00	0.00	1	
omedy Science Fiction	0.00	0.00	0.00	0	Animation Fantasy Family Adventure C
ly Adventure TV Movie	0.00	0.00	0.00	0	Animation Fantasy Fami
mily Comedy Adventure	0.00	0.00	0.00	0	Animation Fantasy Fa
edy Romance Adventure	0.00	0.00	0.00	0	Animation Fantasy Family Com
ntasy Family TV Movie	0.00	0.00	0.00	0	Animation Fa
amily TV Movie Comedy	0.00	0.00	0.00	0	Animation Fantasy F
ation Fantasy History	0.00	0.00	0.00	0	Anim
mation Fantasy Horror	0.01	1.00	0.02	1	Ani
Fantasy Horror Action	0.00	0.00	0.00	1	Animation
ence Fiction TV Movie	0.00	0.00	0.00	0	Animation Fantasy Horror Action Thriller Sci
Fantasy Horror Drama	0.00	0.00	0.00	0	Animation
Fantasy Horror Family	0.00	0.00	0.00	0	Animation
orror Science Fiction	0.00	0.00	0.00	0	Animation Fantasy H
tion Action Adventure	0.00	0.00	0.00	0	Animation Fantasy Horror Science Fic
imation Fantasy Music	0.00	0.00	0.00	0	An
venture Comedy Family	0.00	0.00	0.00	0	Animation Fantasy Music Ad
Fantasy Music Romance	0.00	0.00	0.00	0	Animation
Music Science Fiction	0.00	0.00	0.00	0	Animation Fantasy
ation Fantasy Mystery	0.00	0.00	0.00	2	Anim
Fantasy Mystery Drama	0.00	0.00	0.00	0	Animation
Mystery Horror Action	0.00	0.00	0.00	0	Animation Fantasy
ntasy Mystery Romance	0.00	0.00	0.00	0	Animation Fa
Romance Action Crime	0.00	0.00	0.00	0	Animation Fantasy Mystery
ation Fantasy Romance	0.00	0.00	0.00	3	Anim
Romance Comedy Family	0.00	0.00	0.00	0	Animation Fantasy
antasy Romance Horror	0.00	0.00	0.00	0	Animation F
Romance Music Family	0.00	0.00	0.00	0	Animation Fantasy
					Animation Fa

antasy Science Fiction	0.00	1.00	0.00	1	Animation Fantasy Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Animation Fantasy S
cience Fiction Horror	0.00	0.00	0.00	0	Animation Fantasy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Anima
tion Fantasy TV Movie	1.00	0.33	0.50	3	Anima
tion Fantasy Thriller	0.00	0.00	0.00	1	
Animation History	0.00	1.00	0.01	6	Animation History Ac
tion Adventure Family	0.20	1.00	0.33	1	Animation History Action
Drama Fantasy Romance	0.00	0.00	0.00	0	Animation History
Action Drama TV Movie	0.00	0.00	0.00	0	Animat
ion History Adventure	0.00	0.00	0.00	0	Animation History Adve
nture Science Fiction	0.00	0.00	0.00	0	Ani
mation History Comedy	0.00	0.00	0.00	0	Animation History
Comedy Drama Mystery	0.00	0.00	0.00	0	Animatio
n History Documentary	0.00	0.00	0.00	1	Animation History
Documentary TV Movie	0.00	0.00	0.00	0	An
imation History Drama	0.00	0.00	0.00	2	Animation
History Drama Family	0.00	0.00	0.00	0	Animat
ion History Drama War	0.00	0.00	0.00	0	Ani
mation History Family	0.00	0.00	0.00	0	Anim
ation History Fantasy	0.00	0.00	0.00	0	
Animation History War	0.00	0.00	0.00	1	Animat
ion History War Drama	0.00	0.00	0.00	0	
Animation Horror	0.01	0.50	0.03	8	An
imation Horror Action	0.00	0.00	0.00	0	Animatio
n Horror Action Crime	0.00	0.00	0.00	0	Animation Horror A
ction Science Fiction	0.00	0.00	0.00	0	Anima
tion Horror Adventure	0.00	0.00	0.00	0	Animation Horror Ad

venture Comedy Family	0.00	0.00	0.00	0	Animation Ho
rror Comedy Adventure	0.00	0.00	0.00	0	Animation
Horror Comedy Fantasy	0.00	0.00	0.00	0	A
nimation Horror Drama	0.00	0.00	0.00	0	Animation
Horror Family Fantasy	0.00	0.00	0.00	0	Ani
mation Horror Fantasy	0.00	0.00	0.00	1	Animation
Horror Fantasy Action	0.00	0.00	0.00	0	Animation Hor
ror Fantasy Adventure	0.00	0.00	0.00	1	Animation
Horror Fantasy Family	0.00	0.00	0.00	0	A
nimation Horror Music	0.00	0.00	0.00	0	Ani
mation Horror Mystery	0.00	0.00	0.00	0	Animation H
orror Mystery Fantasy	0.00	0.00	0.00	0	Animation Horror My
stery Science Fiction	0.00	0.00	0.00	0	Animation H
orror Science Fiction	0.00	0.00	0.00	2	Animation Horror Science Fi
ction Action Thriller	0.00	0.00	0.00	0	Animation Horror Sc
ience Fiction Mystery	0.00	0.00	0.00	0	Animation Horror Sci
ence Fiction Thriller	0.00	0.00	0.00	0	Anim
ation Horror TV Movie	0.00	0.00	0.00	0	Anim
ation Horror Thriller	0.00	0.00	0.00	0	
Animation Music	0.01	0.67	0.01	9	A
nimation Music Action	0.00	0.00	0.00	0	Animation M
usic Adventure Comedy	0.00	0.00	0.00	0	Animation Music Adventure
Comedy Family Fantasy	0.00	0.00	0.00	0	A
nimation Music Comedy	0.00	0.00	0.00	1	Animation Music Comedy Adventure
Drama Family Fantasy	0.00	0.00	0.00	0	Animati
on Music Comedy Drama	0.00	0.00	0.00	0	Animation Music
Comedy Fantasy Family	0.00	0.00	0.00	0	
Animation Music Crime	0.00	0.00	0.00	0	Animat

ion Music Documentary	0.00	0.00	0.00	0	Animation Mus
ic Documentary Comedy	0.00	0.00	0.00	0	
Animation Music Drama	0.00	0.00	0.00	1	Animatio
n Music Drama History	0.00	0.00	0.00	0	A
nimation Music Family	0.00	0.00	0.00	1	Animatio
n Music Family Comedy	0.00	0.00	0.00	0	Animation
Music Family TV Movie	0.00	0.00	0.00	0	An
imation Music Fantasy	0.00	0.00	0.00	1	Animation
Music Fantasy Comedy	0.00	0.00	0.00	0	An
imation Music Romance	0.00	0.00	0.00	0	Animation
Music Romance Family	0.00	0.00	0.00	0	Ani
mation Music TV Movie	0.00	0.00	0.00	0	
Animation Mystery	0.01	0.80	0.01	5	Ani
mation Mystery Action	0.00	0.00	0.00	0	Animation
Mystery Action Crime	0.00	0.00	0.00	0	Animat
ion Mystery Adventure	0.00	0.00	0.00	0	Animation
Mystery Comedy Family	0.00	0.00	0.00	0	An
imation Mystery Crime	0.00	0.00	0.00	0	Animation Mystery C
rime Action Adventure	0.00	0.00	0.00	0	
Animation Mystery Documentary Horror Science Fiction History Thriller TV Movie War A					
ction Adventure Crime	0.00	0.00	0.00	0	Animation Mystery D
rama Horror Adventure	0.00	0.00	0.00	0	Ani
mation Mystery Family	0.00	0.00	0.00	0	Animation
Mystery Family Horror	0.00	0.00	0.00	0	Anim
ation Mystery Fantasy	0.00	0.00	0.00	1	Animation M
ystery Fantasy Horror	0.00	0.00	0.00	1	Ani
mation Mystery Horror	0.00	0.00	0.00	1	Animation My
stery Science Fiction	0.00	0.00	0.00	0	Animation Mystery S
cience Fiction Action	0.00	0.00	0.00	0	Animation Mystery

Science Fiction Drama	0.00	0.00	0.00	0	Animation Mystery Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Animation Mystery S
cience Fiction Horror	0.00	0.00	0.00	0	Anima
tion Mystery TV Movie	0.00	0.00	0.00	0	
Animation Romance	0.00	0.00	0.00	2	Animation Romance Ad
venture Drama Fantasy	0.00	0.00	0.00	0	Ani
mation Romance Comedy	0.00	0.67	0.00	3	Animation
Romance Comedy Drama	0.00	0.00	0.00	1	Animation Romance
Comedy Drama Fantasy	0.00	0.00	0.00	1	Animation R
omance Comedy Fantasy	0.00	0.00	0.00	0	Animation Romance C
omedy Science Fiction	0.00	0.00	0.00	0	An
imation Romance Drama	0.00	0.33	0.00	3	Animation
Romance Drama Comedy	0.00	0.00	0.00	2	Animation
Romance Drama Fantasy	0.00	0.00	0.00	0	Animation
Romance Drama History	0.00	0.00	0.00	0	Animation R
omance Family Fantasy	0.00	0.00	0.00	0	Anim
ation Romance Fantasy	0.00	0.00	0.00	0	Animation Romance
Fantasy Action Drama	0.00	0.00	0.00	0	Animation R
omance Fantasy Family	0.00	0.00	0.00	0	Animation
Romance Fantasy Music	0.00	0.00	0.00	0	An
imation Romance Music	0.00	0.00	0.00	0	Animation Ro
mance Science Fiction	0.00	0.00	0.00	0	Animation Romance Science
Fiction Action Comedy	0.00	0.00	0.00	0	Animation Romance Science
Fiction Comedy Family	0.00	0.00	0.00	0	Animation Romance
Science Fiction Drama	0.00	0.00	0.00	0	Anima
tion Romance TV Movie	0.00	0.00	0.00	0	Anim
ation Science Fiction	0.03	0.59	0.06	29	Animation S
cience Fiction Action	0.04	0.45	0.07	11	Animation Science Fic

tion Action Adventure	0.00	0.00	0.00	1	
					Animation Science Fiction Actio
n Adventure Drama War	0.00	0.00	0.00	0	
					Animation Science Fiction Ac
tion Adventure Family	0.00	0.00	0.00	0	
					Animation Science Fiction Act
ion Adventure Romance	0.00	0.00	0.00	0	
					Animation Science
Fiction Action Comedy	0.00	0.00	0.00	0	
					Animation Science Fiction
Action Comedy Romance	0.00	0.00	0.00	0	
					Animation Science
Fiction Action Drama	0.00	0.00	0.00	1	
					Animation Science Fiction A
ction Drama Adventure	0.00	0.00	0.00	0	
					Animation Science
Fiction Action Family	0.00	0.00	0.00	0	
					Animation Science Fiction Action Fam
ily Adventure Fantasy	0.00	0.00	0.00	0	
					Animation Science
Fiction Action Horror	0.00	0.00	0.00	0	
					Animation Science F
iction Action Mystery	0.00	0.00	0.00	0	
					Animation Science Fiction Action Ro
mance Adventure Drama	0.00	0.00	0.00	0	
					Animation Science Fi
ction Action Thriller	0.00	0.00	0.00	0	
					Animation Scie
nce Fiction Adventure	0.01	0.75	0.03	4	
					Animation Science Fic
tion Adventure Comedy	0.00	0.00	0.00	0	
					Animation Science Fiction A
dventure Crime Comedy	0.00	0.00	0.00	0	
					Animation Science Fi
ction Adventure Drama	0.00	0.00	0.00	1	
					Animation Science Fiction Ad
venture Drama Fantasy	0.00	0.00	0.00	0	
					Animation Science Fiction Adventure Drama F
antasy Romance Action	0.00	0.00	0.00	0	
					Animation Science Fic
tion Adventure Family	0.00	0.00	0.00	1	
					Animation Science Fiction Ad
venture Family Action	0.00	0.00	0.00	0	
					Animation Science Fict
ion Adventure Fantasy	0.00	0.00	0.00	0	
					Animation Science Fiction Ad
venture Fantasy Music	0.00	0.00	0.00	0	
					Animation Science Fic
tion Adventure Horror	0.00	0.00	0.00	1	
					Animation S
cience Fiction Comedy	0.00	0.00	0.00	3	
					Animation Science
Fiction Comedy Action	0.00	0.00	0.00	0	
					Animation Science Fic
tion Comedy Adventure	0.00	0.00	0.00	0	
					Animation Science

Fiction Comedy Drama	0.00	0.00	0.00	0	Animation Science
Fiction Comedy Family	0.00	0.00	0.00	0	Animation Science Fiction C
Comedy Family TV Movie	0.00	0.00	0.00	0	Animation
Science Fiction Crime	0.00	0.00	0.00	0	Animation Science Fict
ion Crime Mystery War	0.00	0.00	0.00	0	Animation Scienc
e Fiction Documentary	0.00	0.00	0.00	0	Animation
Science Fiction Drama	0.00	0.00	0.00	1	Animation Science
Fiction Drama Comedy	0.00	0.00	0.00	0	Animation Science
Fiction Drama Fantasy	0.00	0.00	0.00	0	Animation Scienc
e Fiction Drama Music	0.00	0.00	0.00	0	Animation Science
Fiction Drama Romance	0.00	0.00	0.00	0	Animation S
Science Fiction Family	0.00	0.00	0.00	0	Animation Science
Fiction Family Action	0.00	0.00	0.00	0	Animation Science Fiction
Family Action Mystery	0.00	0.00	0.00	0	Animation Science Fiction F
amily Action TV Movie	0.00	0.00	0.00	0	Animation Science Fic
tion Family Adventure	0.00	0.00	0.00	0	Animation Science Fiction F
amily Adventure Drama	0.00	0.00	0.00	0	Animation Science Fiction Family Adv
enture Fantasy Comedy	0.00	0.00	0.00	0	Animation Science
Fiction Family Comedy	0.00	0.00	0.00	0	Animation Science Fiction F
amily TV Movie Action	0.00	0.00	0.00	0	Animation Sc
ience Fiction Fantasy	0.00	1.00	0.00	2	Animation Science F
iction Fantasy Action	0.00	0.00	0.00	0	Animation Science Fiction Fantas
y Action Drama Comedy	0.00	0.00	0.00	0	Animation Science Fict
ion Fantasy Adventure	0.00	0.00	0.00	1	Animation Science Fiction Fantasy Ad
venture Family Comedy	0.00	0.00	0.00	0	Animation Science F
iction Fantasy Family	0.00	0.00	0.00	0	Animation S
Science Fiction Horror	0.00	0.00	0.00	1	Animation Science Fictio
n Horror Action Crime	0.00	0.00	0.00	0	Animation Science Fi

ction Horror TV Movie	0.00	0.00	0.00	0	Animation
Science Fiction Music	0.00	0.00	0.00	0	Animation Science
Fiction Mystery Crime	0.00	0.00	0.00	0	Animation Science
Fiction Mystery Drama	0.00	0.00	0.00	0	Animation Sc
ience Fiction Romance	0.00	0.00	0.00	1	Animation Science
Fiction Romance Drama	0.00	0.00	0.00	0	Animation Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Animation Science Fiction TV Movie Adventure Family Comedy F
antasy Mystery Action	0.00	0.00	0.00	0	Animation Sci
ence Fiction Thriller	0.00	1.00	0.00	1	Animation Science Fic
tion Thriller Mystery	0.00	0.00	0.00	0	Animatio
n Science Fiction War	0.00	0.00	0.00	1	Animation Science
Fiction War Adventure	0.00	0.00	0.00	1	Animation Scie
nce Fiction War Drama	0.00	0.00	0.00	0	Animation Sc
ience Fiction Western	0.00	0.00	0.00	0	Animation T
Animation TV Movie	0.00	0.00	0.00	3	Animation TV Movi
V Movie Action Comedy	0.00	0.00	0.00	0	Animation TV Movie Adventure Comedy F
e Action Comedy Crime	0.00	0.00	0.00	1	Anim
amily Science Fiction	0.00	0.00	0.00	0	Animation T
ation TV Movie Comedy	0.00	0.00	0.00	1	Animation TV Movie Comedy Scie
V Movie Comedy Family	0.00	0.00	0.00	1	Anim
nce Fiction Adventure	0.00	0.00	0.00	0	Animation TV
ation TV Movie Family	0.00	0.00	0.00	3	Anima
Movie Fantasy Family	0.00	0.00	0.00	0	Animation TV
tion TV Movie History	0.00	0.00	0.00	0	Animation Thr
Movie Science Fiction	0.00	0.00	0.00	0	
Animation Thriller	0.00	1.00	0.01	5	
ation Thriller Action	0.00	0.00	0.00	0	
mation Thriller Crime	0.00	0.00	0.00	0	

iller Science Fiction	0.00	0.00	0.00	0	
Animation War	0.00	0.00	0.00	0	Animation War
Action Fantasy Drama	0.00	0.00	0.00	0	Animation War A
ction Science Fiction	0.00	0.00	0.00	0	Animation War Adventure S
cience Fiction Action	0.00	0.00	0.00	0	
Animation War Comedy	0.00	0.00	0.00	0	Anim
ation War Documentary	0.00	0.00	0.00	0	
Animation War Drama	0.00	0.00	0.00	1	Anima
tion War Drama Family	0.00	0.00	0.00	0	Animat
ion War Drama Romance	0.00	0.00	0.00	0	
Animation War History	0.00	0.00	0.00	0	Animatio
n War Science Fiction	0.00	0.00	0.00	0	
Animation Western	0.00	0.00	0.00	2	Animation
Western Action Crime	0.00	0.00	0.00	0	Animation Western A
ction Science Fiction	0.00	0.00	0.00	0	Ani
mation Western Comedy	0.00	0.00	0.00	0	Animation Western Scie
nce Fiction Adventure	0.00	0.00	0.00	0	
Comedy	0.44	0.71	0.54	700	Comedy Actio
n Adventure Animation	0.00	0.00	0.00	0	Co
medy Action Animation	0.00	0.00	0.00	1	Comedy Actio
n Animation Adventure	0.00	0.00	0.00	0	Comedy Action Animation Adventure Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Comedy A
ction Animation Drama	0.00	0.00	0.00	0	Comedy Action Animatio
n Science Fiction War	0.00	0.00	0.00	0	Comedy A
ction Crime Animation	0.00	0.00	0.00	0	Comed
y Adventure Animation	0.00	0.00	0.00	1	Comedy Advent
ure Animation Fantasy	0.00	0.00	0.00	0	Comedy Adventure Ani
mation Fantasy Family	0.00	0.00	0.00	1	Comedy Adventure

Documentary Animation	0.00	0.00	0.00	2	
tasy Animation Family	0.00	0.00	0.00	0	Comedy Adventure Fan
Comedy Animation	0.03	0.05	0.04	21	
medy Animation Action	0.00	0.00	0.00	3	Co
ntasy Science Fiction	0.00	0.00	0.00	0	Comedy Animation Action Adventure Fa
on Adventure Thriller	0.00	0.00	0.00	0	Comedy Animation Acti
ction Science Fiction	0.00	0.00	0.00	0	Comedy Animation A
ation Action Thriller	0.00	0.00	0.00	0	Comedy Anim
y Animation Adventure	0.00	0.00	0.00	2	Comed
tion Adventure Family	0.00	0.00	0.00	1	Comedy Anima
enture Family Fantasy	0.00	0.00	0.00	0	Comedy Animation Adv
ion Adventure Fantasy	0.00	0.00	0.00	0	Comedy Animat
tion Adventure Horror	0.00	0.00	0.00	0	Comedy Anima
on Adventure TV Movie	0.00	0.00	0.00	0	Comedy Animati
Family Drama Mystery	0.00	0.00	0.00	0	Comedy Animation Adventure TV Movie
Animation Documentary	0.00	0.00	0.00	0	Comedy
omedy Animation Drama	0.00	1.00	0.00	1	C
nimation Drama Action	0.00	0.00	0.00	0	Comedy A
imation Drama Romance	0.00	0.00	0.00	0	Comedy An
medy Animation Family	0.00	0.00	0.00	1	Co
tion Family Adventure	0.00	0.00	0.00	1	Comedy Anima
ily Adventure Western	0.00	0.00	0.00	0	Comedy Animation Fam
amily Science Fiction	0.00	0.00	0.00	0	Comedy Animation F
edy Animation Fantasy	0.00	0.00	0.00	0	Com
ion Fantasy Adventure	0.00	0.00	0.00	0	Comedy Animat
tasy Adventure Horror	0.00	0.00	0.00	0	Comedy Animation Fan
nture Science Fiction	0.00	0.00	0.00	0	Comedy Animation Fantasy Adve
ntasy Science Fiction	0.00	0.00	0.00	0	Comedy Animation Fa
					Comedy Animation Hi

story Science Fiction	0.00	0.00	0.00	0	
					Co
medy Animation Horror	0.00	0.00	0.00	0	
					Comedy An
imation Horror Action	0.00	0.00	0.00	0	
					C
omedy Animation Music	0.00	0.00	0.00	0	
					Comedy Ani
mation Music TV Movie	0.00	0.00	0.00	0	
					Com
edy Animation Mystery	0.00	0.00	0.00	1	
					Com
edy Animation Romance	0.00	0.00	0.00	0	
					Comedy Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Comedy Animation S
cience Fiction Action	0.00	0.00	0.00	0	
					Comedy Animation Scie
nce Fiction Adventure	0.00	0.00	0.00	0	
					Comedy Animation
Science Fiction Drama	0.00	0.00	0.00	0	
					Come
dy Animation TV Movie	0.00	0.00	0.00	1	
					Com
edy Animation Western	0.00	0.00	0.00	0	
					C
omedy Crime Animation	0.00	0.00	0.00	0	
					Comedy Documentary Advent
ure Animation Fantasy	0.00	0.00	0.00	0	
					C
omedy Drama Animation	0.00	1.00	0.00	3	
					Comedy Drama Fant
asy Romance Animation	0.00	0.00	0.00	0	
					Comedy Dr
ama Romance Animation	0.00	0.00	0.00	0	
					Comedy Drama Roman
ce Animation TV Movie	0.00	0.00	0.00	0	
					Co
medy Family Animation	0.00	0.00	0.00	1	
					Comedy Famili
y Animation Adventure	0.00	0.00	0.00	1	
					Comedy Family Animat
ion Adventure Fantasy	0.00	0.00	0.00	0	
					Comedy Family An
imation Fantasy Music	0.00	0.00	0.00	1	
					Comedy Family Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Comedy Family Anima
tion TV Movie Fantasy	0.00	0.00	0.00	0	
					Comedy Fam
ily Fantasy Animation	0.00	0.00	0.00	0	
					Comedy Family Fantasy Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Comedy Fantasy A
nimation Drama Family	0.00	0.00	0.00	0	
					Comedy Fant

asy Animation Romance	0.00	0.00	0.00	0	
					Comedy Fan
tasy Horror Animation	0.00	0.00	0.00	0	
					Co
medy Horror Animation	0.00	0.00	0.00	0	
					C
omedy Music Animation	0.00	0.00	0.00	0	
					Com
edy Romance Animation	0.00	0.00	0.00	1	
					Comedy Roma
nce Animation Fantasy	0.00	0.00	0.00	0	
					Comedy Scie
nce Fiction Animation	0.00	0.00	0.00	1	
					Comedy Science Fiction Animation TV Movie Act
ion Adventure Fantasy	0.00	0.00	0.00	0	
					Comedy Science Fict
ion Fantasy Animation	0.00	0.00	0.00	0	
					Come
dy TV Movie Animation	0.00	0.00	0.00	0	
Crime	0.03	0.68	0.06	19	
					Crime Actio
n Adventure Animation	0.00	0.00	0.00	0	
					C
rime Action Animation	0.00	0.00	0.00	0	
					C
rime Animation Action	0.00	0.00	0.00	0	
					Crime Animation Action
Fantasy Mystery Drama	0.00	0.00	0.00	0	
					C
rime Animation Comedy	0.00	0.00	0.00	0	
					Crime Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Crime Comedy Drama
Documentary Animation	0.00	0.00	0.00	0	
					Crime TV Mo
vie Animation Mystery	0.00	0.00	0.00	0	
Documentary	0.22	0.47	0.30	95	
Documentary Animation	0.02	0.50	0.03	14	
					Documen
tary Animation Comedy	0.00	1.00	0.00	1	
					Documentary Ani
mation Comedy History	0.00	0.00	0.00	0	
					Docume
ntary Animation Drama	0.00	0.00	0.00	0	
					Documentary A
nimation Drama Comedy	0.00	0.00	0.00	0	
					Documentary
Animation Drama Music	0.00	0.00	0.00	0	
					Documen
tary Animation Family	0.00	0.00	0.00	0	
					Documentary Ani
mation Family History	0.00	0.00	0.00	0	
					Document

ary Animation History	0.00	0.00	0.00	0	
					Documentary
Animation History War	0.00	0.00	0.00	0	
					Docume
ntary Animation Music	0.00	0.00	0.00	0	
					Documentary Ani
mation Music TV Movie	0.00	0.00	0.00	0	
					Document
ary Animation Mystery	0.00	0.00	0.00	0	
					Documentary Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Documenta
ry Animation TV Movie	0.00	0.00	0.00	0	
					Documentary Animation TV
Movie Science Fiction	0.00	0.00	0.00	0	
					Docu
mentary Animation War	0.00	0.00	0.00	0	
					Documentary Co
medy Family Animation	0.00	0.00	0.00	0	
					Docume
ntary Crime Animation	0.00	0.00	0.00	0	
					Docume
ntary Drama Animation	0.00	0.00	0.00	0	
					Documen
tary Family Animation	0.00	0.00	0.00	0	
					Document
ary History Animation	0.00	0.00	0.00	1	
					Documentary
History War Animation	0.00	0.00	0.00	0	
					Docume
ntary Music Animation	0.00	0.00	0.00	0	
					Documentary Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Documenta
ry TV Movie Animation	0.00	0.00	0.00	0	
					Documentary War Histor
y Animation Adventure	0.25	1.00	0.40	1	
					Document
ary Western Animation	0.00	0.00	0.00	0	
Drama	0.18	0.39	0.25	181	
					D
rama Action Animation	0.00	0.00	0.00	0	
					Drama Ac
tion Animation Comedy	0.00	0.00	0.00	0	
					Dram
a Adventure Animation	0.00	0.00	0.00	1	
Drama Animation	0.01	0.80	0.02	5	
					D
rama Animation Action	0.00	0.00	0.00	0	
					Drama Ani
mation Action Fantasy	0.00	0.00	0.00	0	
					Dram
a Animation Adventure	0.00	0.00	0.00	0	
					Drama Anima

tion Adventure Action	0.00	0.00	0.00	0	
					Drama Animat
ion Adventure History	0.00	0.00	0.00	0	
					D
rama Animation Comedy	0.00	1.00	0.00	1	
					Drama
Animation Documentary	0.00	0.00	0.00	1	
					D
rama Animation Family	0.00	0.00	0.00	0	
					Drama An
imation Family Comedy	0.00	0.00	0.00	0	
					Dr
ama Animation Fantasy	0.00	0.00	0.00	1	
					Dr
ama Animation History	0.00	0.00	0.00	0	
					Drama Animat
ion History Adventure	0.00	0.00	0.00	0	
Drama Animation Music	0.00	0.00	0.00	1	
					Dr
ama Animation Mystery	0.00	0.00	0.00	0	
					Drama Ani
mation Romance Comedy	0.00	0.00	0.00	0	
					Drama Anim
ation Romance Fantasy	0.00	0.00	0.00	0	
					Drama Animation Ro
mance Science Fiction	0.00	0.00	0.00	0	
					Drama Animatio
n Romance War History	0.00	0.00	0.00	0	
					Drama Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Drama Animation Science Fic
tion Adventure Action	0.00	0.00	0.00	0	
					Dra
ma Animation TV Movie	0.00	0.00	0.00	0	
					D
rama Comedy Animation	0.00	0.00	0.00	0	
					Drama Co
medy Animation Family	0.00	0.00	0.00	0	
					Drama Comedy Documentary Animation Ho
rror Thriller Mystery	0.00	0.00	0.00	0	
					Drama Comedy Document
ary History Animation	0.00	0.00	0.00	0	
					Drama Com
edy Fantasy Animation	0.00	0.00	0.00	0	
Drama Crime Animation	0.00	0.00	0.00	0	
					Drama
Documentary Animation	0.00	0.00	0.00	2	
					D
rama Family Animation	0.00	0.00	0.00	1	
					Drama Family Animation Sc
ience Fiction History	0.00	0.00	0.00	0	
					Drama Family A
nimation War TV Movie	0.00	0.00	0.00	0	
					Drama F

amily Music Animation	0.00	0.00	0.00	0	
					Dr
ama Fantasy Animation	0.00	0.00	0.00	0	
					Drama Fantasy Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Dr
ama History Animation	0.00	0.00	0.00	0	
					D
rama Horror Animation	0.00	0.00	0.00	1	
					Drama Hor
ror Animation Mystery	0.00	0.00	0.00	0	
Drama Music Animation	0.00	0.00	0.00	0	
					Drama Mys
tery Comedy Animation	0.00	0.00	0.00	0	
					Drama Rom
ance Action Animation	0.00	0.00	0.00	0	
					Dr
ama Romance Animation	0.00	0.00	0.00	2	
					Drama Romance Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Drama Roma
nce Fantasy Animation	0.00	0.00	0.00	0	
					Drama Romance Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Drama Science Fic
tion Action Animation	0.00	0.00	0.00	0	
					Drama Scie
nce Fiction Animation	0.00	0.00	0.00	2	
					Dra
ma Thriller Animation	0.00	0.00	0.00	0	
					Drama War Action Scie
nce Fiction Animation	0.00	0.00	0.00	0	
Family	0.44	0.78	0.56	584	
					Family Action Adven
ture Comedy Animation	0.00	0.00	0.00	0	
					Family Ac
tion Animation Comedy	0.00	0.00	0.00	0	
					Famil
y Adventure Animation	0.00	0.00	0.00	4	
					Family Adventure Ani
mation Action Fantasy	0.00	0.00	0.00	0	
					Family Adven
ture Animation Comedy	0.00	0.00	0.00	1	
					Family Adventure Ani
mation Comedy Fantasy	0.00	0.00	0.00	0	
					Family Adventure Animation C
omedy Fantasy Mystery	0.00	0.00	0.00	0	
					Family Adventure Ani
mation Comedy Mystery	0.00	0.00	0.00	0	
					Family Adventure A
nimation Drama Action	0.00	0.00	0.00	0	
					Family Adventure An
imation Drama Romance	0.00	0.00	0.00	0	
					Family Advent

ure Animation Fantasy	0.00	0.00	0.00	0	
			Family Adventure Animation Fantasy A		
ction Science Fiction	0.00	0.00	0.00	0	
			Family Adventure Animation Fa		
ntasy Science Fiction	0.00	0.00	0.00	0	
			Family Advent		
ure Animation Romance	0.00	0.00	0.00	0	
			Family Advent		
ure Fantasy Animation	0.00	0.00	0.00	0	
Family Animation	0.04	0.59	0.07	27	
			Family Anima		
tion Action Adventure	0.00	0.00	0.00	0	
			Family Animation Action Adventure Fa		
ntasy Science Fiction	0.00	0.00	0.00	0	
			Family Animation Action Comedy Science Ficti		
on TV Movie Adventure	0.00	0.00	0.00	0	
			Family Animation A		
ction Science Fiction	0.00	0.00	0.00	0	
			Famil		
y Animation Adventure	0.01	0.82	0.02	11	
			Family Anima		
tion Adventure Action	0.00	0.00	0.00	0	
			Family Animation Adv		
enture Action Fantasy	0.00	0.00	0.00	0	
			Family Anima		
tion Adventure Comedy	0.00	0.00	0.00	2	
			Family Animation Adv		
enture Comedy Fantasy	0.00	0.00	0.00	1	
			Family Animation Adve		
nture Comedy TV Movie	0.00	0.00	0.00	0	
			Family Animation Ad		
venture Drama History	0.00	0.00	0.00	0	
			Family Animat		
ion Adventure Fantasy	0.00	0.00	0.00	0	
			Family Animation Adventure Fantasy C		
omedy Science Fiction	0.00	0.00	0.00	0	
			Family Animation Adventure Fa		
ntasy Science Fiction	0.00	0.00	0.00	0	
			Family Anim		
ation Adventure Music	0.00	0.00	0.00	0	
			Family Animation Adventure Sc		
ience Fiction Fantasy	0.00	0.00	0.00	0	
			Fa		
mily Animation Comedy	0.01	0.20	0.01	5	
			Family Anima		
tion Comedy Adventure	0.11	0.50	0.18	2	
			Family Animation Com		
edy Adventure Fantasy	0.00	0.00	0.00	0	
			Family A		
nimation Comedy Drama	0.00	0.00	0.00	0	
			Family Ani		
mation Comedy Fantasy	0.00	0.00	0.00	2	
			Family Animation		
Comedy Fantasy Action	0.00	0.00	0.00	0	
			Family Animation Comedy Fa		

antasy Adventure Music	0.00	0.00	0.00	0	Family Animation Comedy
Fantasy Music Mystery	0.00	0.00	0.00	0	Family Ani
mation Comedy Mystery	0.00	0.00	0.00	0	Family Animation Comedy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Family Anim
ation Comedy TV Movie	0.00	0.00	0.00	1	Family Animation Come
dy TV Movie Adventure	0.00	0.00	0.00	0	Family Ani
mation Comedy Western	0.00	0.00	0.00	0	Family A
nimation Crime Comedy	0.00	0.00	0.00	0	F
amily Animation Drama	0.00	0.00	0.00	0	Family A
nimation Drama Comedy	0.00	0.00	0.00	0	Fam
ily Animation Fantasy	0.01	0.50	0.02	8	Family Animat
ion Fantasy Adventure	0.00	0.00	0.00	0	Family Animation Fan
tasy Adventure Comedy	0.00	0.00	0.00	0	Family Animation Fantasy A
dventure Comedy Music	0.00	0.00	0.00	0	Family Animation Fant
asy Adventure Mystery	0.00	0.00	0.00	0	Family Animation Fantasy Adve
nture Science Fiction	0.00	0.00	0.00	0	Family Ani
mation Fantasy Comedy	0.01	1.00	0.01	1	Family An
imation Fantasy Music	0.00	0.00	0.00	0	Family Animation Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Family Anima
tion Fantasy TV Movie	0.00	0.00	0.00	0	Fam
ily Animation History	0.00	0.00	0.00	0	Family Anim
ation Horror TV Movie	0.00	0.00	0.00	0	F
amily Animation Music	0.00	0.00	0.00	0	Family Anim
ation Music Adventure	0.00	0.00	0.00	0	Fam
ily Animation Mystery	0.00	0.00	0.00	0	Family Animation
Mystery Comedy Horror	0.00	0.00	0.00	0	Family Animation Rom
ance Adventure Comedy	0.00	0.00	0.00	0	Family An
imation Romance Drama	0.00	0.00	0.00	0	Family Anim

ation Romance Fantasy	0.00	0.00	0.00	0	
					Family Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Family Animation S
cience Fiction Comedy	0.00	0.00	0.00	1	
					Fami
ly Animation TV Movie	0.00	0.00	0.00	0	
					Family Anim
ation TV Movie Comedy	0.00	0.00	0.00	0	
					Family Comed
y Adventure Animation	0.23	0.33	0.27	9	
					Family Comedy Adven
ture Animation Action	0.00	0.00	0.00	0	
					Family Comedy Advent
ure Animation Fantasy	0.00	0.00	0.00	1	
					Family Comedy Adve
nture Animation Music	0.00	0.00	0.00	0	
					Family Comedy Adventure Anim
ation Science Fiction	0.00	0.00	0.00	1	
					Family Comedy Adventure Animation Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	
					Fa
mily Comedy Animation	0.00	0.00	0.00	1	
					Family Comedy Ani
mation Action Western	0.00	0.00	0.00	0	
					Family Comed
y Animation Adventure	0.00	0.00	0.00	0	
					Family Comedy Animat
ion Adventure Fantasy	0.00	0.00	0.00	0	
					Family Comedy Animat
ion Adventure Mystery	0.00	0.00	0.00	0	
					Family Comedy Animation Adven
ture TV Movie Fantasy	0.00	0.00	0.00	0	
					Family Com
edy Animation Fantasy	0.00	0.00	0.00	0	
					Family Comedy Animat
ion Fantasy Adventure	0.00	0.00	0.00	0	
					Family Comedy Animation S
cience Fiction Action	0.00	0.00	0.00	0	
					Family Comedy Crim
e Adventure Animation	0.00	0.00	0.00	0	
					Family Comedy Crime Adven
ture Animation Action	0.00	0.00	0.00	0	
					Family C
omedy Crime Animation	0.00	0.00	0.00	0	
					Family Comedy Musi
c Adventure Animation	0.00	0.00	0.00	0	
					Family C
omedy Music Animation	0.00	0.00	0.00	0	
					Family Com
edy Mystery Animation	0.00	0.00	0.00	0	
					F
amily Drama Animation	0.00	0.00	0.00	0	
					Family
Drama Music Animation	0.00	0.00	0.00	0	
					Family Fa

mily Animation Comedy	0.00	0.00	0.00	0	
ture Comedy Animation	0.00	0.00	0.00	0	Family Fantasy Adven
ily Fantasy Animation	0.00	0.00	0.00	3	Fam
y Animation Adventure	0.01	1.00	0.01	1	Family Fantas
ation Adventure Music	0.00	0.00	0.00	0	Family Fantasy Anim
on Adventure TV Movie	0.00	0.00	0.00	0	Family Fantasy Animati
imation Drama Romance	0.00	0.00	0.00	0	Family Fantasy An
asy Animation Mystery	0.00	0.00	0.00	0	Family Fant
asy Animation Romance	0.00	0.00	0.00	1	Family Fant
sy Animation TV Movie	0.00	0.00	0.00	0	Family Fanta
sic Mystery Adventure	0.00	0.00	0.00	0	Family Fantasy Animation TV Movie Mu
y Adventure Animation	0.00	0.00	0.00	0	Family Fantasy Comed
tasy Comedy Animation	0.00	0.00	0.00	0	Family Fan
edy Romance Animation	0.00	0.00	0.00	0	Family Fantasy Com
c Adventure Animation	0.00	0.00	0.00	0	Family Fantasy Musi
sic Romance Animation	0.00	0.00	0.00	0	Family Fantasy Mu
amily Music Animation	0.00	0.00	0.00	0	F
y Adventure Animation	0.00	0.00	0.00	0	Family Music Comed
usic Comedy Animation	0.00	0.00	0.00	0	Family M
Comedy Action Fantasy	0.00	0.00	0.00	0	Family Mystery Animation
ly TV Movie Animation	0.00	0.00	0.00	0	Fami
e Animation Adventure	0.00	0.00	0.00	0	Family TV Movi
Animation Documentary	0.00	0.00	0.00	0	Family TV Movie
ovie Comedy Animation	0.00	0.00	0.00	0	Family TV M
Fantasy	0.21	0.48	0.29	218	
n Adventure Animation	0.00	0.00	0.00	0	Fantasy Actio
ture Animation Family	0.00	0.00	0.00	0	Fantasy Action Adven
ation Science Fiction	0.00	0.00	0.00	0	Fantasy Action Adventure Anim
					Fantasy Action Anim

ation Science Fiction	0.00	0.00	0.00	0	
					Fantas
y Adventure Animation	0.00	0.00	0.00	1	
					Fantasy Adventure An
imation Comedy Family	0.02	1.00	0.03	1	
					Fantasy Adventure Animation
Drama Science Fiction	0.00	0.00	0.00	0	
					Fantasy Adven
ture Animation Family	0.01	1.00	0.01	2	
					Fantasy Adventur
e Drama Animation War	0.00	0.00	0.00	0	
					Fantasy Adven
ture Family Animation	0.00	0.00	0.00	1	
Fantasy Animation	0.00	0.50	0.00	2	
					Fan
tasy Animation Action	0.00	0.00	0.00	1	
					Fantasy Anima
tion Action Adventure	0.00	0.00	0.00	0	
					Fantasy Animation Action Adventure Science
Fiction Drama Romance	0.00	0.00	0.00	0	
					Fantasy Animation A
ction Science Fiction	0.00	0.00	0.00	1	
					Fantasy Animation A
ction Thriller Family	0.00	0.00	0.00	0	
					Fantas
y Animation Adventure	0.00	0.00	0.00	0	
					Fantasy Anim
ation Adventure Crime	0.00	0.00	0.00	0	
					Fantasy Anim
ation Adventure Drama	0.00	0.00	0.00	0	
					Fantasy Anima
tion Adventure Family	0.00	0.00	0.00	0	
					Fantasy Animation Adventure Family S
cience Fiction Comedy	0.00	0.00	0.00	0	
					Fantasy Animation Adve
nture Family TV Movie	0.00	0.00	0.00	0	
					Fantasy Animation Adv
enture Music TV Movie	0.00	0.00	0.00	0	
					Fantasy Animati
on Adventure TV Movie	0.00	0.00	0.00	0	
					Fan
tasy Animation Comedy	0.00	0.00	0.00	1	
					Fantasy An
imation Comedy Family	0.00	0.00	0.00	2	
					Fantasy Animation Comedy Fam
ily Mystery Adventure	0.00	0.00	0.00	0	
					Fantasy A
nimation Comedy Music	0.00	0.00	0.00	0	
					Fantasy Ani
mation Comedy Romance	0.00	0.00	0.00	0	
					Fantasy Anim
ation Drama Adventure	0.00	0.00	0.00	0	
					Fantasy Animati
on Drama Music Action	0.00	0.00	0.00	0	
					Fantasy Animation

Drama Science Fiction	0.00	0.00	0.00	0	Fan
tasy Animation Family	0.00	0.00	0.00	2	Fantasy Anima
tion Family Adventure	0.00	0.00	0.00	0	Fantasy Animation F
Comedy Mystery Horror	0.00	0.00	0.00	0	Fantasy Animation F
amily Music Adventure	0.00	0.00	0.00	0	Fantasy Anim
ation Family TV Movie	0.00	0.00	0.00	0	Fan
tasy Animation Horror	0.00	0.00	0.00	0	Fantasy An
imation Horror Action	0.00	0.00	0.00	1	Fantasy Animation Horror Action T
hriller Mystery Crime	0.00	0.00	0.00	0	Fantasy Anim
ation Science Fiction	0.00	0.00	0.00	2	Fantasy Animation Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Fantasy Animation S
cience Fiction Family	0.00	0.00	0.00	0	Fantasy Animation Science
Fiction Family Comedy	0.00	0.00	0.00	0	Fan
tasy Comedy Animation	0.00	0.00	0.00	0	Fantasy Comed
y Animation Adventure	0.00	0.00	0.00	0	Fantasy Co
medy Animation Family	0.00	0.00	0.00	0	Fantasy Comedy Animation S
cience Fiction Family	0.00	0.00	0.00	0	Fantasy Drama Science Fiction Rom
ance Family Animation	0.00	0.00	0.00	0	Fan
tasy Family Animation	0.00	0.00	0.00	0	Fantasy Famil
y Animation Adventure	0.00	0.00	0.00	0	Fantasy Fami
ly Animation TV Movie	0.00	0.00	0.00	0	Fa
ntasy Music Animation	0.00	0.00	0.00	0	Fantasy M
usic Comedy Animation	0.00	0.00	0.00	0	Fant
asy Mystery Animation	0.00	0.00	0.00	0	Fantasy Science Fic
tion Action Animation	0.00	0.00	0.00	0	Fantasy Scie
nce Fiction Animation	0.00	0.00	0.00	0	Fantasy Science Fic
tion Family Animation	0.00	0.00	0.00	0	
History	0.03	0.42	0.05	24	History Adven

ture Comedy Animation	0.00	0.00	0.00	0	
History Animation	0.00	0.00	0.00	3	
y Animation Adventure	0.00	0.00	0.00	0	Histor
ation Adventure Drama	0.00	0.00	0.00	0	History Anim
tory Animation Family	0.00	0.00	0.00	0	His
tion Family Adventure	0.00	0.00	0.00	0	History Anima
ily Fantasy Adventure	0.00	0.00	0.00	0	History Animation Fam
ory Animation Fantasy	0.00	0.00	0.00	0	Hist
Documentary Animation	0.00	0.00	0.00	0	History
tory Horror Animation	0.00	0.00	0.00	0	His
Horror	0.19	0.27	0.22	89	
ntasy Science Fiction	0.00	0.00	0.00	0	Horror Action Animation Fa
Horror Animation	0.00	1.00	0.01	2	
rror Animation Action	0.00	0.00	0.00	0	Ho
rror Animation Comedy	0.00	0.00	0.00	0	Ho
nimation Comedy Music	0.00	0.00	0.00	0	Horror A
ror Animation Mystery	0.00	0.00	0.00	0	Hor
ystery Crime Thriller	0.00	0.00	0.00	0	Horror Animation M
ation Science Fiction	0.00	0.00	0.00	0	Horror Anim
cience Fiction Action	0.00	0.00	0.00	0	Horror Animation S
or Animation TV Movie	0.00	0.00	0.00	0	Horr
or Animation Thriller	0.00	0.00	0.00	0	Horr
tion Thriller Fantasy	0.00	0.00	0.00	0	Horror Anima
rror Comedy Animation	0.00	0.00	0.00	0	Ho
dy Animation TV Movie	0.00	0.00	0.00	0	Horror Come
tion TV Movie Mystery	0.00	0.00	0.00	0	Horror Comedy Anima
asy Romance Animation	0.00	0.00	0.00	0	Horror Comedy Fant
orror Drama Animation	0.00	0.00	0.00	1	H
					Hor

ror Fantasy Animation	0.00	0.00	0.00	0	
					Horror Fan
tasy Animation Family	0.00	0.00	0.00	0	
					Horror Mys
tery Animation Comedy	0.00	0.00	0.00	0	
					Horror Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Horr
or Thriller Animation	0.00	0.00	0.00	0	
Music	0.23	0.38	0.29	105	
Musical	0.00	0.00	0.00	0	
					Music Adventure
Animation Crime Drama	0.00	0.00	0.00	0	
Music Animation	0.00	0.00	0.00	6	
					M
usic Animation Comedy	0.00	0.00	0.00	1	
					Music
Animation Documentary	0.00	0.00	0.00	0	
Music Animation Drama	0.00	0.00	0.00	0	
					M
usic Animation Family	0.00	0.00	0.00	1	
					Music Ani
mation Family Fantasy	0.00	0.00	0.00	0	
					Mu
sic Animation Romance	0.00	0.00	0.00	0	
					Music An
imation Romance Drama	0.00	0.00	0.00	0	
					Music
Documentary Animation	0.00	0.00	0.00	0	
					Music Documentary Hi
story Animation Drama	0.00	0.00	0.00	0	
					Music D
rama Animation Comedy	0.00	0.00	0.00	0	
					M
usic Family Animation	0.00	0.00	0.00	0	
					Music Fam
ily Fantasy Animation	0.00	0.00	0.00	0	
					Mu
sic Fantasy Animation	0.00	0.00	0.00	0	
Mystery	0.06	0.22	0.09	27	
Mystery Animation	0.00	0.00	0.00	0	
					My
stery Animation Crime	0.00	0.00	0.00	0	
					Mystery Ani
mation Drama Thriller	0.00	0.00	0.00	0	
					Myst
ery Animation Fantasy	0.00	0.00	0.00	0	
					Mys
tery Animation Horror	0.00	0.00	0.00	0	
					Mystery Anim

ation Horror Thriller	0.00	0.00	0.00	0	
					Mystery Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Mystery Animation Sci
ence Fiction Thriller	0.00	0.00	0.00	0	
					My
stery Crime Animation	0.00	0.00	0.00	0	
					Mystery Crime Thr
iller Drama Animation	0.00	0.00	0.00	0	
					Mystery Dr
ama Fantasy Animation	0.00	0.00	0.00	0	
					Mystery Family Anima
tion Adventure Comedy	0.00	0.00	0.00	0	
					Mys
tery Horror Animation	0.00	0.00	0.00	0	
					Mystery Romance Drama Ani
mation Crime TV Movie	0.00	0.00	0.00	0	
Romance	0.11	0.29	0.16	49	
					Romance Ac
tion Animation Comedy	0.00	0.00	0.00	0	
					Romanc
e Adventure Animation	0.00	0.00	0.00	0	
					Romance Adventure Fantasy Science Fic
tion Action Animation	0.00	0.00	0.00	0	
Romance Animation	0.00	1.00	0.00	1	
					Rom
ance Animation Comedy	0.00	0.00	0.00	0	
					Romance Animation
Comedy Family Fantasy	0.00	0.00	0.00	0	
					Romance Ani
mation Comedy History	0.00	0.00	0.00	0	
					Ro
mance Animation Drama	0.00	0.00	0.00	1	
					Romance A
nimation Drama Comedy	0.00	0.00	0.00	0	
					Romance An
imation Drama Fantasy	0.00	0.00	0.00	0	
					Romance Animation Fam
ily Adventure Fantasy	0.00	0.00	0.00	0	
					Romance Animation
Family Fantasy Comedy	0.00	0.00	0.00	0	
					Romance Animat
ion Fantasy Adventure	0.00	0.00	0.00	0	
					Romance An
imation History Drama	0.00	0.00	0.00	0	
					Rom
ance Animation Horror	0.00	0.00	0.00	0	
					Roman
ce Animation TV Movie	0.00	0.00	0.00	0	
					Rom
ance Comedy Animation	0.00	0.00	0.00	1	
					Ro
mance Drama Animation	0.00	0.00	0.00	0	
					Roma

nce Fantasy Animation	0.00	0.00	0.00	1	
Science Fiction	0.31	0.56	0.40	149	
n Adventure Animation	0.00	0.00	0.00	0	Science Fiction Actio
tion Action Animation	0.00	0.00	0.00	1	Science Fic
n Animation Adventure	0.03	1.00	0.05	1	Science Fiction Actio
tion Animation Comedy	0.00	0.00	0.00	0	Science Fiction Ac
tion Comedy Adventure	0.00	0.00	0.00	0	Science Fiction Action Anima
ction Animation Drama	0.00	0.00	0.00	0	Science Fiction A
ion Animation Fantasy	0.00	0.00	0.00	1	Science Fiction Act
tion Fantasy TV Movie	0.00	0.00	0.00	0	Science Fiction Action Anima
tion Comedy Animation	0.00	0.00	0.00	0	Science Fiction Ac
ction Crime Animation	0.00	0.00	0.00	1	Science Fiction A
n Adventure Animation	0.00	0.00	0.00	1	Science Fictio
mation Family Fantasy	0.00	0.00	0.00	0	Science Fiction Adventure Ani
ture Family Animation	0.00	0.00	0.00	0	Science Fiction Adven
re TV Movie Animation	0.00	0.00	0.00	0	Science Fiction Adventu
nce Fiction Animation	0.00	0.00	0.00	4	Scie
tion Animation Action	0.00	0.00	0.00	3	Science Fic
tion Action Adventure	0.00	0.00	0.00	0	Science Fiction Anima
on Adventure TV Movie	0.00	0.00	0.00	0	Science Fiction Animation Acti
Action Adventure War	0.00	0.00	0.00	0	Science Fiction Animation
imation Action Comedy	0.00	0.00	0.00	0	Science Fiction An
Crime Comedy Thriller	0.00	0.00	0.00	0	Science Fiction Animation Action
Action Drama Fantasy	0.00	0.00	0.00	0	Science Fiction Animation
ation Action Thriller	0.00	0.00	0.00	0	Science Fiction Anim
Animation Action War	0.00	0.00	0.00	0	Science Fiction
n Animation Adventure	0.00	0.00	0.00	0	Science Fictio
tion Adventure Action	0.00	0.00	0.00	0	Science Fiction Anima
					Science Fiction Animat

ion Adventure Fantasy	0.00	0.00	0.00	0	Science Fic
tion Animation Comedy	0.00	0.00	0.00	0	Science Fiction Animation
Comedy Fantasy Drama	0.00	0.00	0.00	0	Science Fi
ction Animation Drama	0.00	0.00	0.00	0	Science Fiction A
nimation Drama Action	1.00	1.00	1.00	1	Science Fic
tion Animation Family	0.00	0.00	0.00	0	Science Fiction Anim
ation Family TV Movie	0.00	0.00	0.00	0	Science Fiction Animation Fantasy Action Horror Advent
ure Thriller TV Movie	0.00	0.00	0.00	0	Science Fiction Animat
ion Fantasy Adventure	0.00	0.00	0.00	0	Science Fiction Animation
Fantasy Comedy Music	0.00	0.00	0.00	0	Science Ficti
on Animation TV Movie	0.00	0.00	0.00	0	Science Fic
tion Comedy Animation	0.00	0.00	0.00	0	Science Fi
ction Drama Animation	0.00	0.00	0.00	0	Science Fiction Dr
ama Fantasy Animation	0.00	0.00	0.00	0	Science Fict
ion Fantasy Animation	0.00	0.00	0.00	0	Science Fiction Fantasy Anim
ation Action TV Movie	0.00	0.00	0.00	0	Science Fiction
Fantasy War Animation	0.00	0.00	0.00	0	Science Fiction Myste
ry Thriller Animation	0.00	0.00	0.00	0	Science Fiction Rom
ance Action Animation	0.00	0.00	0.00	0	Science Fiction Thriller Drama Mys
tery Animation Horror	0.00	0.00	0.00	0	
TV Movie	0.11	0.24	0.15	49	TV Movie Actio
n Adventure Animation	0.00	0.00	0.00	0	TV Movie Action Adventure Animation Comedy
Drama Fantasy Family	0.00	0.00	0.00	0	TV Movie Action Adven
ture Animation Family	0.00	0.00	0.00	0	TV Movie Action Adventure Animation F
amily Science Fiction	0.00	0.00	0.00	0	TV Movie Adventure Animation
Comedy Family Fantasy	0.00	0.00	0.00	0	TV Movie Anima
tion Action Adventure	0.00	0.00	0.00	0	TV Movie Animation Ac
tion Adventure Comedy	0.00	0.00	0.00	0	TV M

ovie Animation Family	0.00	0.00	0.00	0	
ily Adventure Fantasy	0.00	0.00	0.00	0	TV Movie Animation Fam
vie Animation Fantasy	0.00	0.00	0.00	0	TV Mo
medy Animation Family	0.00	0.00	0.00	0	TV Movie Co
mation Family Fantasy	0.00	0.00	0.00	0	TV Movie Comedy Ani
tasy Family Animation	0.00	0.00	0.00	0	TV Movie Comedy Fan
sy Thriller Animation	0.00	0.00	0.00	0	TV Movie Crime Drama Fanta
imation Fantasy Music	0.00	0.00	0.00	0	TV Movie Family An
medy Horror Animation	0.00	0.00	0.00	0	TV Movie Family Co
Thriller	0.01	0.33	0.02	18	
ller Action Animation	0.00	0.00	0.00	0	Thri
ation Science Fiction	0.00	0.00	0.00	0	Thriller Action Anim
nce Fiction Adventure	0.00	0.00	0.00	0	Thriller Action Animation Scie
ntasy Science Fiction	0.00	0.00	0.00	0	Thriller Adventure Animation Comedy Fa
Thriller Animation	0.00	0.00	0.00	0	
Comedy Horror Mystery	0.00	0.00	0.00	0	Thriller Animation Action
nimation Action Crime	0.00	0.00	0.00	0	Thriller A
nimation Action Drama	0.00	0.00	0.00	0	Thriller A
Action Drama Romance	0.00	0.00	0.00	0	Thriller Animation
cience Fiction Horror	0.00	0.00	0.00	0	Thriller Animation Crime S
n Documentary Mystery	0.00	0.00	0.00	0	Thriller Animatio
mation Family Fantasy	0.00	0.00	0.00	0	Thriller Ani
ler Animation Fantasy	0.00	0.00	0.00	0	Thril
mation Horror Fantasy	0.00	0.00	0.00	0	Thriller Ani
ler Animation Mystery	0.00	0.00	0.00	0	Thril
ation Science Fiction	0.00	0.00	0.00	0	Thriller Anim
iller Drama Animation	0.00	0.00	0.00	0	Thr
ler Fantasy Animation	0.00	0.00	0.00	0	Thril
					Thri

ller Horror Animation	0.00	0.00	0.00	0	
ama Animation Mystery	0.00	0.00	0.00	0	Thriller War Dr
War	0.03	0.50	0.05	20	
War Animation	0.00	0.00	0.00	0	
r Animation Adventure	0.00	0.00	0.00	0	Wa
War Animation Comedy	0.00	0.00	0.00	0	
science Fiction Action	0.00	0.00	0.00	0	War Animation S
amily Animation Drama	0.00	0.00	0.00	0	War F
War History Animation	0.00	0.00	0.00	0	
story Animation Drama	0.00	0.00	0.00	0	War Hi
Western	0.01	0.64	0.02	22	
rn Animation TV Movie	0.00	0.00	0.00	0	Weste
stern Drama Animation	0.00	0.00	0.00	0	We
micro avg	0.03	0.72	0.05	7354	
macro avg	0.01	0.06	0.01	7354	
weighted avg	0.60	0.72	0.62	7354	
samples avg	0.11	0.67	0.17	7354	

```
In [79]: import time
from sklearn.multioutput import MultiOutputClassifier
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import classification_report, hamming_loss

# 1. Configure the ensemble Random Forest base estimator
# Using 100 trees with controlled max_depth to balance accuracy and execution time
base_rf = RandomForestClassifier(
    n_estimators=100,
    max_depth=20,
    random_state=42,
    class_weight='balanced', # Matches the linear model's handling of rare genres
    n_jobs=-1                # Uses all available CPU cores for individual tree bu
)

# 2. Wrap inside the MultiOutput meta-estimator container
rf_model_pipeline = MultiOutputClassifier(base_rf)

# 3. Fit the ensemble across your sparse train feature space
print("--- Training Multi-Output Random Forest Ensemble ---")
```

```
start_time = time.time()
rf_model_pipeline.fit(X_train_sparse, Y_train)
print(f"Random Forest training complete! Elapsed time: {time.time() - start_time:.2f} seconds")

# 4. Generate predictions on the held-out test split matrix
Y_pred_rf = rf_model_pipeline.predict(X_test_sparse)

# 5. Extract and print the final metrics for your chapter text
hl_rf = hamming_loss(Y_test, Y_pred_rf)
print(f"\nRandom Forest Hamming Loss Score: {hl_rf:.5f}")
print("\n--- Detailed Class Performance Overview (Random Forest) ---")
print(classification_report(Y_test, Y_pred_rf, target_names=Y_train.columns, zero_d
```

--- Training Multi-Output Random Forest Ensemble ---

Random Forest training complete! Elapsed time: 543.64 seconds.

Random Forest Hamming Loss Score: 0.00114

--- Detailed Class Performance Overview (Random Forest) ---

precision recall f1-score support

Action	0.26	0.22	0.24	133	
n Adventure Animation		0.00	0.00	0.00	6
ture Animation Comedy		0.00	0.00	0.00	0
Crime Science Fiction		0.00	0.00	0.00	0
Comedy Crime TV Movie		0.00	0.00	0.00	0
Comedy Drama Romance		0.00	0.00	0.00	0
imation Comedy Family		0.00	0.00	0.00	1
Comedy Family Fantasy		0.00	0.00	0.00	0
ence Fiction TV Movie		0.00	0.00	0.00	0
mation Comedy Fantasy		0.00	0.00	0.00	0
antasy Thriller Crime		0.00	0.00	0.00	0
imation Comedy Horror		0.00	0.00	0.00	0
Comedy Romance Family		0.00	0.00	0.00	0
omedy Science Fiction		0.00	0.00	0.00	0
nture Animation Crime		0.00	0.00	0.00	0
ence Fiction Thriller		0.00	0.00	0.00	0
Crime Science Fiction		0.00	0.00	0.00	0
nture Animation Drama		0.00	0.00	0.00	0
Drama Family Fantasy		0.00	0.00	0.00	0
ture Animation Family		0.00	0.00	0.00	2
mation Family Fantasy		0.00	0.00	0.00	0
ntasy Science Fiction		0.00	0.00	0.00	0
ation Family TV Movie		0.00	0.00	0.00	1

ure Animation Fantasy	0.00	0.00	0.00	3	Action Adventure Animation Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Action Adventure Animation Fantasy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Action Adven
ture Animation Horror	0.00	0.00	0.00	0	Action Adventure Animation
Horror Fantasy Drama	0.00	0.00	0.00	0	Action Adventure Anim
ation Science Fiction	1.00	0.20	0.33	5	Action Adventure Animation
Science Fiction Drama	0.00	0.00	0.00	0	Action Adventure Animation Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Action Adventure Animation Science Fi
ction Fantasy Mystery	0.00	0.00	0.00	0	Action Adventu
re Animation TV Movie	0.00	0.00	0.00	0	Action Adventure Ani
mation TV Movie Crime	0.00	0.00	0.00	0	Action Adventure Animation
TV Movie Crime Drama	0.00	0.00	0.00	0	Action Adven
ture Comedy Animation	0.00	0.00	0.00	0	Action Adventure Comedy Dr
ama Fantasy Animation	0.00	0.00	0.00	0	Action Adventure Comedy TV Movie Science Fict
ion Fantasy Animation	0.00	0.00	0.00	0	Action Adve
nture Drama Animation	0.00	0.00	0.00	0	Action Adven
ture Family Animation	0.00	0.00	0.00	0	Action Adventure Fam
ily Fantasy Animation	0.00	0.00	0.00	0	Action Advent
ure Fantasy Animation	0.00	0.00	0.00	1	Action Adventure Fantasy Anim
ation Science Fiction	0.00	0.00	0.00	0	Action Advent
ure History Animation	0.00	0.00	0.00	0	Action Adventure Mystery Comedy Drama Crime Anim
ation TV Movie Family	0.00	0.00	0.00	0	Action Adventure Scie
nce Fiction Animation	0.00	0.00	0.00	0	Action Adventure Science Ficti
on Animation TV Movie	0.00	0.00	0.00	0	Action Ad
venture War Animation	0.00	0.00	0.00	0	
Action Animation	0.00	0.00	0.00	9	Actio
n Animation Adventure	0.00	0.00	0.00	0	Action Anima
tion Adventure Comedy	0.00	0.00	0.00	0	Action Animation Adventure Comedy Documentary Science Fiction Drama Family

Western TV Movie War	0.00	0.00	0.00	0	Action Animation Adventure
Comedy Family Fantasy	0.00	0.00	0.00	0	Action Animation Adv
enture Comedy Fantasy	0.00	0.00	0.00	0	Action Animation Adventure Comedy Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Action Animation Adventure Co
medy Fantasy TV Movie	0.00	0.00	0.00	1	Action Animation Adve
nture Comedy TV Movie	0.00	0.00	0.00	0	Action Animation Adventure Crime S
cience Fiction Family	0.00	0.00	0.00	0	Action Anim
ation Adventure Drama	0.00	0.00	0.00	1	Action Animat
ion Adventure Fantasy	0.00	0.00	0.00	1	Action Animation Adventure Fan
tasy Mystery Thriller	0.00	0.00	0.00	0	Action Animat
ion Adventure Romance	0.00	0.00	0.00	0	Action Animation Adve
nture Science Fiction	0.00	0.00	0.00	0	Action Animation Adventure Science F
iction Comedy Fantasy	0.00	0.00	0.00	0	Action Animation Adventure Science
Fiction Drama Romance	0.00	0.00	0.00	0	Action Animation Adventure Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Action Animation Adventure S
cience Fiction Horror	0.00	0.00	0.00	0	Ac
tion Animation Comedy	0.00	0.00	0.00	1	Action A
nimation Comedy Crime	0.00	0.00	0.00	0	Action Animation Come
dy Documentary Family	0.00	0.00	0.00	0	Action Animati
on Comedy Drama Crime	0.00	0.00	0.00	0	Action Animation Comedy
Drama Fantasy Horror	0.00	0.00	0.00	0	Action An
imation Comedy Family	0.00	0.00	0.00	0	Action Animation Co
medy Family Adventure	0.00	0.00	0.00	1	Action Animation Comedy Family Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Action Animation Comedy F
amily Science Fiction	0.00	0.00	0.00	0	Action Animation
Comedy Fantasy Horror	0.00	0.00	0.00	0	Action Animation Com
edy Music Romance War	0.00	0.00	0.00	0	Action Ani
mation Comedy Romance	0.00	0.00	0.00	0	Action Animation C

Comedy Science Fiction	0.00	0.00	0.00	1	Action Animation
Comedy TV Movie Crime	0.00	0.00	0.00	0	Action Animation Comedy
TV Movie Crime Drama	0.00	0.00	0.00	0	Action Ani
mation Comedy Western	0.00	0.00	0.00	0	Action An
imation Crime Mystery	0.00	0.00	0.00	0	Action Animation
Crime Mystery Romance	0.00	0.00	0.00	0	Action Animation
Crime Science Fiction	0.00	0.00	0.00	0	Action Ani
mation Crime Thriller	0.00	0.00	0.00	0	A
ction Animation Drama	0.00	0.00	0.00	1	Action An
imation Drama Fantasy	0.00	0.00	0.00	1	Action Animation
Drama Fantasy Romance	0.00	0.00	0.00	0	Action An
imation Drama History	0.00	0.00	0.00	0	Action Animation
Drama Science Fiction	0.00	0.00	0.00	0	Action Animation Dram
a Science Fiction War	0.00	0.00	0.00	0	Action Ani
mation Drama Thriller	0.00	0.00	0.00	0	Ac
tion Animation Family	0.00	0.00	0.00	1	Action Animation Family Adve
nture Science Fiction	0.00	0.00	0.00	0	Act
ion Animation Fantasy	0.00	0.00	0.00	2	Action Animat
ion Fantasy Adventure	0.00	0.00	0.00	0	Action Ani
mation Fantasy Comedy	0.00	0.00	0.00	0	Action An
imation Fantasy Drama	0.00	0.00	0.00	0	Action Ani
mation Fantasy Horror	0.00	0.00	0.00	0	Action Animati
on Fantasy Horror War	0.00	0.00	0.00	0	Action Animation Fa
ntasy Science Fiction	0.00	0.00	0.00	2	Act
ion Animation History	0.00	0.00	0.00	0	Ac
tion Animation Horror	0.00	0.00	0.00	0	Action Animation
Music Science Fiction	0.00	0.00	0.00	0	Act
ion Animation Romance	0.00	0.00	0.00	0	Action Animation Ro

mance Science Fiction	0.00	0.00	0.00	0	Action Anim
ation Science Fiction	0.00	0.00	0.00	9	Action Animation Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Action Animation Science Fic
tion Comedy Adventure	0.00	0.00	0.00	0	Action Animation
Science Fiction Crime	0.00	0.00	0.00	1	Action Animation
Science Fiction Drama	0.00	0.00	0.00	0	Action Animation Sc
ience Fiction Fantasy	0.00	0.00	0.00	1	Action Animation S
cience Fiction Horror	0.00	0.00	0.00	0	Action Animation Sc
ience Fiction Mystery	0.00	0.00	0.00	0	Action Animation Sci
ence Fiction TV Movie	0.00	0.00	0.00	1	Action Animation Sci
ence Fiction Thriller	0.00	0.00	0.00	0	Action Animatio
n Science Fiction War	0.00	0.00	0.00	2	Acti
on Animation Thriller	0.00	0.00	0.00	0	Action Anim
ation Thriller Comedy	0.00	0.00	0.00	0	Ac
tion Comedy Animation	0.00	0.00	0.00	0	Action Comed
y Animation Adventure	0.00	0.00	0.00	0	Action Co
medy Animation Family	0.00	0.00	0.00	0	Action Comedy Ani
mation Thriller Crime	0.00	0.00	0.00	0	Action Comedy Science Fictio
n Adventure Animation	0.00	0.00	0.00	0	Action Comedy Scie
nce Fiction Animation	0.00	0.00	0.00	0	Action Drama Adventure Science
Fiction War Animation	0.00	0.00	0.00	0	A
ction Drama Animation	0.00	0.00	0.00	0	Action Drama Animati
on Adventure Thriller	0.00	0.00	0.00	0	Action Drama Scie
nce Fiction Animation	0.00	0.00	0.00	0	Act
ion Fantasy Animation	0.00	0.00	0.00	1	Action Fa
ntasy Animation Drama	0.00	0.00	0.00	0	Action Fan
tasy Animation Horror	0.00	0.00	0.00	0	Action Fan
tasy Horror Animation	0.00	0.00	0.00	0	Action

History War Animation	0.00	0.00	0.00	0	
					Action Horror Science Fiction Animation
					Action Mystery Romance Thriller Animation
					Act
ion Romance Animation	0.00	0.00	0.00	0	
					Action Science Fiction Animation
				1	
					Action Science Fiction Animation Adventure
				0	
					Action Science Fiction Animation War
				0	
					Action Science Fiction Comedy Fantasy Animation
				0	
					Action War Science Fiction Animation
				0	
					Act
ion Western Animation	0.00	0.00	0.00	0	
Adventure	0.19	0.45	0.27	173	
					Advent
ture Action Animation	0.00	0.00	0.00	0	
					Adventure Action Animation
tion Animation Comedy	0.00	0.00	0.00	0	
					Adventure Action Animation Comedy Fantasy
				0	
					Adventure Action Animation Drama Family Fantasy Science Fiction
				0	
					Adventure Action Animation Drama Fantasy Science Fiction
				0	
					Adventure Action Animation Family Fantasy Science Fiction
				1	
					Adventure Act
ion Animation Fantasy	0.00	0.00	0.00	0	
					Adventure Action Animation Science Fiction
				1	
					Adventure Act
ion Fantasy Animation	0.00	0.00	0.00	0	
					Adventure Action Fantasy Science Fiction Animation
				0	
Adventure Animation	0.00	0.00	0.00	2	
					Advent
ture Animation Action	0.00	0.00	0.00	3	
					Adventure Animation Action Comedy Family Science Fiction
				0	
					Adventure Animation Action Fantasy
				0	
					Adventure Animation Action Fantasy Science Fiction
				0	
					Adventure Animation Action History Family
				0	
					Adventure Animation Action Science Fiction
				1	
					Adventure Animation

ation Action TV Movie	0.00	0.00	0.00	0	Adven
ture Animation Comedy	0.00	0.00	0.00	2	Adventure A
nimation Comedy Crime	0.00	0.00	0.00	0	Adventure Animation
Comedy Drama Fantasy	0.00	0.00	0.00	0	Adventure An
imation Comedy Family	0.00	0.00	0.00	6	Adventure Animation
Comedy Family Fantasy	0.00	0.00	0.00	1	Adventure Animation Comedy
Family Fantasy Music	0.00	0.00	0.00	0	Adventure Animation Comedy Fa
mily Fantasy TV Movie	0.00	0.00	0.00	0	Adventure Animatio
n Comedy Family Music	0.00	0.00	0.00	1	Adventure Animation Comedy F
amily Science Fiction	0.00	0.00	0.00	2	Adventure Ani
mation Comedy Fantasy	0.00	0.00	0.00	1	Adventure Ani
mation Comedy History	0.00	0.00	0.00	0	Adventure Anim
ation Comedy TV Movie	0.00	0.00	0.00	0	Adventure Animation
Comedy TV Movie Crime	0.00	0.00	0.00	0	Adventure Ani
mation Comedy Western	0.00	0.00	0.00	0	Adventure Animatio
n Crime Drama Mystery	0.00	0.00	0.00	0	Adventure Animation Crime F
amily Fantasy Romance	0.00	0.00	0.00	0	Adve
nture Animation Drama	0.00	0.00	0.00	1	Adventure A
nimation Drama Family	0.00	0.00	0.00	1	Adventure Animation
Drama Family Fantasy	0.00	0.00	0.00	0	Adventure Animation Drama F
amily Mystery Fantasy	0.00	0.00	0.00	0	Adventure An
imation Drama Fantasy	0.00	0.00	0.00	0	Adventure Animation
Drama Fantasy Romance	0.00	0.00	0.00	0	Adventure An
imation Drama History	0.00	0.00	0.00	0	Adventure An
imation Drama Romance	0.00	0.00	0.00	0	Adventure Animation
Drama Science Fiction	0.00	0.00	0.00	0	Adven
ture Animation Family	0.00	0.00	0.00	6	Adventure An
imation Family Comedy	0.00	0.00	0.00	0	Adventure Animation

Family Comedy Fantasy	0.00	0.00	0.00	1	
Adventure Animation Family Comedy My					
stery Science Fiction	0.00	0.00	0.00	0	
Adventure Animation Family C					
omedy Science Fiction	0.00	0.00	0.00	0	
Adventure Animation Family Comedy Sc					
ience Fiction Fantasy	0.00	0.00	0.00	0	
Adventure Ani					
mation Family Fantasy	0.00	0.00	0.00	7	
Adventure Animation Family Fa					
ntasy Science Fiction	0.00	0.00	0.00	0	
Adventure Ani					
mation Family History	0.00	0.00	0.00	0	
Adventure Ani					
mation Family Romance	0.00	0.00	0.00	0	
Adventure Animation F					
amily Science Fiction	0.00	0.00	0.00	0	
Adventure Animation Family Science F					
iction Comedy Fantasy	0.00	0.00	0.00	0	
Adventure Animation Family Sci					
ence Fiction TV Movie	0.00	0.00	0.00	0	
Adventure Anim					
ation Family TV Movie	0.00	0.00	0.00	0	
Advent					
ure Animation Fantasy	0.00	0.00	0.00	0	
Adventure Ani					
mation Fantasy Family	0.00	0.00	0.00	0	
Adventure Anim					
ation Fantasy Mystery	0.00	0.00	0.00	0	
Adventure Animation Fa					
ntasy Science Fiction	0.00	0.00	0.00	0	
Adventure Animation					
Music Fantasy Comedy	0.00	0.00	0.00	0	
Advent					
ure Animation Mystery	0.00	0.00	0.00	0	
Adventure Animation Ro					
mance Science Fiction	0.00	0.00	0.00	0	
Adventure Anim					
ation Science Fiction	0.00	0.00	0.00	3	
Adventure Animation S					
cience Fiction Action	0.00	0.00	0.00	0	
Adventure Animation Science					
Fiction Action Comedy	0.00	0.00	0.00	0	
Adven					
ture Comedy Animation	0.00	0.00	0.00	1	
Adventure Co					
medy Animation Family	0.00	0.00	0.00	0	
Adventure C					
omedy Crime Animation	0.00	0.00	0.00	0	
Adventure C					
omedy Drama Animation	0.00	0.00	0.00	0	
Adventure Co					
medy Family Animation	0.00	0.00	0.00	0	
Adventure Comedy Hor					
ror History Animation	0.00	0.00	0.00	0	
Adventure Comedy Rom					

ance Animation Family	0.00	0.00	0.00	0	
					Adve
nture Drama Animation	0.00	0.00	0.00	0	
					Adventure Dr
ama Animation Fantasy	0.00	0.00	0.00	0	
					Adventure D
rama Comedy Animation	0.00	0.00	0.00	0	
					Adventure D
rama Family Animation	0.00	0.00	0.00	0	
					Adventure Dr
ama Fantasy Animation	0.00	0.00	0.00	0	
					Adventure Drama Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Adven
ture Family Animation	0.00	0.00	0.00	1	
					Adventure Family An
imation Action Comedy	0.00	0.00	0.00	0	
					Adventure Family Animation Comedy Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Adventure Fam
ily Animation Fantasy	0.00	0.00	0.00	1	
					Adventure Family Animation Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Adventure Fam
ily Fantasy Animation	0.00	0.00	0.00	0	
					Adventure Family Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Advent
ure Fantasy Animation	0.00	0.00	0.00	3	
					Adventure Fantasy An
imation Action Family	0.00	0.00	0.00	0	
					Adventure Fan
tasy Animation Comedy	0.00	0.00	0.00	0	
					Adventure Fantasy An
imation Comedy Family	0.00	0.00	0.00	0	
					Adventure Fantasy Animation C
omedy Science Fiction	0.00	0.00	0.00	0	
					Adventure Fantasy A
nimation Drama Family	0.00	0.00	0.00	0	
					Adventure Fan
tasy Animation Family	0.00	0.00	0.00	0	
					Adventure Fantasy Ani
mation Mystery Family	0.00	0.00	0.00	0	
					Adventure Fantasy Animation S
cience Fiction Family	0.00	0.00	0.00	0	
					Adventure Fan
tasy Family Animation	0.00	0.00	0.00	0	
					Adventure Fantasy Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Adventure M
usic Animation Family	0.00	0.00	0.00	1	
					Adventure Mystery Co
medy Animation Family	0.00	0.00	0.00	0	
					Adventure Mys
tery Family Animation	0.00	0.00	0.00	0	
					Adventure Scie

nce Fiction Animation	0.00	0.00	0.00	0	
					Adventure Science Fi
ction Drama Animation	0.00	0.00	0.00	0	
					Adventure TV M
ovie Action Animation	0.00	0.00	0.00	0	
Animation	1.00	0.89	0.94	3567	
Animation Action	0.00	0.00	0.00	9	
					Anima
tion Action Adventure	0.00	0.00	0.00	4	
					Animation Ac
tion Adventure Comedy	0.00	0.00	0.00	2	
					Animation Action A
dventure Comedy Crime	0.00	0.00	0.00	0	
					Animation Action Adventure Comedy Drama F
amily Science Fiction	0.00	0.00	0.00	0	
					Animation Action Ad
venture Comedy Family	0.00	0.00	0.00	1	
					Animation Action Adventure
Comedy Family Fantasy	0.00	0.00	0.00	0	
					Animation Action Adventure Comedy
Family Fantasy Music	0.00	0.00	0.00	0	
					Animation Action Adventure Comedy Family Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Animation Action Adventure Comedy F
amily Science Fiction	0.00	0.00	0.00	0	
					Animation Action Adv
enture Comedy Fantasy	0.00	0.00	0.00	0	
					Animation Action Adventure C
omedy Fantasy Romance	0.00	0.00	0.00	0	
					Animation Action Adventure C
omedy Science Fiction	0.00	0.00	0.00	0	
					Animation Action A
dventure Crime Comedy	0.00	0.00	0.00	0	
					Animation Action Adventure
Crime Comedy TV Movie	0.00	0.00	0.00	0	
					Animation A
ction Adventure Drama	0.00	0.00	0.00	0	
					Animation Action Adventu
re Drama Crime Comedy	0.00	0.00	0.00	0	
					Animation Action Adventure Drama F
amily Science Fiction	0.00	0.00	0.00	0	
					Animation Action Ad
venture Drama Fantasy	0.00	0.00	0.00	1	
					Animation Action Adventure Drama Fantasy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	
					Animation Action Ad
venture Drama Mystery	0.00	0.00	0.00	0	
					Animation Action Adventure
Drama Science Fiction	0.00	0.00	0.00	0	
					Animation Ac
tion Adventure Family	0.00	0.00	0.00	2	
					Animation Action Ad
venture Family Comedy	0.00	0.00	0.00	0	
					Animation Action Adv

enture Family Fantasy	0.00	0.00	0.00	0	Animation Action Adventure F
amily Fantasy Romance	0.00	0.00	0.00	0	Animation Action Adventure F
amily Science Fiction	0.00	0.00	0.00	0	Animation Action Adve
nture Family TV Movie	0.00	0.00	0.00	0	Animation Act
ion Adventure Fantasy	0.00	0.00	0.00	3	Animation Action Adv
enture Fantasy Comedy	0.00	0.00	0.00	0	Animation Action Adventure
Fantasy Drama Horror	0.00	0.00	0.00	0	Animation Action Adv
enture Fantasy Family	0.00	0.00	0.00	0	Animation Action Adve
nture Fantasy Romance	0.00	0.00	0.00	0	Animation Action Adventure Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Animation Action Adven
ture Fantasy Thriller	0.00	0.00	0.00	0	Animation Action Adventure
History Drama Fantasy	0.00	0.00	0.00	0	Animation Action Adve
nture History Fantasy	0.00	0.00	0.00	0	Animation Act
ion Adventure Mystery	0.00	0.00	0.00	0	Animation Action Adventure Romance
Drama Science Fiction	0.00	0.00	0.00	0	Animation Action Adve
nture Science Fiction	0.00	0.00	0.00	4	Animation Action Adventure S
cience Fiction Comedy	0.00	0.00	0.00	0	Animation Action Adventure Science
Fiction Comedy Music	0.00	0.00	0.00	0	Animation Acti
on Adventure TV Movie	0.00	0.00	0.00	0	Animation Acti
on Adventure Thriller	0.00	0.00	0.00	0	Animation Action Adven
ture Thriller Romance	0.00	0.00	0.00	0	Animation Action Adventur
e War Science Fiction	0.00	0.00	0.00	0	An
imation Action Comedy	0.00	0.00	0.00	5	Animation Action Comedy Adv
enture TV Movie Crime	0.00	0.00	0.00	0	Animatio
n Action Comedy Crime	0.00	0.00	0.00	0	Animation Action
Comedy Crime Mystery	0.00	0.00	0.00	0	Animation Action
Comedy Crime TV Movie	0.00	0.00	0.00	0	Animation Acti
on Comedy Drama Crime	0.00	0.00	0.00	0	Animation

Action Comedy Family	0.00	0.00	0.00	1	
amily Science Fiction	0.00	0.00	0.00	0	Animation Action Comedy F
Fantasy Horror Music	0.00	0.00	0.00	0	Animation Action Comedy
omedy Fantasy Mystery	0.00	0.00	0.00	0	Animation Action C
Action Comedy Horror	0.00	0.00	0.00	0	Animation
Action Comedy Mystery	0.00	0.00	0.00	0	Animation
mance Science Fiction	0.00	0.00	0.00	0	Animation Action Comedy Ro
ction Comedy TV Movie	0.00	0.00	0.00	0	Animation A
nimation Action Crime	0.00	0.00	0.00	1	A
on Action Crime Drama	0.00	0.00	0.00	1	Animati
n Action Crime Horror	0.00	0.00	0.00	0	Animatio
Action Crime Mystery	0.00	0.00	0.00	1	Animation
tery Romance Thriller	0.00	0.00	0.00	0	Animation Action Crime Mys
Crime Romance Mystery	0.00	0.00	0.00	0	Animation Action
on Action Documentary	0.00	0.00	0.00	0	Animati
nimation Action Drama	0.00	0.00	0.00	2	A
ama Adventure Fantasy	0.00	0.00	0.00	0	Animation Action Dr
vie Romance Adventure	0.00	0.00	0.00	0	Animation Action Drama Family TV Mo
Action Drama Fantasy	0.00	0.00	0.00	0	Animation
ntasy Horror Thriller	0.00	0.00	0.00	0	Animation Action Drama Fa
Action Drama History	0.00	0.00	0.00	0	Animation
Action Drama Mystery	0.00	0.00	0.00	0	Animation
nce Fiction Music War	0.00	0.00	0.00	0	Animation Action Drama Romance Scie
Drama Science Fiction	0.00	0.00	0.00	1	Animation Action
tion Action Drama War	0.00	0.00	0.00	0	Anima
imation Action Family	0.00	0.00	0.00	0	An
Action Family Comedy	0.00	0.00	0.00	0	Animation
amily Science Fiction	0.00	0.00	0.00	0	Animation Action F
					Ani

mation Action Fantasy	0.00	0.00	0.00	8	Animation Act
ion Fantasy Adventure	0.00	0.00	0.00	3	Animation
Action Fantasy Comedy	0.00	0.00	0.00	0	Animation Action Fan
tasy Comedy Adventure	0.00	0.00	0.00	0	Animation Action Fantasy Co
medy Adventure Family	0.00	0.00	0.00	0	Animation Action Fantasy Co
medy Family Adventure	0.00	0.00	0.00	0	Animation
Action Fantasy Drama	0.00	0.00	0.00	1	Animation Action
Fantasy Drama Romance	0.00	0.00	0.00	0	Animation
Action Fantasy Family	0.00	0.00	0.00	0	Animation Action Fantasy Fa
mily Adventure Comedy	0.00	0.00	0.00	0	Animation
Action Fantasy Horror	0.00	0.00	0.00	0	Animation A
ction Fantasy Mystery	0.00	0.00	0.00	0	Animation Action Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Animation Action Fantasy S
cience Fiction Family	0.00	0.00	0.00	0	Animation Ac
tion Fantasy Thriller	0.00	0.00	0.00	2	Ani
mation Action History	0.00	0.00	0.00	0	Animation Action
History Fantasy Drama	0.00	0.00	0.00	0	Animation Action History My
stery Science Fiction	0.00	0.00	0.00	0	An
imation Action Horror	0.00	0.00	0.00	0	Animation Action H
orror Science Fiction	0.00	0.00	0.00	1	Ani
mation Action Mystery	0.00	0.00	0.00	0	Animation
Action Mystery Crime	0.00	0.00	0.00	0	Animation Action My
stery Science Fiction	0.00	0.00	0.00	0	Animation A
ction Romance Fantasy	0.00	0.00	0.00	1	Animation Action Ro
mance Science Fiction	0.00	0.00	0.00	0	Animation A
ction Science Fiction	0.00	0.00	0.00	11	Animation Action Scie
nce Fiction Adventure	0.00	0.00	0.00	1	Animation Action S
cience Fiction Comedy	0.00	0.00	0.00	0	Animation Action Science F

iction Comedy Fantasy	0.00	0.00	0.00	0	
					Animation Action
Science Fiction Drama	0.00	0.00	0.00	1	
					Animation Action Science
Fiction Drama Fantasy	0.00	0.00	0.00	0	
					Animation Action Science Fic
tion Family Adventure	0.00	0.00	0.00	0	
					Animation Action Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	
					Animation Action Science
Fiction Romance Music	0.00	0.00	0.00	0	
					Animation Action Science
Fiction TV Movie War	0.00	0.00	0.00	0	
					Animation Action Sci
ence Fiction Thriller	0.00	0.00	0.00	0	
					Animation Action Science Fic
tion Thriller Mystery	0.00	0.00	0.00	0	
					Animation Actio
n Science Fiction War	0.00	0.00	0.00	1	
					Anim
ation Action Thriller	1.00	0.50	0.67	2	
					Animation
Action Thriller Crime	0.00	0.00	0.00	0	
					Animation Action Thriller
Crime Science Fiction	0.00	0.00	0.00	0	
					Animation Action Thr
iller Fantasy Romance	0.00	0.00	0.00	0	
					Animation Action Thr
iller Science Fiction	0.00	0.00	0.00	0	
Animation Action War	0.00	0.00	0.00	0	
Animation Adventure	0.00	0.00	0.00	18	
					Anima
tion Adventure Action	0.00	0.00	0.00	3	
					Animation Adventure Action
Comedy Family Fantasy	0.00	0.00	0.00	0	
					Animation Adventure
Action Comedy Fantasy	0.00	0.00	0.00	0	
					Animation A
dventure Action Drama	0.00	0.00	0.00	0	
					Animation Adventure
Action Drama Fantasy	0.00	0.00	0.00	0	
					Animation Ad
venture Action Family	0.00	0.00	0.00	0	
					Animation Adventure Ac
tion Family Drama War	0.00	0.00	0.00	0	
					Animation Adv
enture Action Fantasy	0.00	0.00	0.00	1	
					Animation Adventure Action Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Animation Adventure A
ction Science Fiction	0.00	0.00	0.00	1	
					Anima
tion Adventure Comedy	0.00	0.00	0.00	7	
					Animation Adventur

e Comedy Action Crime	0.00	0.00	0.00	0	
					Animation Adventu
re Comedy Crime Drama	0.00	0.00	0.00	0	
					Animation Adventur
e Comedy Drama Family	0.00	0.00	0.00	1	
					Animation Ad
venture Comedy Family	0.00	0.00	0.00	9	
					Animation Adventur
e Comedy Family Drama	0.00	0.00	0.00	0	
					Animation Adventure
Comedy Family Fantasy	0.00	0.00	0.00	1	
					Animation Adventure Comedy
Family Fantasy Music	0.00	0.00	0.00	0	
					Animation Adventure Comedy Fa
mily Fantasy TV Movie	0.00	0.00	0.00	0	
					Animation Adventur
e Comedy Family Music	0.00	0.00	0.00	0	
					Animation Adventure Comedy F
amily Science Fiction	0.00	0.00	0.00	1	
					Animation Adv
enture Comedy Fantasy	0.00	0.00	0.00	2	
					Animation Adventure
Comedy Fantasy Action	0.00	0.00	0.00	0	
					Animation Adventure Comedy
Fantasy Action Family	0.00	0.00	0.00	0	
					Animation Adventure
Comedy Fantasy Family	0.00	0.00	0.00	0	
					Animation Adventure Comedy Fantasy S
cience Fiction Family	0.00	0.00	0.00	0	
					Animation Adventure Comedy Fa
ntasy TV Movie Family	0.00	0.00	0.00	0	
					Animation A
dventure Comedy Music	0.00	0.00	0.00	0	
					Animation Adventure C
omedy Science Fiction	0.00	0.00	0.00	0	
					Animation Adventure Comedy Science F
iction Fantasy Family	0.00	0.00	0.00	0	
					Animation Adve
nture Comedy TV Movie	0.00	0.00	0.00	0	
					Animation Adventure Comedy TV Movie Science F
iction Fantasy Action	0.00	0.00	0.00	0	
					Animation Adventure Co
medy TV Movie Western	0.00	0.00	0.00	0	
					Animation Adv
enture Comedy Western	0.00	0.00	0.00	0	
					Animation Adventure
Crime Comedy TV Movie	0.00	0.00	0.00	0	
					Animation Adventure Crime Mystery TV Movie
Action Comedy Family	0.00	0.00	0.00	1	
					Animation Adventur
e Documentary History	0.00	0.00	0.00	0	
					Anim
ation Adventure Drama	0.00	0.00	0.00	2	
					Animation A
dventure Drama Family	0.00	0.00	0.00	2	
					Animation Adventure

Drama Family Fantasy	0.00	0.00	0.00	0	
ence Fiction TV Movie	0.00	0.00	0.00	0	Animation Adventure Drama Family Fantasy Sci
Drama Family TV Movie	0.00	0.00	0.00	0	Animation Adventure
venture Drama Fantasy	0.00	0.00	0.00	0	Animation Ad
Drama Fantasy Family	0.00	0.00	0.00	0	Animation Adventure
Drama Fantasy Mystery	0.00	0.00	0.00	0	Animation Adventure
venture Drama Romance	0.00	0.00	0.00	0	Animation Ad
science Fiction Action	0.00	0.00	0.00	0	Animation Adventure Drama S
Drama TV Movie Family	0.00	0.00	0.00	0	Animation Adventure
n Adventure Drama War	0.00	0.00	0.00	0	Animatio
tion Adventure Family	0.00	0.00	0.00	15	Anima
omedy Science Fiction	0.00	0.00	0.00	0	Animation Adventure Family Action C
venture Family Comedy	0.00	0.00	0.00	4	Animation Ad
Family Comedy Fantasy	0.00	0.00	0.00	1	Animation Adventure
Family Comedy Romance	0.00	0.00	0.00	0	Animation Adventure
dventure Family Drama	0.00	0.00	0.00	1	Animation A
enture Family Fantasy	0.00	0.00	0.00	5	Animation Adv
Family Fantasy Action	0.00	0.00	0.00	0	Animation Adventure
Family Fantasy Comedy	0.00	0.00	0.00	2	Animation Adventure
ntasy Comedy TV Movie	0.00	0.00	0.00	0	Animation Adventure Family Fa
Family Fantasy Drama	0.00	0.00	0.00	0	Animation Adventure
Family Fantasy Horror	0.00	0.00	0.00	0	Animation Adventure
Family Fantasy Music	0.00	0.00	0.00	1	Animation Adventure
ntasy Science Fiction	0.00	0.00	0.00	1	Animation Adventure Family Fa
science Fiction Comedy	0.00	0.00	0.00	1	Animation Adventure Family Fantasy S
re Family History War	0.00	0.00	0.00	0	Animation Adventu
dventure Family Music	0.00	0.00	0.00	1	Animation A
re Family Music Drama	0.00	0.00	0.00	0	Animation Adventu
					Animation Adventure Fa

mily Mystery TV Movie	0.00	0.00	0.00	0	Animation Adv
enture Family Romance	0.00	0.00	0.00	1	Animation Adventure F
amily Science Fiction	0.00	0.00	0.00	2	Animation Adventure Family Science F
iction Comedy Fantasy	0.00	0.00	0.00	0	Animation Adventure Family Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Animation Adventure Family Science F
iction Fantasy Comedy	0.00	0.00	0.00	0	Animation Adve
nture Family TV Movie	0.00	0.00	0.00	1	Animation Adv
enture Family Western	0.00	0.00	0.00	0	Animat
ion Adventure Fantasy	0.00	0.00	0.00	3	Animation Adv
enture Fantasy Action	0.00	0.00	0.00	0	Animation Adventure
Fantasy Action Family	0.00	0.00	0.00	1	Animation Adv
enture Fantasy Comedy	0.00	0.00	0.00	1	Animation Adventure
Fantasy Comedy Family	0.00	0.00	0.00	2	Animation Ad
venture Fantasy Drama	0.00	0.00	0.00	2	Animation Adventure
Fantasy Drama Family	0.00	0.00	0.00	0	Animation Adv
enture Fantasy Family	0.00	0.00	0.00	2	Animation Adventure Fantasy
Family Action Comedy	0.00	0.00	0.00	0	Animation Adventure
Fantasy Family Comedy	0.00	0.00	0.00	0	Animation Adventure Fantasy F
amily Science Fiction	0.00	0.00	0.00	0	Animation Adventure Fa
ntasy Family TV Movie	0.00	0.00	0.00	0	Animation Adv
enture Fantasy Horror	0.00	0.00	0.00	0	Animation Adventure
Fantasy Horror Family	0.00	0.00	0.00	0	Animation Adve
nture Fantasy Mystery	0.00	0.00	0.00	0	Animation Adventure Fantasy My
stery Science Fiction	0.00	0.00	0.00	0	Animation Adve
nture Fantasy Romance	0.00	0.00	0.00	1	Animation Adventure Fa
ntasy Science Fiction	0.00	0.00	0.00	1	Animation Adventure Fantasy Science
Fiction Comedy Family	0.00	0.00	0.00	0	Animation Adventure Fantasy S
cience Fiction Family	0.00	0.00	0.00	2	Animat

ion Adventure History	0.00	0.00	0.00	0	
					Animation Ad
venture History Drama	0.00	0.00	0.00	0	
					Anim
ation Adventure Music	0.00	0.00	0.00	0	
					Animation Advent
ure Music Documentary	0.00	0.00	0.00	0	
					Animation Adventure
Music Family Fantasy	0.00	0.00	0.00	0	
					Animation Adventure Music
Family Fantasy Comedy	0.00	0.00	0.00	0	
					Animat
ion Adventure Mystery	0.00	0.00	0.00	0	
					Animation Adventure M
ystery Crime TV Movie	0.00	0.00	0.00	0	
					Animation Adventure Mystery Crime TV Movie
Action Comedy Family	0.00	0.00	0.00	0	
					Animation Adventure M
ystery Family Fantasy	0.00	0.00	0.00	0	
					Animation Adve
nture Mystery Fantasy	0.00	0.00	0.00	0	
					Animation Adventure Mystery Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
					Animation Adventure Mystery
Science Fiction Crime	0.00	0.00	0.00	0	
					Animation Adven
ture Mystery TV Movie	0.00	0.00	0.00	0	
					Animat
ion Adventure Romance	0.00	0.00	0.00	0	
					Animation Adve
nture Science Fiction	0.00	0.00	0.00	2	
					Animation Adventure Science Fiction
Drama Family Fantasy	0.00	0.00	0.00	0	
					Animation Adventure Science Fiction
Family Fantasy Comedy	0.00	0.00	0.00	0	
					Animation Adventure Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	
					Animation Adventure Science Fiction F
antasy History Horror	0.00	0.00	0.00	0	
					Animation Adventure Science F
iction Fantasy Horror	0.00	0.00	0.00	0	
					Animation Adventure Sci
ence Fiction TV Movie	0.00	0.00	0.00	2	
					Animation Adventure Science Fiction TV Movie Action Comedy
Crime Family Mystery	0.00	0.00	0.00	0	
					Animati
on Adventure TV Movie	0.00	0.00	0.00	1	
					Animation Adventure
TV Movie Family Music	0.00	0.00	0.00	0	
Animation Comedy	0.04	0.16	0.06	45	
					An
imation Comedy Action	0.00	0.00	0.00	2	
					Animation Co
medy Action Adventure	0.00	0.00	0.00	0	
					Animation Comedy Ac

tion Adventure Family	0.00	0.00	0.00	0	
					Animation Comedy Action Adventure Family Fa
antasy Science Fiction	0.00	0.00	0.00	0	
					Animation Comedy Action Adventure Music Sc
ience Fiction Romance	0.00	0.00	0.00	0	
					Animatio
n Comedy Action Drama	0.00	0.00	0.00	0	
					Animation Comedy Action Drama F
amily Science Fiction	0.00	0.00	0.00	0	
					Animation Comedy
Action Drama Romance	0.00	0.00	0.00	0	
					Animation
Comedy Action Family	0.00	0.00	0.00	0	
					Anima
tion Comedy Adventure	0.00	0.00	0.00	3	
					Animation Co
medy Adventure Action	0.00	0.00	0.00	1	
					Animation Comedy Ad
venture Action Family	0.00	0.00	0.00	0	
					Animation Comedy
Adventure Crime Music	0.00	0.00	0.00	0	
					Animation Comedy A
dventure Drama Family	0.00	0.00	0.00	0	
					Animation Comedy Adventure Drama Fantasy Music Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	
					Animation Co
medy Adventure Family	0.00	0.00	0.00	6	
					Animation Comedy Adv
enture Family Fantasy	0.00	0.00	0.00	0	
					Animation Comedy Adv
enture Family Mystery	0.00	0.00	0.00	0	
					Animation Comedy Adventure F
amily Science Fiction	0.00	0.00	0.00	0	
					Animation Com
edy Adventure Fantasy	0.00	0.00	0.00	1	
					Animation Comedy Adventure F
antasy Romance Family	0.00	0.00	0.00	0	
					Animation Comedy Adventure Fa
antasy Science Fiction	0.00	0.00	0.00	0	
					Animation C
omedy Adventure Music	0.00	0.00	0.00	0	
					Animation Comedy Adve
nture Science Fiction	0.00	0.00	0.00	0	
					Animation Come
dy Adventure TV Movie	0.00	0.00	0.00	1	
					Animation
Comedy Adventure War	0.00	0.00	0.00	0	
					A
nimation Comedy Crime	0.00	0.00	0.00	0	
					Animation Comedy
Crime Family Fantasy	0.00	0.00	0.00	0	
					Animati
on Comedy Crime Music	0.00	0.00	0.00	0	
					Animation Comedy Documentary
Drama Fantasy Mystery	0.00	0.00	0.00	0	
					A

Animation Comedy Drama	0.00	0.00	0.00	8	Animation Comedy
Drama Action TV Movie	0.00	0.00	0.00	0	Animation Comedy
Drama Action Thriller	0.00	0.00	0.00	0	Animation C
Comedy Drama Adventure	0.00	0.00	0.00	0	Animatio
n Comedy Drama Family	0.00	0.00	0.00	2	Animation Come
dy Drama Family Music	0.00	0.00	0.00	0	Animation Comedy
Drama Fantasy Romance	0.00	0.00	0.00	0	Animatio
n Comedy Drama Horror	0.00	0.00	0.00	0	Animati
on Comedy Drama Music	0.00	0.00	0.00	0	Animation Comed
y Drama Music Romance	0.00	0.00	0.00	0	Animation
Comedy Drama Mystery	0.00	0.00	0.00	0	Animation Comedy
Drama Mystery Fantasy	0.00	0.00	0.00	0	Animation
Comedy Drama Romance	0.00	0.00	0.00	1	Animation Comedy Drama
Science Fiction Music	0.00	0.00	0.00	0	Animation Comedy Drama Thriller Mystery Ro
mance Science Fiction	0.00	0.00	0.00	0	An
imation Comedy Family	0.00	0.00	0.00	28	Animation
Comedy Family Action	0.00	0.00	0.00	0	Animation Comedy Family Act
ion Adventure Fantasy	0.00	0.00	0.00	0	Animation Co
medy Family Adventure	0.00	0.00	0.00	6	Animation Comedy Fa
mily Adventure Action	0.00	0.00	0.00	0	Animation Comedy Fam
ily Adventure Fantasy	0.00	0.00	0.00	0	Animation Comedy Family Adve
nture Fantasy Mystery	0.00	0.00	0.00	0	Animation Comedy F
amily Adventure Music	0.00	0.00	0.00	0	Animation Comedy Family Ad
venture Music Romance	0.00	0.00	0.00	0	Animation Comedy Fam
y Crime Music Western	0.00	0.00	0.00	0	Animation Comedy Fam
ily Documentary Drama	0.00	0.00	0.00	0	Animation
Comedy Family Fantasy	0.00	0.00	0.00	7	Animation Comedy
Family Fantasy Action	0.00	0.00	0.00	0	Animation Comedy Fam

ily Fantasy Adventure	0.00	0.00	0.00	1	Animation Comedy
Family Fantasy Horror	0.00	0.00	0.00	0	Animation Comedy
Family Fantasy Music	0.00	0.00	0.00	0	Animation Comedy Family Fa
ntasy Science Fiction	0.00	0.00	0.00	1	Animation Comedy Family Fantasy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Animation Comedy Fa
mily Fantasy TV Movie	0.00	0.00	0.00	1	Animation Comedy Family Fantasy TV
Movie Science Fiction	0.00	0.00	0.00	0	Animation
Comedy Family Horror	0.00	0.00	0.00	0	Animatio
n Comedy Family Music	0.00	0.00	0.00	2	Animation Comedy
Family Music Fantasy	0.00	0.00	0.00	0	Animation Comedy Family
Music Science Fiction	0.00	0.00	0.00	0	Animation
Comedy Family Mystery	0.00	0.00	0.00	1	Animation Comedy Family Myst
ery Adventure Fantasy	0.00	0.00	0.00	0	Animation Comedy F
amily Mystery Fantasy	0.00	0.00	0.00	0	Animation
Comedy Family Romance	0.00	0.00	0.00	0	Animation Comedy F
amily Science Fiction	0.00	0.00	0.00	2	Animation Comedy Family Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Animation Comedy Family Science Fiction Adve
nture TV Movie Action	0.00	0.00	0.00	0	Animation Comedy Family Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Animation Comedy Family Science Fiction
TV Movie Action Drama	0.00	0.00	0.00	0	Animation C
omedy Family TV Movie	0.00	0.00	0.00	1	Animation Comedy Family TV Movie A
ction Adventure Drama	0.00	0.00	0.00	0	Ani
mation Comedy Fantasy	0.00	0.00	0.00	5	Animation Comedy Fantasy Ac
tion Adventure Family	0.00	0.00	0.00	2	Animation Com
edy Fantasy Adventure	0.00	0.00	0.00	0	Animation Comedy Fan
tasy Adventure Family	0.00	0.00	0.00	1	Animation Comedy Fantasy Ad
venture Family Horror	0.00	0.00	0.00	0	Animation
Comedy Fantasy Drama	0.00	0.00	0.00	1	Animation

Comedy Fantasy Family	0.00	0.00	0.00	0	Animation Comedy
Fantasy Family Drama	0.00	0.00	0.00	0	Animation Comedy
Fantasy Family Music	0.00	0.00	0.00	0	Animation
Comedy Fantasy Horror	0.00	0.00	0.00	0	Animation Comedy Fantasy H
orror Science Fiction	0.00	0.00	0.00	0	Animation
Comedy Fantasy Music	0.00	0.00	0.00	0	Animation C
omedy Fantasy Mystery	0.00	0.00	0.00	1	Animation C
omedy Fantasy Romance	0.00	0.00	0.00	0	Animation Comedy Fa
antasy Science Fiction	0.00	0.00	0.00	0	Ani
mation Comedy History	0.00	0.00	0.00	0	Animation Comedy History Fantasy War Western Action Adve
nture Science Fiction	0.00	0.00	0.00	0	An
imation Comedy Horror	0.00	0.00	0.00	4	Animation Comedy
Horror Drama Fantasy	0.00	0.00	0.00	0	Animation
Comedy Horror Family	0.00	0.00	0.00	0	Animation
Comedy Horror Mystery	0.00	0.00	0.00	0	Animation C
omedy Horror TV Movie	0.00	0.00	0.00	0	A
nimation Comedy Music	0.00	0.00	0.00	1	Animatio
n Comedy Music Action	0.00	0.00	0.00	0	Animation C
omedy Music Adventure	0.00	0.00	0.00	0	Animati
on Comedy Music Drama	0.00	0.00	0.00	0	Animatio
n Comedy Music Family	0.00	0.00	0.00	0	Animation Comedy
Music Family Fantasy	0.00	0.00	0.00	0	Animation Comedy Music Fantasy Ac
tion Adventure Family	0.00	0.00	0.00	0	Animation Comedy
Music Science Fiction	0.00	0.00	0.00	0	Animation Comedy Music S
cience Fiction Horror	0.00	0.00	0.00	0	Animation
Comedy Music TV Movie	0.00	0.00	0.00	0	Ani
mation Comedy Mystery	0.00	0.00	0.00	0	Animation
Comedy Mystery Family	0.00	0.00	0.00	0	Animation Comedy M

mystery Fantasy Horror	0.00	0.00	0.00	0	Animation Co
medy Mystery Thriller	0.00	0.00	0.00	0	Animation Comedy My
stery Thriller Horror	0.00	0.00	0.00	0	Ani
mation Comedy Romance	0.00	0.00	0.00	2	Animation
Comedy Romance Drama	0.00	0.00	0.00	2	Animation
Comedy Romance Family	0.00	0.00	0.00	0	Animation
Comedy Romance Music	0.00	0.00	0.00	1	Animation Comedy Ro
mance Science Fiction	0.00	0.00	0.00	0	Animation C
omedy Science Fiction	0.00	0.00	0.00	6	Animation Comedy Science Fiction Acti
on Adventure TV Movie	0.00	0.00	0.00	0	Animation Comedy Science F
iction Action Fantasy	0.00	0.00	0.00	0	Animation Comedy Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Animation Comedy Science Fiction Adve
nture Action TV Movie	0.00	0.00	0.00	0	Animation Comedy
Science Fiction Drama	0.00	0.00	0.00	0	Animation Comedy S
cience Fiction Family	0.00	0.00	0.00	0	Animation Comedy Science Fic
tion Family Adventure	0.00	0.00	0.00	1	Animation Comedy Science Fiction Family Fanta
sy Adventure TV Movie	0.00	0.00	0.00	0	Animation Comedy Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Animation Comedy Science Fi
ction Fantasy Romance	0.00	0.00	0.00	0	Animation Comedy Science
Fiction Mystery Drama	0.00	0.00	0.00	0	Animation Comedy Sc
ience Fiction Romance	0.00	0.00	0.00	0	Animation Comedy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Anim
ation Comedy TV Movie	0.00	0.00	0.00	2	Animation C
omedy TV Movie Family	0.00	0.00	0.00	0	Animation Com
edy Thriller TV Movie	0.00	0.00	0.00	0	Ani
mation Comedy Western	0.00	0.00	0.00	2	
Animation Crime	0.00	0.00	0.00	2	A
nimation Crime Action	0.00	0.00	0.00	0	Animation C

rime Action Adventure	0.00	0.00	0.00	0	Animati
on Crime Action Drama	0.00	0.00	0.00	0	Animation
Crime Action Thriller	0.00	0.00	0.00	0	Animation Cr
ime Adventure Mystery	0.00	0.00	0.00	0	A
nimation Crime Comedy	0.00	0.00	0.00	0	Animat
ion Crime Documentary	0.00	0.00	0.00	0	
Animation Crime Drama	0.00	0.00	0.00	0	Animation Crime Dr
ama Mystery Adventure	0.00	0.00	0.00	0	Animation Crime
Family Comedy Mystery	0.00	0.00	0.00	0	Animation Crime Fantasy My
stery Science Fiction	0.00	0.00	0.00	0	An
imation Crime Mystery	0.00	0.00	0.00	0	Animation
Crime Mystery Action	0.00	0.00	0.00	1	Animation Crime Mystery
Comedy Romance Horror	0.00	0.00	0.00	0	Animation Crime My
stery TV Movie Comedy	0.00	0.00	0.00	0	Animation C
rime Mystery Thriller	0.00	0.00	0.00	0	
Animation Documentary	0.00	0.00	0.00	6	Animation Docume
ntary Comedy TV Movie	0.00	0.00	0.00	0	Animation Docu
mentary Crime Mystery	0.00	0.00	0.00	0	Animat
ion Documentary Drama	0.00	0.00	0.00	0	Animation Docu
mentary Drama Fantasy	0.00	0.00	0.00	0	Animation
Documentary Drama War	0.00	0.00	0.00	0	Animatio
n Documentary History	0.00	0.00	0.00	0	Animation Documentary Horror Ro
mance Fantasy Mystery	0.00	0.00	0.00	0	Animat
ion Documentary Music	0.00	0.00	0.00	0	Anim
ation Documentary War	0.00	0.00	0.00	0	
Animation Drama	0.00	0.00	0.00	32	A
nimation Drama Action	0.00	0.00	0.00	1	Animation Drama Action Comedy Thriller My
stery Horror TV Movie	0.00	0.00	0.00	0	Animation

Drama Action Fantasy	0.00	0.00	0.00	0	Animation
Drama Action Mystery	0.00	0.00	0.00	0	Animation Drama A
ction Science Fiction	0.00	0.00	0.00	1	Animation
Drama Action Thriller	0.00	0.00	0.00	0	Anima
tion Drama Action War	0.00	0.00	0.00	0	Anim
ation Drama Adventure	0.00	0.00	0.00	2	Animation D
rama Adventure Family	0.00	0.00	0.00	0	Animation Drama Adve
nture Family TV Movie	0.00	0.00	0.00	0	Animation Dr
ama Adventure Fantasy	0.00	0.00	0.00	0	Animation Drama Adv
enture Fantasy Comedy	0.00	0.00	0.00	0	Animation Drama Adventure Science F
iction Romance Action	0.00	0.00	0.00	0	A
nimation Drama Comedy	0.00	0.00	0.00	4	Animation
Drama Comedy Fantasy	0.00	0.00	0.00	0	Animation Drama
Comedy Horror Fantasy	0.00	0.00	0.00	0	Animati
on Drama Comedy Music	0.00	0.00	0.00	0	Animation
Drama Comedy Mystery	0.00	0.00	0.00	0	Animation
Drama Comedy Romance	0.00	0.00	0.00	0	Animation Drama C
omedy Science Fiction	0.00	0.00	0.00	0	Animation Drama Comedy Sc
ience Fiction Mystery	0.00	0.00	0.00	0	
Animation Drama Crime	0.00	0.00	0.00	0	Animati
on Drama Crime Action	0.00	0.00	0.00	0	A
nimation Drama Family	0.00	0.00	0.00	7	Animation D
rama Family Adventure	0.00	0.00	0.00	0	Animatio
n Drama Family Comedy	0.00	0.00	0.00	0	Animation
Drama Family Fantasy	0.00	0.00	0.00	1	Animation
Drama Family History	0.00	0.00	0.00	0	Animation
Drama Family Mystery	0.00	0.00	0.00	0	An
imation Drama Fantasy	0.00	0.00	0.00	5	Animation Drama

Fantasy Action Music	0.00	0.00	0.00	0	Animation Dr
ama Fantasy Adventure	0.00	0.00	0.00	0	Animation
Drama Fantasy Comedy	0.00	0.00	0.00	0	Animation
Drama Fantasy Family	0.00	0.00	0.00	0	Animation
Drama Fantasy Horror	0.00	0.00	0.00	1	Animation Drama Fantasy My
stery Science Fiction	0.00	0.00	0.00	0	Animation
Drama Fantasy Romance	0.00	0.00	0.00	2	Animat
ion Drama Fantasy War	0.00	0.00	0.00	0	An
imation Drama History	0.00	0.00	0.00	3	Animation
Drama History Horror	0.00	0.00	0.00	0	Animation Drama History Romance Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Animat
ion Drama History War	0.00	0.00	0.00	1	A
nimation Drama Horror	0.00	0.00	0.00	1	Animation Drama Horror Music Thr
iller Science Fiction	0.00	0.00	0.00	0	Animation
Drama Horror TV Movie	0.00	0.00	0.00	0	Animation Drama
Horror Thriller Crime	0.00	0.00	0.00	0	
Animation Drama Music	0.00	0.00	0.00	4	Animation Drama
Music Fantasy History	0.00	0.00	0.00	0	Animatio
n Drama Music Romance	0.00	0.00	0.00	0	Animation Drama
Music Romance Fantasy	0.00	0.00	0.00	0	An
imation Drama Mystery	0.00	0.00	0.00	1	Animation Dr
ama Mystery Adventure	0.00	0.00	0.00	0	Animation
Drama Mystery Fantasy	0.00	0.00	0.00	2	Animation
Drama Mystery Romance	0.00	0.00	0.00	1	Animation Drama Mystery Sc
ience Fiction Romance	0.00	0.00	0.00	0	An
imation Drama Romance	0.00	0.00	0.00	6	Animation
Drama Romance Comedy	0.00	0.00	0.00	1	Animation
Drama Romance Fantasy	0.00	0.00	0.00	0	Animation Drama Ro

mance History Fantasy	0.00	0.00	0.00	0	
					Animation Drama Ro
mance Mystery Fantasy	0.00	0.00	0.00	0	
					Animation Drama Ro
mance Science Fiction	0.00	0.00	0.00	0	
					Animation Drama Romance Science F
iction Fantasy Action	0.00	0.00	0.00	0	
					Animation
Drama Science Fiction	0.00	0.00	0.00	3	
					Animation Drama S
cience Fiction Action	0.00	0.00	0.00	0	
					Animation Drama S
cience Fiction Family	0.00	0.00	0.00	0	
					Animation Drama Sci
ence Fiction Thriller	0.00	0.00	0.00	0	
					Animation
Drama TV Movie Family	0.00	0.00	0.00	0	
					Ani
mation Drama Thriller	0.00	0.00	0.00	0	
					Animation
Drama Thriller Horror	0.00	0.00	0.00	0	
					Animation D
rama Thriller Mystery	0.00	0.00	0.00	0	
Animation Drama War	0.00	0.00	0.00	1	
					Animat
ion Drama War History	0.00	0.00	0.00	0	
					Animation Dram
a War Science Fiction	0.00	0.00	0.00	0	
					Animati
on Drama War TV Movie	0.00	0.00	0.00	0	
					Animation Drama Western
Family Action History	0.00	0.00	0.00	0	
Animation Family	0.09	0.46	0.15	85	
					An
imation Family Action	0.00	0.00	0.00	0	
					Animation Fa
mily Action Adventure	0.00	0.00	0.00	1	
					Animation Family Action Ad
venture Drama Fantasy	0.00	0.00	0.00	0	
					Animation Family Act
ion Adventure Fantasy	0.00	0.00	0.00	0	
					Animation Family Action Adv
enture Fantasy Comedy	0.00	0.00	0.00	0	
					Animation Family Action Adve
nture Science Fiction	0.00	0.00	0.00	0	
					Animation Family Act
ion Adventure Western	0.00	0.00	0.00	0	
					Animation
Family Action Comedy	0.00	0.00	0.00	1	
					Animation Family Action Com
edy Adventure Fantasy	0.00	0.00	0.00	0	
					Animation
Family Action Fantasy	0.00	0.00	0.00	1	
					Animation Family Act

ion Fantasy Adventure	0.00	0.00	0.00	0	Animation Family
Action Fantasy Comedy	0.00	0.00	0.00	0	Animation Family A
ction Science Fiction	0.00	0.00	0.00	0	Animation F
amily Action TV Movie	0.00	0.00	0.00	0	Anima
tion Family Adventure	0.00	0.00	0.00	11	Animation Fa
mily Adventure Action	0.00	0.00	0.00	0	Animation Fa
mily Adventure Comedy	0.00	0.00	0.00	5	Animation Family Adv
enture Comedy Fantasy	0.00	0.00	0.00	1	Animation Family Adventure Comedy Fa
ntasy Science Fiction	0.00	0.00	0.00	2	Animation Family A
dventure Comedy Music	0.00	0.00	0.00	0	Animation Family Adventure C
omedy Science Fiction	0.00	0.00	0.00	0	Animation F
amily Adventure Crime	0.00	0.00	0.00	0	Animation F
amily Adventure Drama	0.00	0.00	0.00	0	Animation Family Ad
venture Drama Fantasy	0.00	0.00	0.00	1	Animation Fam
ily Adventure Fantasy	0.00	0.00	0.00	2	Animation Family Adv
enture Fantasy Action	0.00	0.00	0.00	0	Animation Family Ad
venture Fantasy Music	0.00	0.00	0.00	0	Animation Family Adve
nture Fantasy Romance	0.00	0.00	0.00	0	Animation Family Adventure Fa
ntasy Science Fiction	0.00	0.00	0.00	1	Animation Family Adventure
Music Mystery Comedy	0.00	0.00	0.00	0	Animation Family Adventure
Music Science Fiction	0.00	0.00	0.00	0	Animation Fam
ily Adventure Mystery	0.00	0.00	0.00	0	Animation Fam
ily Adventure Romance	0.00	0.00	0.00	0	Animation Family Adve
nture Science Fiction	0.00	0.00	0.00	0	Animation Family Adventure Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Animation Family Adventure Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Animation Fami
ly Adventure TV Movie	0.00	0.00	0.00	0	Animation Fam
ily Adventure Western	0.00	0.00	0.00	0	An

imation Family Comedy	0.00	0.00	0.00	8	Animation
Family Comedy Action	0.00	0.00	0.00	0	Animation Fa
mily Comedy Adventure	0.00	0.00	0.00	0	Animation Family Co
medy Adventure Action	0.00	0.00	0.00	0	Animation Family Com
edy Adventure Fantasy	0.00	0.00	0.00	0	Animation Family Com
edy Adventure Mystery	0.00	0.00	0.00	0	Animation Family Com
edy Adventure Romance	0.00	0.00	0.00	0	Animatio
n Family Comedy Drama	0.00	0.00	0.00	1	Animation
Family Comedy Fantasy	0.00	0.00	0.00	4	Animation Family Com
edy Fantasy Adventure	0.00	0.00	0.00	0	Animation Family Comed
y Fantasy Drama Music	0.00	0.00	0.00	0	Animation Family C
omedy Fantasy Romance	0.00	0.00	0.00	0	Animation Family Co
medy Fantasy TV Movie	0.00	0.00	0.00	0	Animation
Family Comedy Horror	0.00	0.00	0.00	0	Animatio
n Family Comedy Music	0.00	0.00	0.00	1	Animation Family
Comedy Music TV Movie	0.00	0.00	0.00	1	Animation
Family Comedy Mystery	0.00	0.00	0.00	1	Animation Family Comedy My
stery Science Fiction	0.00	0.00	0.00	0	Animation Family C
omedy Science Fiction	0.00	0.00	0.00	0	Animation Family Comedy Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Animation Family Comedy Science Fiction Adve
nture TV Movie Action	0.00	0.00	0.00	0	Animation Family Comedy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Animation F
amily Comedy TV Movie	0.00	0.00	0.00	2	Animation Family Comedy TV Movie Action Adve
nture Science Fiction	0.00	0.00	0.00	0	Animation Family C
omedy TV Movie Horror	0.00	0.00	0.00	0	Animat
ion Family Comedy War	0.00	0.00	0.00	0	Animati
on Family Documentary	0.00	0.00	0.00	0	A
nimation Family Drama	0.00	0.00	0.00	2	Animation

Family Drama Fantasy	0.00	0.00	0.00	0	Ani
mation Family Fantasy	0.00	0.00	0.00	24	Animation Family Fan
tasy Action Adventure	0.00	0.00	0.00	0	Animation Fam
ily Fantasy Adventure	0.00	0.00	0.00	6	Animation Family Fan
tasy Adventure Comedy	0.00	0.00	0.00	0	Animation Family Fantasy Adv
enture Comedy Romance	0.00	0.00	0.00	0	Animation Family Fa
ntasy Adventure Drama	0.00	0.00	0.00	0	Animation Family Fan
tasy Adventure Horror	0.00	0.00	0.00	0	Animation Family Fa
ntasy Adventure Music	0.00	0.00	0.00	0	Animation Family Fantasy Adve
nture Science Fiction	0.00	0.00	0.00	0	Animation
Family Fantasy Comedy	0.00	0.00	0.00	1	Animation Family Fan
tasy Comedy Adventure	0.00	0.00	0.00	1	Animation Family Fantasy Com
edy Adventure Mystery	0.00	0.00	0.00	0	Animation Family Fantasy C
omedy Science Fiction	0.00	0.00	0.00	0	Animation Family Fanta
sy Documentary Comedy	0.00	0.00	0.00	0	Animation
Family Fantasy Drama	0.00	0.00	0.00	0	Animation Family Fa
ntasy Drama Adventure	0.00	0.00	0.00	0	Animation F
amily Fantasy History	0.00	0.00	0.00	0	Animation
Family Fantasy Horror	0.00	0.00	0.00	0	Animation
Family Fantasy Music	0.00	0.00	0.00	0	Animation Family Fantasy Music Adventure C
omedy Science Fiction	0.00	0.00	0.00	0	Animation F
amily Fantasy Romance	0.00	0.00	0.00	1	Animation Family Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Animation Fa
mily Fantasy TV Movie	0.00	0.00	0.00	1	Animation Family Fanta
sy TV Movie Adventure	0.00	0.00	0.00	1	Ani
mation Family History	0.00	0.00	0.00	0	Animati
on Family History War	0.00	0.00	0.00	0	Animation
Family Horror Fantasy	0.00	0.00	0.00	0	A

animation Family Music	0.00	0.00	0.00	5	Animation F
amily Music Adventure	0.00	0.00	0.00	0	Animatio
n Family Music Comedy	0.00	0.00	0.00	0	Animation Fami
ly Music Comedy Drama	0.00	0.00	0.00	0	Animation
Family Music Fantasy	0.00	0.00	0.00	0	Animation Family Music Fan
tasy Comedy Adventure	0.00	0.00	0.00	0	Animation Family Music Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Animation
Family Music Romance	0.00	0.00	0.00	0	Animation Family
Music Science Fiction	0.00	0.00	0.00	0	Animation
Family Music TV Movie	0.00	0.00	0.00	0	Ani
mation Family Mystery	0.00	0.00	0.00	0	Animation Family Mys
tery Adventure Comedy	0.00	0.00	0.00	0	Animation
Family Mystery Comedy	0.00	0.00	0.00	0	Animation Family Mystery C
omedy Science Fiction	0.00	0.00	0.00	0	Animation Family M
ystery Comedy Western	0.00	0.00	0.00	0	Animation Family M
ystery Fantasy Comedy	0.00	0.00	0.00	0	Ani
mation Family Romance	0.00	0.00	0.00	0	Animati
on Family Romance War	0.00	0.00	0.00	0	Animation F
amily Science Fiction	0.00	0.00	0.00	1	Animation Family S
cience Fiction Action	0.00	0.00	0.00	0	Animation Family Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Animation Family Science Fic
tion Adventure Comedy	0.00	0.00	0.00	0	Animation Family S
cience Fiction Comedy	0.00	0.00	0.00	0	Animation Family Science Fict
ion Fantasy Adventure	0.00	0.00	0.00	0	Animation Family Science Fiction TV Movie Ac
tion Comedy Adventure	0.00	0.00	0.00	0	Animation Family Science Fi
ction TV Movie Comedy	0.00	0.00	0.00	0	Anim
ation Family TV Movie	0.00	0.00	0.00	4	Animation Family TV M
ovie Action Adventure	0.00	0.00	0.00	0	Animation Fami

ly TV Movie Adventure	0.00	0.00	0.00	0	Animation Family TV Movie Adv
enture Comedy Fantasy	0.00	0.00	0.00	0	Animation Family TV Mo
vie Adventure Fantasy	0.00	0.00	0.00	0	Animation F
amily TV Movie Comedy	0.00	0.00	0.00	0	Animation Family TV Movie C
omedy Science Fiction	0.00	0.00	0.00	0	Animation Fa
mily TV Movie Fantasy	0.00	0.00	0.00	2	Animation Family TV Mo
vie Fantasy Adventure	0.00	0.00	0.00	0	Animation
Family TV Movie Music	0.00	0.00	0.00	0	Animation Family TV
Movie Mystery Comedy	0.00	0.00	0.00	0	Animation Family TV
Movie Mystery Fantasy	0.00	0.00	0.00	0	Ani
mation Family Western	0.00	0.00	0.00	0	Animation Family Wes
tern Adventure Comedy	0.00	0.00	0.00	0	
Animation Fantasy	0.17	0.03	0.05	33	Ani
mation Fantasy Action	0.00	0.00	0.00	5	Animation Fan
tasy Action Adventure	0.00	0.00	0.00	3	Animation Fantasy Action Co
medy Adventure Family	0.00	0.00	0.00	2	Animation Fantasy Action Comedy Science Fic
tion Adventure Family	0.00	0.00	0.00	0	Animation Fantas
y Action Drama Comedy	0.00	0.00	0.00	0	Animation Fantasy
Action Family Horror	0.00	0.00	0.00	0	Animation
Fantasy Action Horror	0.00	0.00	0.00	0	Animation Fantasy Action Music Ad
venture Comedy Family	0.00	0.00	0.00	0	Animation F
antasy Action Mystery	0.00	0.00	0.00	0	Animation Fantasy A
ction Science Fiction	0.00	0.00	0.00	0	Animat
ion Fantasy Adventure	0.00	0.00	0.00	9	Animation Fan
tasy Adventure Action	0.00	0.00	0.00	4	Animation Fantasy A
dventure Action Drama	0.00	0.00	0.00	0	Animation Fan
tasy Adventure Comedy	0.00	0.00	0.00	2	Animation Fa
ntasy Adventure Drama	0.00	0.00	0.00	1	Animation Fan

tasy Adventure Family	0.00	0.00	0.00	2	Animation Fantasy Ad
venture Family Action	0.00	0.00	0.00	0	Animation Fantasy Adventure F
amily Science Fiction	0.00	0.00	0.00	1	Animation Fant
asy Adventure Mystery	0.00	0.00	0.00	0	Ani
mation Fantasy Comedy	0.00	0.00	0.00	2	Animation Fantasy Comedy Ac
tion Adventure Family	0.00	0.00	0.00	0	Animation Fan
tasy Comedy Adventure	0.00	0.00	0.00	0	Animation Fantasy Comedy Adve
nture Science Fiction	0.00	0.00	0.00	0	Animation
Fantasy Comedy Family	0.00	0.00	0.00	1	Animation Fantasy Comedy Fa
mily Action Adventure	0.00	0.00	0.00	0	Animation Fantasy Comedy Rom
ance Adventure Horror	0.00	0.00	0.00	0	An
imation Fantasy Drama	0.00	0.00	0.00	4	Animation Fa
ntasy Drama Adventure	0.00	0.00	0.00	0	Animation Fantasy D
rama Adventure Family	0.00	0.00	0.00	0	Animation Fant
asy Drama Documentary	0.00	0.00	0.00	0	Animation
Fantasy Drama Family	0.00	0.00	0.00	0	Animation
Fantasy Drama Horror	0.00	0.00	0.00	0	Animation Fantasy
Drama Horror Thriller	0.00	0.00	0.00	0	Animatio
n Fantasy Drama Music	0.00	0.00	0.00	0	Animation Fantasy
Drama Mystery Action	0.00	0.00	0.00	0	Animation Fantas
y Drama Mystery Music	0.00	0.00	0.00	0	Animation Fantasy
Drama Science Fiction	0.00	0.00	0.00	0	Animation Fantasy
Drama Thriller Horror	0.00	0.00	0.00	0	Ani
mation Fantasy Family	0.00	0.00	0.00	8	Animation
Fantasy Family Action	0.00	0.00	0.00	0	Animation Fantasy Fa
mily Action Adventure	0.00	0.00	0.00	0	Animation Fantasy Family Ac
tion Adventure Comedy	0.00	0.00	0.00	0	Animation Fan
tasy Family Adventure	0.00	0.00	0.00	1	Animation Fantasy Family Adventure C

Comedy Science Fiction	0.00	0.00	0.00	0	
					Animation Fantasy Fami
ly Adventure TV Movie	0.00	0.00	0.00	0	
					Animation Fantasy Fa
mily Comedy Adventure	0.00	0.00	0.00	0	
					Animation Fantasy Family Com
edy Romance Adventure	0.00	0.00	0.00	0	
					Animation Fa
ntasy Family TV Movie	0.00	0.00	0.00	0	
					Animation Fantasy F
amily TV Movie Comedy	0.00	0.00	0.00	0	
					Anim
ation Fantasy History	0.00	0.00	0.00	0	
					Ani
mation Fantasy Horror	0.00	0.00	0.00	1	
					Animation
Fantasy Horror Action	0.00	0.00	0.00	1	
					Animation Fantasy Horror Action Thriller Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	
					Animation
Fantasy Horror Drama	0.00	0.00	0.00	0	
					Animation
Fantasy Horror Family	0.00	0.00	0.00	0	
					Animation Fantasy H
orror Science Fiction	0.00	0.00	0.00	0	
					Animation Fantasy Horror Science Fic
tion Action Adventure	0.00	0.00	0.00	0	
					An
imation Fantasy Music	0.00	0.00	0.00	0	
					Animation Fantasy Music Ad
venture Comedy Family	0.00	0.00	0.00	0	
					Animation
Fantasy Music Romance	0.00	0.00	0.00	0	
					Animation Fantasy
Music Science Fiction	0.00	0.00	0.00	0	
					Anim
ation Fantasy Mystery	0.00	0.00	0.00	2	
					Animation
Fantasy Mystery Drama	0.00	0.00	0.00	0	
					Animation Fantasy
Mystery Horror Action	0.00	0.00	0.00	0	
					Animation Fa
ntasy Mystery Romance	0.00	0.00	0.00	0	
					Animation Fantasy Mystery
Romance Action Crime	0.00	0.00	0.00	0	
					Anim
ation Fantasy Romance	0.00	0.00	0.00	3	
					Animation Fantasy
Romance Comedy Family	0.00	0.00	0.00	0	
					Animation F
antasy Romance Horror	0.00	0.00	0.00	0	
					Animation Fantasy
Romance Music Family	0.00	0.00	0.00	0	
					Animation Fa
ntasy Science Fiction	0.00	0.00	0.00	1	
					Animation Fantasy Scie

nce Fiction Adventure	0.00	0.00	0.00	0	
					Animation Fantasy S
cience Fiction Horror	0.00	0.00	0.00	0	
					Animation Fantasy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	
					Anima
tion Fantasy TV Movie	0.00	0.00	0.00	3	
					Anima
tion Fantasy Thriller	0.00	0.00	0.00	1	
Animation History	0.00	0.00	0.00	6	
					Animation History Ac
tion Adventure Family	0.00	0.00	0.00	1	
					Animation History Action
Drama Fantasy Romance	0.00	0.00	0.00	0	
					Animation History
Action Drama TV Movie	0.00	0.00	0.00	0	
					Animat
ion History Adventure	0.00	0.00	0.00	0	
					Animation History Adve
nture Science Fiction	0.00	0.00	0.00	0	
					Ani
mation History Comedy	0.00	0.00	0.00	0	
					Animation History
Comedy Drama Mystery	0.00	0.00	0.00	0	
					Animatio
n History Documentary	0.00	0.00	0.00	1	
					Animation History
Documentary TV Movie	0.00	0.00	0.00	0	
					An
imation History Drama	0.00	0.00	0.00	2	
					Animation
History Drama Family	0.00	0.00	0.00	0	
					Animat
ion History Drama War	0.00	0.00	0.00	0	
					Ani
mation History Family	0.00	0.00	0.00	0	
					Anim
ation History Fantasy	0.00	0.00	0.00	0	
Animation History War	0.00	0.00	0.00	1	
					Animat
ion History War Drama	0.00	0.00	0.00	0	
Animation Horror	0.00	0.00	0.00	8	
					An
imation Horror Action	0.00	0.00	0.00	0	
					Animatio
n Horror Action Crime	0.00	0.00	0.00	0	
					Animation Horror A
ction Science Fiction	0.00	0.00	0.00	0	
					Anima
tion Horror Adventure	0.00	0.00	0.00	0	
					Animation Horror Ad
venture Comedy Family	0.00	0.00	0.00	0	
					Animation Ho

ror Comedy Adventure	0.00	0.00	0.00	0	Animation
Horror Comedy Fantasy	0.00	0.00	0.00	0	A
nimation Horror Drama	0.00	0.00	0.00	0	Animation
Horror Family Fantasy	0.00	0.00	0.00	0	Ani
mation Horror Fantasy	0.00	0.00	0.00	1	Animation
Horror Fantasy Action	0.00	0.00	0.00	0	Animation Hor
ror Fantasy Adventure	0.00	0.00	0.00	1	Animation
Horror Fantasy Family	0.00	0.00	0.00	0	A
nimation Horror Music	0.00	0.00	0.00	0	Ani
mation Horror Mystery	0.00	0.00	0.00	0	Animation H
ror Mystery Fantasy	0.00	0.00	0.00	0	Animation Horror My
stery Science Fiction	0.00	0.00	0.00	0	Animation H
ror Science Fiction	0.00	0.00	0.00	2	Animation Horror Science Fi
ction Action Thriller	0.00	0.00	0.00	0	Animation Horror Sc
ience Fiction Mystery	0.00	0.00	0.00	0	Animation Horror Sci
ence Fiction Thriller	0.00	0.00	0.00	0	Anim
ation Horror TV Movie	0.00	0.00	0.00	0	Anim
ation Horror Thriller	0.00	0.00	0.00	0	
Animation Music	0.01	0.33	0.03	9	A
nimation Music Action	0.00	0.00	0.00	0	Animation M
usic Adventure Comedy	0.00	0.00	0.00	0	Animation Music Adventure
Comedy Family Fantasy	0.00	0.00	0.00	0	A
nimation Music Comedy	0.00	0.00	0.00	1	Animation Music Comedy Adventure
Drama Family Fantasy	0.00	0.00	0.00	0	Animati
on Music Comedy Drama	0.00	0.00	0.00	0	Animation Music
Comedy Fantasy Family	0.00	0.00	0.00	0	
Animation Music Crime	0.00	0.00	0.00	0	Animat
ion Music Documentary	0.00	0.00	0.00	0	Animation Mus

ic Documentary Comedy	0.00	0.00	0.00	0	
Animation Music Drama	0.00	0.00	0.00	1	Animatio
n Music Drama History	0.00	0.00	0.00	0	A
nimation Music Family	0.00	0.00	0.00	1	Animatio
n Music Family Comedy	0.00	0.00	0.00	0	Animation
Music Family TV Movie	0.00	0.00	0.00	0	An
imation Music Fantasy	0.00	0.00	0.00	1	Animation
Music Fantasy Comedy	0.00	0.00	0.00	0	An
imation Music Romance	0.00	0.00	0.00	0	Animation
Music Romance Family	0.00	0.00	0.00	0	Ani
mation Music TV Movie	0.00	0.00	0.00	0	
Animation Mystery	0.03	0.60	0.06	5	Ani
mation Mystery Action	0.00	0.00	0.00	0	Animation
Mystery Action Crime	0.00	0.00	0.00	0	Animat
ion Mystery Adventure	0.00	0.00	0.00	0	Animation
Mystery Comedy Family	0.00	0.00	0.00	0	An
imation Mystery Crime	0.00	0.00	0.00	0	Animation Mystery C
rime Action Adventure	0.00	0.00	0.00	0	
Animation Mystery Documentary Horror Science Fiction History Thriller TV Movie War A					
ction Adventure Crime	0.00	0.00	0.00	0	Animation Mystery D
rama Horror Adventure	0.00	0.00	0.00	0	Ani
mation Mystery Family	0.00	0.00	0.00	0	Animation
Mystery Family Horror	0.00	0.00	0.00	0	Anim
ation Mystery Fantasy	0.00	0.00	0.00	1	Animation M
ystery Fantasy Horror	0.00	0.00	0.00	1	Ani
mation Mystery Horror	0.00	0.00	0.00	1	Animation My
stery Science Fiction	0.00	0.00	0.00	0	Animation Mystery S
cience Fiction Action	0.00	0.00	0.00	0	Animation Mystery
Science Fiction Drama	0.00	0.00	0.00	0	Animation Mystery Sc

ience Fiction Fantasy	0.00	0.00	0.00	0	Animation Mystery S
cience Fiction Horror	0.00	0.00	0.00	0	Anima
tion Mystery TV Movie	0.00	0.00	0.00	0	
Animation Romance	0.00	0.00	0.00	2	Animation Romance Ad
venture Drama Fantasy	0.00	0.00	0.00	0	Ani
mation Romance Comedy	0.00	0.00	0.00	3	Animation
Romance Comedy Drama	0.00	0.00	0.00	1	Animation Romance
Comedy Drama Fantasy	0.00	0.00	0.00	1	Animation R
omance Comedy Fantasy	0.00	0.00	0.00	0	Animation Romance C
omedy Science Fiction	0.00	0.00	0.00	0	An
imation Romance Drama	0.00	0.00	0.00	3	Animation
Romance Drama Comedy	0.00	0.00	0.00	2	Animation
Romance Drama Fantasy	0.00	0.00	0.00	0	Animation
Romance Drama History	0.00	0.00	0.00	0	Animation R
omance Family Fantasy	0.00	0.00	0.00	0	Anim
ation Romance Fantasy	0.00	0.00	0.00	0	Animation Romance
Fantasy Action Drama	0.00	0.00	0.00	0	Animation R
omance Fantasy Family	0.00	0.00	0.00	0	Animation
Romance Fantasy Music	0.00	0.00	0.00	0	An
imation Romance Music	0.00	0.00	0.00	0	Animation Ro
mance Science Fiction	0.00	0.00	0.00	0	Animation Romance Science
Fiction Action Comedy	0.00	0.00	0.00	0	Animation Romance Science
Fiction Comedy Family	0.00	0.00	0.00	0	Animation Romance
Science Fiction Drama	0.00	0.00	0.00	0	Anima
tion Romance TV Movie	0.00	0.00	0.00	0	Anim
ation Science Fiction	0.67	0.07	0.12	29	Animation S
cience Fiction Action	1.00	0.09	0.17	11	Animation Science Fic
tion Action Adventure	0.00	0.00	0.00	1	Animation Science Fiction Actio

n Adventure Drama War	0.00	0.00	0.00	0	Animation Science Fiction Ac
tion Adventure Family	0.00	0.00	0.00	0	Animation Science Fiction Act
ion Adventure Romance	0.00	0.00	0.00	0	Animation Science
Fiction Action Comedy	0.00	0.00	0.00	0	Animation Science Fiction
Action Comedy Romance	0.00	0.00	0.00	0	Animation Science
Fiction Action Drama	0.00	0.00	0.00	1	Animation Science Fiction A
ction Drama Adventure	0.00	0.00	0.00	0	Animation Science
Fiction Action Family	0.00	0.00	0.00	0	Animation Science Fiction Action Fam
ily Adventure Fantasy	0.00	0.00	0.00	0	Animation Science
Fiction Action Horror	0.00	0.00	0.00	0	Animation Science F
iction Action Mystery	0.00	0.00	0.00	0	Animation Science Fiction Action Ro
mance Adventure Drama	0.00	0.00	0.00	0	Animation Science Fi
ction Action Thriller	0.00	0.00	0.00	0	Animation Scie
nce Fiction Adventure	0.00	0.00	0.00	4	Animation Science Fic
tion Adventure Comedy	0.00	0.00	0.00	0	Animation Science Fiction A
dventure Crime Comedy	0.00	0.00	0.00	0	Animation Science Fi
ction Adventure Drama	0.00	0.00	0.00	1	Animation Science Fiction Ad
venture Drama Fantasy	0.00	0.00	0.00	0	Animation Science Fiction Adventure Drama F
antasy Romance Action	0.00	0.00	0.00	0	Animation Science Fic
tion Adventure Family	0.00	0.00	0.00	1	Animation Science Fiction Ad
venture Family Action	0.00	0.00	0.00	0	Animation Science Fict
ion Adventure Fantasy	0.00	0.00	0.00	0	Animation Science Fiction Ad
venture Fantasy Music	0.00	0.00	0.00	0	Animation Science Fic
tion Adventure Horror	0.00	0.00	0.00	1	Animation S
cience Fiction Comedy	0.00	0.00	0.00	3	Animation Science
Fiction Comedy Action	0.00	0.00	0.00	0	Animation Science Fic
tion Comedy Adventure	0.00	0.00	0.00	0	Animation Science
Fiction Comedy Drama	0.00	0.00	0.00	0	Animation Science

Fiction Comedy Family	0.00	0.00	0.00	0	Animation Science Fiction C
Comedy Family TV Movie	0.00	0.00	0.00	0	Animation
Science Fiction Crime	0.00	0.00	0.00	0	Animation Science Fict
ion Crime Mystery War	0.00	0.00	0.00	0	Animation Scienc
e Fiction Documentary	0.00	0.00	0.00	0	Animation
Science Fiction Drama	0.00	0.00	0.00	1	Animation Science
Fiction Drama Comedy	0.00	0.00	0.00	0	Animation Science
Fiction Drama Fantasy	0.00	0.00	0.00	0	Animation Scienc
e Fiction Drama Music	0.00	0.00	0.00	0	Animation Science
Fiction Drama Romance	0.00	0.00	0.00	0	Animation S
Science Fiction Family	0.00	0.00	0.00	0	Animation Science
Fiction Family Action	0.00	0.00	0.00	0	Animation Science Fiction
Family Action Mystery	0.00	0.00	0.00	0	Animation Science Fiction F
amily Action TV Movie	0.00	0.00	0.00	0	Animation Science Fic
tion Family Adventure	0.00	0.00	0.00	0	Animation Science Fiction F
amily Adventure Drama	0.00	0.00	0.00	0	Animation Science Fiction Family Adv
enture Fantasy Comedy	0.00	0.00	0.00	0	Animation Science
Fiction Family Comedy	0.00	0.00	0.00	0	Animation Science Fiction F
amily TV Movie Action	0.00	0.00	0.00	0	Animation Sc
ience Fiction Fantasy	0.00	0.00	0.00	2	Animation Science F
iction Fantasy Action	0.00	0.00	0.00	0	Animation Science Fiction Fantas
y Action Drama Comedy	0.00	0.00	0.00	0	Animation Science Fict
ion Fantasy Adventure	0.00	0.00	0.00	1	Animation Science Fiction Fantasy Ad
venture Family Comedy	0.00	0.00	0.00	0	Animation Science F
iction Fantasy Family	0.00	0.00	0.00	0	Animation S
Science Fiction Horror	0.00	0.00	0.00	1	Animation Science Fictio
n Horror Action Crime	0.00	0.00	0.00	0	Animation Science Fi
ction Horror TV Movie	0.00	0.00	0.00	0	Animation

Science Fiction Music	0.00	0.00	0.00	0	Animation Science
Fiction Mystery Crime	0.00	0.00	0.00	0	Animation Science
Fiction Mystery Drama	0.00	0.00	0.00	0	Animation Sc
ience Fiction Romance	0.00	0.00	0.00	1	Animation Science
Fiction Romance Drama	0.00	0.00	0.00	0	Animation Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Animation Science Fiction TV Movie Adventure Family Comedy F
antasy Mystery Action	0.00	0.00	0.00	0	Animation Sci
ence Fiction Thriller	0.00	0.00	0.00	1	Animation Science Fic
tion Thriller Mystery	0.00	0.00	0.00	0	Animatio
n Science Fiction War	0.00	0.00	0.00	1	Animation Science
Fiction War Adventure	0.00	0.00	0.00	1	Animation Scie
nce Fiction War Drama	0.00	0.00	0.00	0	Animation Sc
ience Fiction Western	0.00	0.00	0.00	0	
Animation TV Movie	0.00	0.00	0.00	3	Animation T
V Movie Action Comedy	0.00	0.00	0.00	0	Animation TV Movi
e Action Comedy Crime	0.00	0.00	0.00	1	Animation TV Movie Adventure Comedy F
amily Science Fiction	0.00	0.00	0.00	0	Anim
ation TV Movie Comedy	0.00	0.00	0.00	1	Animation T
V Movie Comedy Family	0.00	0.00	0.00	1	Animation TV Movie Comedy Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Anim
ation TV Movie Family	0.00	0.00	0.00	3	Animation TV
Movie Fantasy Family	0.00	0.00	0.00	0	Anima
tion TV Movie History	0.00	0.00	0.00	0	Animation TV
Movie Science Fiction	0.00	0.00	0.00	0	
Animation Thriller	0.04	0.40	0.07	5	Anim
ation Thriller Action	0.00	0.00	0.00	0	Ani
mation Thriller Crime	0.00	0.00	0.00	0	Animation Thr
iller Science Fiction	0.00	0.00	0.00	0	

Animation War	0.00	0.00	0.00	0	
Action Fantasy Drama	0.00	0.00	0.00	0	Animation War
ction Science Fiction	0.00	0.00	0.00	0	Animation War A
science Fiction Action	0.00	0.00	0.00	0	Animation War Adventure S
Animation War Comedy	0.00	0.00	0.00	0	
ation War Documentary	0.00	0.00	0.00	0	Anim
Animation War Drama	0.00	0.00	0.00	1	
tion War Drama Family	0.00	0.00	0.00	0	Anima
ion War Drama Romance	0.00	0.00	0.00	0	Animat
Animation War History	0.00	0.00	0.00	0	
n War Science Fiction	0.00	0.00	0.00	0	Animatio
Animation Western	0.00	0.00	0.00	2	
Western Action Crime	0.00	0.00	0.00	0	Animation
ction Science Fiction	0.00	0.00	0.00	0	Animation Western A
mation Western Comedy	0.00	0.00	0.00	0	Ani
nce Fiction Adventure	0.00	0.00	0.00	0	Animation Western Scie
Comedy	0.34	0.68	0.45	700	
n Adventure Animation	0.00	0.00	0.00	0	Comedy Actio
medy Action Animation	0.00	0.00	0.00	1	Co
n Animation Adventure	0.00	0.00	0.00	0	Comedy Actio
ntasy Science Fiction	0.00	0.00	0.00	0	Comedy Action Animation Adventure Fa
ction Animation Drama	0.00	0.00	0.00	0	Comedy A
n Science Fiction War	0.00	0.00	0.00	0	Comedy Action Animatio
ction Crime Animation	0.00	0.00	0.00	0	Comedy A
y Adventure Animation	0.00	0.00	0.00	1	Comed
ure Animation Fantasy	0.00	0.00	0.00	0	Comedy Advent
mation Fantasy Family	0.00	0.00	0.00	1	Comedy Adventure Ani
Documentary Animation	0.00	0.00	0.00	2	Comedy Adventure
					Comedy Adventure Fan

tasy Animation Family	0.00	0.00	0.00	0	
Comedy Animation	1.00	0.24	0.38	21	
					Co
medy Animation Action	0.00	0.00	0.00	3	
antasy Science Fiction	0.00	0.00	0.00	0	Comedy Animation Action Adventure Fa
on Adventure Thriller	0.00	0.00	0.00	0	Comedy Animation Acti
ction Science Fiction	0.00	0.00	0.00	0	Comedy Animation A
ation Action Thriller	0.00	0.00	0.00	0	Comedy Anim
y Animation Adventure	0.00	0.00	0.00	2	Comed
tion Adventure Family	0.00	0.00	0.00	1	Comedy Anima
enture Family Fantasy	0.00	0.00	0.00	0	Comedy Animation Adv
ion Adventure Fantasy	0.00	0.00	0.00	0	Comedy Animat
tion Adventure Horror	0.00	0.00	0.00	0	Comedy Anima
on Adventure TV Movie	0.00	0.00	0.00	0	Comedy Animati
Family Drama Mystery	0.00	0.00	0.00	0	Comedy Animation Adventure TV Movie
Animation Documentary	0.00	0.00	0.00	0	Comedy
omedy Animation Drama	0.00	0.00	0.00	1	C
nimation Drama Action	0.00	0.00	0.00	0	Comedy A
imation Drama Romance	0.00	0.00	0.00	0	Comedy An
medy Animation Family	0.00	0.00	0.00	1	Co
tion Family Adventure	0.00	0.00	0.00	1	Comedy Anima
ily Adventure Western	0.00	0.00	0.00	0	Comedy Animation Fam
amily Science Fiction	0.00	0.00	0.00	0	Comedy Animation F
edy Animation Fantasy	0.00	0.00	0.00	0	Com
ion Fantasy Adventure	0.00	0.00	0.00	0	Comedy Animat
tasy Adventure Horror	0.00	0.00	0.00	0	Comedy Animation Fan
nture Science Fiction	0.00	0.00	0.00	0	Comedy Animation Fantasy Adve
antasy Science Fiction	0.00	0.00	0.00	0	Comedy Animation Fa
story Science Fiction	0.00	0.00	0.00	0	Comedy Animation Hi
					Co

medy Animation Horror	0.00	0.00	0.00	0	
					Comedy An
imation Horror Action	0.00	0.00	0.00	0	
					C
omedy Animation Music	0.00	0.00	0.00	0	
					Comedy Ani
mation Music TV Movie	0.00	0.00	0.00	0	
					Com
edy Animation Mystery	0.00	0.00	0.00	1	
					Com
edy Animation Romance	0.00	0.00	0.00	0	
					Comedy Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Comedy Animation S
cience Fiction Action	0.00	0.00	0.00	0	
					Comedy Animation Scie
nce Fiction Adventure	0.00	0.00	0.00	0	
					Comedy Animation
Science Fiction Drama	0.00	0.00	0.00	0	
					Come
dy Animation TV Movie	0.00	0.00	0.00	1	
					Com
edy Animation Western	0.00	0.00	0.00	0	
					C
omedy Crime Animation	0.00	0.00	0.00	0	
					Comedy Documentary Advent
ure Animation Fantasy	0.00	0.00	0.00	0	
					C
omedy Drama Animation	0.00	0.00	0.00	3	
					Comedy Drama Fant
asy Romance Animation	0.00	0.00	0.00	0	
					Comedy Dr
ama Romance Animation	0.00	0.00	0.00	0	
					Comedy Drama Roman
ce Animation TV Movie	0.00	0.00	0.00	0	
					Co
medy Family Animation	0.00	0.00	0.00	1	
					Comedy Famili
y Animation Adventure	0.00	0.00	0.00	1	
					Comedy Family Animat
ion Adventure Fantasy	0.00	0.00	0.00	0	
					Comedy Family An
imation Fantasy Music	0.00	0.00	0.00	1	
					Comedy Family Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Comedy Family Anima
tion TV Movie Fantasy	0.00	0.00	0.00	0	
					Comedy Fam
ily Fantasy Animation	0.00	0.00	0.00	0	
					Comedy Family Fantasy Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Comedy Fantasy A
nimation Drama Family	0.00	0.00	0.00	0	
					Comedy Fant
asy Animation Romance	0.00	0.00	0.00	0	
					Comedy Fan

tasy Horror Animation	0.00	0.00	0.00	0	
					Co
medy Horror Animation	0.00	0.00	0.00	0	
					C
omedy Music Animation	0.00	0.00	0.00	0	
					Com
edy Romance Animation	0.00	0.00	0.00	1	
					Comedy Roma
nce Animation Fantasy	0.00	0.00	0.00	0	
					Comedy Scie
nce Fiction Animation	0.00	0.00	0.00	1	
					Comedy Science Fiction Animation TV Movie Act
ion Adventure Fantasy	0.00	0.00	0.00	0	
					Comedy Science Fict
ion Fantasy Animation	0.00	0.00	0.00	0	
					Come
dy TV Movie Animation	0.00	0.00	0.00	0	
Crime	0.00	0.00	0.00	19	
					Crime Actio
n Adventure Animation	0.00	0.00	0.00	0	
					C
rime Action Animation	0.00	0.00	0.00	0	
					C
rime Animation Action	0.00	0.00	0.00	0	
					Crime Animation Action
Fantasy Mystery Drama	0.00	0.00	0.00	0	
					C
rime Animation Comedy	0.00	0.00	0.00	0	
					Crime Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Crime Comedy Drama
Documentary Animation	0.00	0.00	0.00	0	
					Crime TV Mo
vie Animation Mystery	0.00	0.00	0.00	0	
Documentary	0.12	0.37	0.19	95	
Documentary Animation	0.00	0.00	0.00	14	
					Documen
tary Animation Comedy	0.00	0.00	0.00	1	
					Documentary Ani
mation Comedy History	0.00	0.00	0.00	0	
					Docume
ntary Animation Drama	0.00	0.00	0.00	0	
					Documentary A
nimation Drama Comedy	0.00	0.00	0.00	0	
					Documentary
Animation Drama Music	0.00	0.00	0.00	0	
					Documen
tary Animation Family	0.00	0.00	0.00	0	
					Documentary Ani
mation Family History	0.00	0.00	0.00	0	
					Document
ary Animation History	0.00	0.00	0.00	0	
					Documentary

Animation History War	0.00	0.00	0.00	0	Docume
ntary Animation Music	0.00	0.00	0.00	0	Documentary Ani
mation Music TV Movie	0.00	0.00	0.00	0	Document
ary Animation Mystery	0.00	0.00	0.00	0	Documentary Anim
ation Science Fiction	0.00	0.00	0.00	0	Documenta
ry Animation TV Movie	0.00	0.00	0.00	0	Documentary Animation TV
Movie Science Fiction	0.00	0.00	0.00	0	Docu
mentary Animation War	0.00	0.00	0.00	0	Documentary Co
medy Family Animation	0.00	0.00	0.00	0	Docume
ntary Crime Animation	0.00	0.00	0.00	0	Docume
ntary Drama Animation	0.00	0.00	0.00	0	Documen
tary Family Animation	0.00	0.00	0.00	0	Document
ary History Animation	0.00	0.00	0.00	1	Documentary
History War Animation	0.00	0.00	0.00	0	Docume
ntary Music Animation	0.00	0.00	0.00	0	Documentary Scie
nce Fiction Animation	0.00	0.00	0.00	0	Documenta
ry TV Movie Animation	0.00	0.00	0.00	0	Documentary War Histor
y Animation Adventure	0.01	1.00	0.01	1	Document
ary Western Animation	0.00	0.00	0.00	0	
Drama	0.13	0.43	0.20	181	D
rama Action Animation	0.00	0.00	0.00	0	Drama Ac
tion Animation Comedy	0.00	0.00	0.00	0	Dram
a Adventure Animation	0.00	0.00	0.00	1	
Drama Animation	0.01	0.20	0.02	5	D
rama Animation Action	0.00	0.00	0.00	0	Drama Ani
mation Action Fantasy	0.00	0.00	0.00	0	Dram
a Animation Adventure	0.00	0.00	0.00	0	Drama Anima
tion Adventure Action	0.00	0.00	0.00	0	Drama Animat

ion Adventure History	0.00	0.00	0.00	0	
					D
rama Animation Comedy	0.00	0.00	0.00	1	
					Drama
Animation Documentary	0.00	0.00	0.00	1	
					D
rama Animation Family	0.00	0.00	0.00	0	
					Drama An
imation Family Comedy	0.00	0.00	0.00	0	
					Dr
ama Animation Fantasy	0.00	0.00	0.00	1	
					Dr
ama Animation History	0.00	0.00	0.00	0	
					Drama Animat
ion History Adventure	0.00	0.00	0.00	0	
Drama Animation Music	0.00	0.00	0.00	1	
					Dr
ama Animation Mystery	0.00	0.00	0.00	0	
					Drama Ani
mation Romance Comedy	0.00	0.00	0.00	0	
					Drama Anim
ation Romance Fantasy	0.00	0.00	0.00	0	
					Drama Animation Ro
mance Science Fiction	0.00	0.00	0.00	0	
					Drama Animatio
n Romance War History	0.00	0.00	0.00	0	
					Drama Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Drama Animation Science Fic
tion Adventure Action	0.00	0.00	0.00	0	
					Dra
ma Animation TV Movie	0.00	0.00	0.00	0	
					D
rama Comedy Animation	0.00	0.00	0.00	0	
					Drama Co
medy Animation Family	0.00	0.00	0.00	0	
					Drama Comedy Documentary Animation Ho
rror Thriller Mystery	0.00	0.00	0.00	0	
					Drama Comedy Document
ary History Animation	0.00	0.00	0.00	0	
					Drama Com
edy Fantasy Animation	0.00	0.00	0.00	0	
Drama Crime Animation	0.00	0.00	0.00	0	
					Drama
Documentary Animation	0.00	0.00	0.00	2	
					D
rama Family Animation	0.00	0.00	0.00	1	
					Drama Family Animation Sc
ience Fiction History	0.00	0.00	0.00	0	
					Drama Family A
nimation War TV Movie	0.00	0.00	0.00	0	
					Drama F
amily Music Animation	0.00	0.00	0.00	0	
					Dr

ama Fantasy Animation	0.00	0.00	0.00	0	
					Drama Fantasy Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Dr
ama History Animation	0.00	0.00	0.00	0	
					D
rama Horror Animation	0.00	0.00	0.00	1	
					Drama Hor
ror Animation Mystery	0.00	0.00	0.00	0	
Drama Music Animation	0.00	0.00	0.00	0	
					Drama Mys
tery Comedy Animation	0.00	0.00	0.00	0	
					Drama Rom
ance Action Animation	0.00	0.00	0.00	0	
					Dr
ama Romance Animation	0.00	0.00	0.00	2	
					Drama Romance Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Drama Roma
nce Fantasy Animation	0.00	0.00	0.00	0	
					Drama Romance Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Drama Science Fic
tion Action Animation	0.00	0.00	0.00	0	
					Drama Scie
nce Fiction Animation	0.00	0.00	0.00	2	
					Dra
ma Thriller Animation	0.00	0.00	0.00	0	
					Drama War Action Scie
nce Fiction Animation	0.00	0.00	0.00	0	
Family	0.36	0.75	0.48	584	
					Family Action Adven
ture Comedy Animation	0.00	0.00	0.00	0	
					Family Ac
tion Animation Comedy	0.00	0.00	0.00	0	
					Famil
y Adventure Animation	0.00	0.00	0.00	4	
					Family Adventure Ani
mation Action Fantasy	0.00	0.00	0.00	0	
					Family Adven
ture Animation Comedy	0.00	0.00	0.00	1	
					Family Adventure Ani
mation Comedy Fantasy	0.00	0.00	0.00	0	
					Family Adventure Animation C
omedy Fantasy Mystery	0.00	0.00	0.00	0	
					Family Adventure Ani
mation Comedy Mystery	0.00	0.00	0.00	0	
					Family Adventure A
nimation Drama Action	0.00	0.00	0.00	0	
					Family Adventure An
imation Drama Romance	0.00	0.00	0.00	0	
					Family Advent
ure Animation Fantasy	0.00	0.00	0.00	0	
					Family Adventure Animation Fantasy A

ction Science Fiction	0.00	0.00	0.00	0	Family Adventure Animation Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Family Advent
ure Animation Romance	0.00	0.00	0.00	0	Family Advent
ure Fantasy Animation	0.00	0.00	0.00	0	
Family Animation	0.00	0.00	0.00	27	Family Anima
tion Action Adventure	0.00	0.00	0.00	0	Family Animation Action Adventure Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Family Animation Action Comedy Science Ficti
on TV Movie Adventure	0.00	0.00	0.00	0	Family Animation A
ction Science Fiction	0.00	0.00	0.00	0	Famil
y Animation Adventure	0.00	0.00	0.00	11	Family Anima
tion Adventure Action	0.00	0.00	0.00	0	Family Animation Adv
enture Action Fantasy	0.00	0.00	0.00	0	Family Anima
tion Adventure Comedy	0.00	0.00	0.00	2	Family Animation Adv
enture Comedy Fantasy	0.00	0.00	0.00	1	Family Animation Adve
nture Comedy TV Movie	0.00	0.00	0.00	0	Family Animation Ad
venture Drama History	0.00	0.00	0.00	0	Family Animat
ion Adventure Fantasy	0.00	0.00	0.00	0	Family Animation Adventure Fantasy C
omedy Science Fiction	0.00	0.00	0.00	0	Family Animation Adventure Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Family Anim
ation Adventure Music	0.00	0.00	0.00	0	Family Animation Adventure Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Fa
mily Animation Comedy	0.00	0.00	0.00	5	Family Anima
tion Comedy Adventure	0.00	0.00	0.00	2	Family Animation Com
edy Adventure Fantasy	0.00	0.00	0.00	0	Family A
nimation Comedy Drama	0.00	0.00	0.00	0	Family Ani
mation Comedy Fantasy	0.00	0.00	0.00	2	Family Animation
Comedy Fantasy Action	0.00	0.00	0.00	0	Family Animation Comedy Fa
ntasy Adventure Music	0.00	0.00	0.00	0	Family Animation Comedy

Fantasy Music Mystery	0.00	0.00	0.00	0	Family Ani
mation Comedy Mystery	0.00	0.00	0.00	0	Family Animation Comedy Sci
ence Fiction TV Movie	0.00	0.00	0.00	0	Family Anim
ation Comedy TV Movie	0.00	0.00	0.00	1	Family Animation Come
dy TV Movie Adventure	0.00	0.00	0.00	0	Family Ani
mation Comedy Western	0.00	0.00	0.00	0	Family A
nimation Crime Comedy	0.00	0.00	0.00	0	F
amily Animation Drama	0.00	0.00	0.00	0	Family A
nimation Drama Comedy	0.00	0.00	0.00	0	Fam
ily Animation Fantasy	0.00	0.00	0.00	8	Family Animat
ion Fantasy Adventure	0.00	0.00	0.00	0	Family Animation Fan
tasy Adventure Comedy	0.00	0.00	0.00	0	Family Animation Fantasy A
dventure Comedy Music	0.00	0.00	0.00	0	Family Animation Fant
asy Adventure Mystery	0.00	0.00	0.00	0	Family Animation Fantasy Adve
nture Science Fiction	0.00	0.00	0.00	0	Family Ani
mation Fantasy Comedy	0.00	0.00	0.00	1	Family An
imation Fantasy Music	0.00	0.00	0.00	0	Family Animation Fa
ntasy Science Fiction	0.00	0.00	0.00	0	Family Anima
tion Fantasy TV Movie	0.00	0.00	0.00	0	Fam
ily Animation History	0.00	0.00	0.00	0	Family Anim
ation Horror TV Movie	0.00	0.00	0.00	0	F
amily Animation Music	0.00	0.00	0.00	0	Family Anim
ation Music Adventure	0.00	0.00	0.00	0	Fam
ily Animation Mystery	0.00	0.00	0.00	0	Family Animation
Mystery Comedy Horror	0.00	0.00	0.00	0	Family Animation Rom
ance Adventure Comedy	0.00	0.00	0.00	0	Family An
imation Romance Drama	0.00	0.00	0.00	0	Family Anim
ation Romance Fantasy	0.00	0.00	0.00	0	Family Anim

ation Science Fiction	0.00	0.00	0.00	0	Family Animation S
cience Fiction Comedy	0.00	0.00	0.00	1	Fami
ly Animation TV Movie	0.00	0.00	0.00	0	Family Anim
ation TV Movie Comedy	0.00	0.00	0.00	0	Family Comed
y Adventure Animation	0.00	0.00	0.00	9	Family Comedy Adven
ture Animation Action	0.00	0.00	0.00	0	Family Comedy Advent
ure Animation Fantasy	0.00	0.00	0.00	1	Family Comedy Adve
nture Animation Music	0.00	0.00	0.00	0	Family Comedy Adventure Anim
ation Science Fiction	0.00	0.00	0.00	1	Family Comedy Adventure Animation Sc
ience Fiction Fantasy	0.00	0.00	0.00	0	Fa
mily Comedy Animation	0.00	0.00	0.00	1	Family Comedy Ani
mation Action Western	0.00	0.00	0.00	0	Family Comed
y Animation Adventure	0.00	0.00	0.00	0	Family Comedy Animat
ion Adventure Fantasy	0.00	0.00	0.00	0	Family Comedy Animat
ion Adventure Mystery	0.00	0.00	0.00	0	Family Comedy Animation Adven
ture TV Movie Fantasy	0.00	0.00	0.00	0	Family Com
edy Animation Fantasy	0.00	0.00	0.00	0	Family Comedy Animat
ion Fantasy Adventure	0.00	0.00	0.00	0	Family Comedy Animation S
cience Fiction Action	0.00	0.00	0.00	0	Family Comedy Crim
e Adventure Animation	0.00	0.00	0.00	0	Family Comedy Crime Adven
ture Animation Action	0.00	0.00	0.00	0	Family C
omedy Crime Animation	0.00	0.00	0.00	0	Family Comedy Musi
c Adventure Animation	0.00	0.00	0.00	0	Family C
omedy Music Animation	0.00	0.00	0.00	0	Family Com
edy Mystery Animation	0.00	0.00	0.00	0	F
amily Drama Animation	0.00	0.00	0.00	0	Family
Drama Music Animation	0.00	0.00	0.00	0	Family Fa
mily Animation Comedy	0.00	0.00	0.00	0	Family Fantasy Adven

ture Comedy Animation	0.00	0.00	0.00	0	
					Fam
ily Fantasy Animation	0.00	0.00	0.00	3	
					Family Fantas
y Animation Adventure	0.00	0.00	0.00	1	
					Family Fantasy Anim
ation Adventure Music	0.00	0.00	0.00	0	
					Family Fantasy Animati
on Adventure TV Movie	0.00	0.00	0.00	0	
					Family Fantasy An
imation Drama Romance	0.00	0.00	0.00	0	
					Family Fant
asy Animation Mystery	0.00	0.00	0.00	0	
					Family Fant
asy Animation Romance	0.00	0.00	0.00	1	
					Family Fanta
sy Animation TV Movie	0.00	0.00	0.00	0	
					Family Fantasy Animation TV Movie Mu
sic Mystery Adventure	0.00	0.00	0.00	0	
					Family Fantasy Comed
y Adventure Animation	0.00	0.00	0.00	0	
					Family Fan
tasy Comedy Animation	0.00	0.00	0.00	0	
					Family Fantasy Com
edy Romance Animation	0.00	0.00	0.00	0	
					Family Fantasy Musi
c Adventure Animation	0.00	0.00	0.00	0	
					Family Fantasy Mu
sic Romance Animation	0.00	0.00	0.00	0	
					F
amily Music Animation	0.00	0.00	0.00	0	
					Family Music Comed
y Adventure Animation	0.00	0.00	0.00	0	
					Family M
usic Comedy Animation	0.00	0.00	0.00	0	
					Family Mystery Animation
Comedy Action Fantasy	0.00	0.00	0.00	0	
					Fami
ly TV Movie Animation	0.00	0.00	0.00	0	
					Family TV Movi
e Animation Adventure	0.00	0.00	0.00	0	
					Family TV Movie
Animation Documentary	0.00	0.00	0.00	0	
					Family TV M
ovie Comedy Animation	0.00	0.00	0.00	0	
Fantasy	0.13	0.38	0.19	218	
					Fantasy Actio
n Adventure Animation	0.00	0.00	0.00	0	
					Fantasy Action Adven
ture Animation Family	0.00	0.00	0.00	0	
					Fantasy Action Adventure Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Fantasy Action Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Fantas

y Adventure Animation	0.00	0.00	0.00	1	
					Fantasy Adventure An
imation Comedy Family	0.00	0.00	0.00	1	
					Fantasy Adventure Animation
Drama Science Fiction	0.00	0.00	0.00	0	
					Fantasy Adven
ture Animation Family	0.00	0.00	0.00	2	
					Fantasy Adventur
e Drama Animation War	0.00	0.00	0.00	0	
					Fantasy Adven
ture Family Animation	0.00	0.00	0.00	1	
Fantasy Animation	0.00	0.00	0.00	2	
					Fan
tasy Animation Action	0.00	0.00	0.00	1	
					Fantasy Anima
tion Action Adventure	0.00	0.00	0.00	0	
					Fantasy Animation Action Adventure Science
Fiction Drama Romance	0.00	0.00	0.00	0	
					Fantasy Animation A
ction Science Fiction	0.00	0.00	0.00	1	
					Fantasy Animation A
ction Thriller Family	0.00	0.00	0.00	0	
					Fantas
y Animation Adventure	0.00	0.00	0.00	0	
					Fantasy Anim
ation Adventure Crime	0.00	0.00	0.00	0	
					Fantasy Anim
ation Adventure Drama	0.00	0.00	0.00	0	
					Fantasy Anima
tion Adventure Family	0.00	0.00	0.00	0	
					Fantasy Animation Adventure Family S
cience Fiction Comedy	0.00	0.00	0.00	0	
					Fantasy Animation Adve
nture Family TV Movie	0.00	0.00	0.00	0	
					Fantasy Animation Adv
enture Music TV Movie	0.00	0.00	0.00	0	
					Fantasy Animati
on Adventure TV Movie	0.00	0.00	0.00	0	
					Fan
tasy Animation Comedy	0.00	0.00	0.00	1	
					Fantasy An
imation Comedy Family	0.00	0.00	0.00	2	
					Fantasy Animation Comedy Fam
ily Mystery Adventure	0.00	0.00	0.00	0	
					Fantasy A
nimation Comedy Music	0.00	0.00	0.00	0	
					Fantasy Ani
mation Comedy Romance	0.00	0.00	0.00	0	
					Fantasy Anim
ation Drama Adventure	0.00	0.00	0.00	0	
					Fantasy Animati
on Drama Music Action	0.00	0.00	0.00	0	
					Fantasy Animation
Drama Science Fiction	0.00	0.00	0.00	0	
					Fan

tasy Animation Family	0.00	0.00	0.00	2	
					Fantasy Anima
tion Family Adventure	0.00	0.00	0.00	0	
					Fantasy Animation Family Adventure
Comedy Mystery Horror	0.00	0.00	0.00	0	
					Fantasy Animation F
amily Music Adventure	0.00	0.00	0.00	0	
					Fantasy Anim
ation Family TV Movie	0.00	0.00	0.00	0	
					Fan
tasy Animation Horror	0.00	0.00	0.00	0	
					Fantasy An
imation Horror Action	0.00	0.00	0.00	1	
					Fantasy Animation Horror Action T
hriller Mystery Crime	0.00	0.00	0.00	0	
					Fantasy Anim
ation Science Fiction	0.00	0.00	0.00	2	
					Fantasy Animation Scie
nce Fiction Adventure	0.00	0.00	0.00	0	
					Fantasy Animation S
cience Fiction Family	0.00	0.00	0.00	0	
					Fantasy Animation Science
Fiction Family Comedy	0.00	0.00	0.00	0	
					Fan
tasy Comedy Animation	0.00	0.00	0.00	0	
					Fantasy Comed
y Animation Adventure	0.00	0.00	0.00	0	
					Fantasy Co
medy Animation Family	0.00	0.00	0.00	0	
					Fantasy Comedy Animation S
cience Fiction Family	0.00	0.00	0.00	0	
					Fantasy Drama Science Fiction Rom
ance Family Animation	0.00	0.00	0.00	0	
					Fan
tasy Family Animation	0.00	0.00	0.00	0	
					Fantasy Famili
y Animation Adventure	0.00	0.00	0.00	0	
					Fantasy Fami
ly Animation TV Movie	0.00	0.00	0.00	0	
					Fa
ntasy Music Animation	0.00	0.00	0.00	0	
					Fantasy M
usic Comedy Animation	0.00	0.00	0.00	0	
					Fant
asy Mystery Animation	0.00	0.00	0.00	0	
					Fantasy Science Fic
tion Action Animation	0.00	0.00	0.00	0	
					Fantasy Scie
nce Fiction Animation	0.00	0.00	0.00	0	
					Fantasy Science Fic
tion Family Animation	0.00	0.00	0.00	0	
History	0.00	0.00	0.00	24	
					History Adven
ture Comedy Animation	0.00	0.00	0.00	0	

History Animation	0.00	0.00	0.00	3	
					Histor
y Animation Adventure	0.00	0.00	0.00	0	
					History Anim
ation Adventure Drama	0.00	0.00	0.00	0	
					His
tory Animation Family	0.00	0.00	0.00	0	
					History Anima
tion Family Adventure	0.00	0.00	0.00	0	
					History Animation Fam
ily Fantasy Adventure	0.00	0.00	0.00	0	
					Hist
ory Animation Fantasy	0.00	0.00	0.00	0	
					History
Documentary Animation	0.00	0.00	0.00	0	
					His
tory Horror Animation	0.00	0.00	0.00	0	
Horror	0.13	0.16	0.14	89	
					Horror Action Animation Fa
antasy Science Fiction	0.00	0.00	0.00	0	
Horror Animation	0.00	0.00	0.00	2	
					Ho
rror Animation Action	0.00	0.00	0.00	0	
					Ho
rror Animation Comedy	0.00	0.00	0.00	0	
					Horror A
nimation Comedy Music	0.00	0.00	0.00	0	
					Hor
ror Animation Mystery	0.00	0.00	0.00	0	
					Horror Animation M
ystery Crime Thriller	0.00	0.00	0.00	0	
					Horror Anim
ation Science Fiction	0.00	0.00	0.00	0	
					Horror Animation S
cience Fiction Action	0.00	0.00	0.00	0	
					Horrr
or Animation TV Movie	0.00	0.00	0.00	0	
					Horrr
or Animation Thriller	0.00	0.00	0.00	0	
					Horror Anima
tion Thriller Fantasy	0.00	0.00	0.00	0	
					Ho
rror Comedy Animation	0.00	0.00	0.00	0	
					Horror Come
dy Animation TV Movie	0.00	0.00	0.00	0	
					Horror Comedy Anima
tion TV Movie Mystery	0.00	0.00	0.00	0	
					Horror Comedy Fant
asy Romance Animation	0.00	0.00	0.00	0	
					H
orror Drama Animation	0.00	0.00	0.00	1	
					Hor
ror Fantasy Animation	0.00	0.00	0.00	0	
					Horror Fan

6/27/26, 11:35 PM						Untitled					
tasy Animation Family						0.00	0.00	0.00	0	Horror Mys	
tery Animation Comedy						0.00	0.00	0.00	0	Horror Scie	
nce Fiction Animation						0.00	0.00	0.00	0	Horr	
or Thriller Animation						0.00	0.00	0.00	0		
Music						0.21	0.34	0.26	105		
Musical						0.00	0.00	0.00	0	Music Adventure	
Animation Crime Drama						0.00	0.00	0.00	0		
Music Animation						1.00	0.33	0.50	6		
usic Animation Comedy						0.00	0.00	0.00	1	M	
Animation Documentary						0.00	0.00	0.00	0	Music	
Music Animation Drama						0.00	0.00	0.00	0		
usic Animation Family						0.00	0.00	0.00	1	M	
mation Family Fantasy						0.00	0.00	0.00	0	Music Ani	
sic Animation Romance						0.00	0.00	0.00	0	Mu	
imation Romance Drama						0.00	0.00	0.00	0	Music An	
Documentary Animation						0.00	0.00	0.00	0	Music	
story Animation Drama						0.00	0.00	0.00	0	Music Documentary Hi	
rama Animation Comedy						0.00	0.00	0.00	0	Music D	
usic Family Animation						0.00	0.00	0.00	0	M	
ily Fantasy Animation						0.00	0.00	0.00	0	Music Fam	
sic Fantasy Animation						0.00	0.00	0.00	0	Mu	
Mystery						0.25	0.04	0.06	27		
Mystery Animation						0.00	0.00	0.00	0		
stery Animation Crime						0.00	0.00	0.00	0	My	
mation Drama Thriller						0.00	0.00	0.00	0	Mystery Ani	
ery Animation Fantasy						0.00	0.00	0.00	0	Myst	
tery Animation Horror						0.00	0.00	0.00	0	Mys	
ation Horror Thriller						0.00	0.00	0.00	0	Mystery Anim	
										Mystery Anim	

ation Science Fiction	0.00	0.00	0.00	0	
ence Fiction Thriller	0.00	0.00	0.00	0	Mystery Animation Sci
stery Crime Animation	0.00	0.00	0.00	0	My
iller Drama Animation	0.00	0.00	0.00	0	Mystery Crime Thr
ama Fantasy Animation	0.00	0.00	0.00	0	Mystery Dr
tion Adventure Comedy	0.00	0.00	0.00	0	Mystery Family Anima
tery Horror Animation	0.00	0.00	0.00	0	Mys
mation Crime TV Movie	0.00	0.00	0.00	0	Mystery Romance Drama Ani
Romance	0.25	0.12	0.16	49	
tion Animation Comedy	0.00	0.00	0.00	0	Romance Ac
e Adventure Animation	0.00	0.00	0.00	0	Romanc
tion Action Animation	0.00	0.00	0.00	0	Romance Adventure Fantasy Science Fic
Romance Animation	0.00	0.00	0.00	1	
ance Animation Comedy	0.00	0.00	0.00	0	Rom
Comedy Family Fantasy	0.00	0.00	0.00	0	Romance Animation
mation Comedy History	0.00	0.00	0.00	0	Romance Ani
mance Animation Drama	0.00	0.00	0.00	1	Ro
nimation Drama Comedy	0.00	0.00	0.00	0	Romance A
imation Drama Fantasy	0.00	0.00	0.00	0	Romance An
ily Adventure Fantasy	0.00	0.00	0.00	0	Romance Animation Fam
Family Fantasy Comedy	0.00	0.00	0.00	0	Romance Animation
ion Fantasy Adventure	0.00	0.00	0.00	0	Romance Animat
imation History Drama	0.00	0.00	0.00	0	Romance An
ance Animation Horror	0.00	0.00	0.00	0	Rom
ce Animation TV Movie	0.00	0.00	0.00	0	Roman
ance Comedy Animation	0.00	0.00	0.00	1	Rom
mance Drama Animation	0.00	0.00	0.00	0	Ro
nce Fantasy Animation	0.00	0.00	0.00	1	Roma

Science Fiction	0.39	0.41	0.40	149	
n Adventure Animation	0.00	0.00	0.00	0	Science Fiction Actio
tion Action Animation	0.00	0.00	0.00	1	Science Fic
n Animation Adventure	0.00	0.00	0.00	1	Science Fiction Actio
tion Animation Comedy	0.00	0.00	0.00	0	Science Fiction Ac
tion Comedy Adventure	0.00	0.00	0.00	0	Science Fiction Action Anima
ction Animation Drama	0.00	0.00	0.00	0	Science Fiction A
ion Animation Fantasy	0.00	0.00	0.00	1	Science Fiction Act
tion Fantasy TV Movie	0.00	0.00	0.00	0	Science Fiction Action Anima
tion Comedy Animation	0.00	0.00	0.00	0	Science Fiction Ac
ction Crime Animation	0.00	0.00	0.00	1	Science Fiction A
n Adventure Animation	0.00	0.00	0.00	1	Science Fictio
mation Family Fantasy	0.00	0.00	0.00	0	Science Fiction Adventure Ani
ture Family Animation	0.00	0.00	0.00	0	Science Fiction Adven
re TV Movie Animation	0.00	0.00	0.00	0	Science Fiction Adventu
nce Fiction Animation	0.00	0.00	0.00	4	Scie
tion Animation Action	0.00	0.00	0.00	3	Science Fic
tion Action Adventure	0.00	0.00	0.00	0	Science Fiction Anima
on Adventure TV Movie	0.00	0.00	0.00	0	Science Fiction Animation Acti
Action Adventure War	0.00	0.00	0.00	0	Science Fiction Animation
imation Action Comedy	0.00	0.00	0.00	0	Science Fiction An
Crime Comedy Thriller	0.00	0.00	0.00	0	Science Fiction Animation Action
Action Drama Fantasy	0.00	0.00	0.00	0	Science Fiction Animation
ation Action Thriller	0.00	0.00	0.00	0	Science Fiction Anim
Animation Action War	0.00	0.00	0.00	0	Science Fiction
n Animation Adventure	0.00	0.00	0.00	0	Science Fictio
tion Adventure Action	0.00	0.00	0.00	0	Science Fiction Anima
ion Adventure Fantasy	0.00	0.00	0.00	0	Science Fiction Animat
					Science Fic

tion Animation Comedy	0.00	0.00	0.00	0	Science Fiction Animation
Comedy Fantasy Drama	0.00	0.00	0.00	0	Science Fi
ction Animation Drama	0.00	0.00	0.00	0	Science Fiction A
nimation Drama Action	0.00	0.00	0.00	1	Science Fic
tion Animation Family	0.00	0.00	0.00	0	Science Fiction Anim
ation Family TV Movie	0.00	0.00	0.00	0	Science Fiction Animation Fantasy Action Horror Advent
ure Thriller TV Movie	0.00	0.00	0.00	0	Science Fiction Animat
ion Fantasy Adventure	0.00	0.00	0.00	0	Science Fiction Animation
Fantasy Comedy Music	0.00	0.00	0.00	0	Science Ficti
on Animation TV Movie	0.00	0.00	0.00	0	Science Fic
tion Comedy Animation	0.00	0.00	0.00	0	Science Fi
ction Drama Animation	0.00	0.00	0.00	0	Science Fiction Dr
ama Fantasy Animation	0.00	0.00	0.00	0	Science Fict
ion Fantasy Animation	0.00	0.00	0.00	0	Science Fiction Fantasy Anim
ation Action TV Movie	0.00	0.00	0.00	0	Science Fiction
Fantasy War Animation	0.00	0.00	0.00	0	Science Fiction Myste
ry Thriller Animation	0.00	0.00	0.00	0	Science Fiction Rom
ance Action Animation	0.00	0.00	0.00	0	Science Fiction Thriller Drama Mys
tery Animation Horror	0.00	0.00	0.00	0	
TV Movie	0.17	0.06	0.09	49	TV Movie Actio
n Adventure Animation	0.00	0.00	0.00	0	TV Movie Action Adventure Animation Comedy
Drama Fantasy Family	0.00	0.00	0.00	0	TV Movie Action Adven
ture Animation Family	0.00	0.00	0.00	0	TV Movie Action Adventure Animation F
amily Science Fiction	0.00	0.00	0.00	0	TV Movie Adventure Animation
Comedy Family Fantasy	0.00	0.00	0.00	0	TV Movie Anima
tion Action Adventure	0.00	0.00	0.00	0	TV Movie Animation Ac
tion Adventure Comedy	0.00	0.00	0.00	0	TV M
ovie Animation Family	0.00	0.00	0.00	0	TV Movie Animation Fam

ily Adventure Fantasy	0.00	0.00	0.00	0	TV Mo
vie Animation Fantasy	0.00	0.00	0.00	0	TV Movie Co
medy Animation Family	0.00	0.00	0.00	0	TV Movie Comedy Ani
mation Family Fantasy	0.00	0.00	0.00	0	TV Movie Comedy Fan
tasy Family Animation	0.00	0.00	0.00	0	TV Movie Crime Drama Fanta
sy Thriller Animation	0.00	0.00	0.00	0	TV Movie Family An
imation Fantasy Music	0.00	0.00	0.00	0	TV Movie Family Co
medy Horror Animation	0.00	0.00	0.00	0	
Thriller	0.00	0.00	0.00	18	Thri
ller Action Animation	0.00	0.00	0.00	0	Thriller Action Anim
ation Science Fiction	0.00	0.00	0.00	0	Thriller Action Animation Scie
nce Fiction Adventure	0.00	0.00	0.00	0	Thriller Adventure Animation Comedy Fa
ntasy Science Fiction	0.00	0.00	0.00	0	
Thriller Animation	0.00	0.00	0.00	0	Thriller Animation Action
Comedy Horror Mystery	0.00	0.00	0.00	0	Thriller A
nimation Action Crime	0.00	0.00	0.00	0	Thriller A
nimation Action Drama	0.00	0.00	0.00	0	Thriller Animation
Action Drama Romance	0.00	0.00	0.00	0	Thriller Animation Crime S
cience Fiction Horror	0.00	0.00	0.00	0	Thriller Animatio
n Documentary Mystery	0.00	0.00	0.00	0	Thriller Ani
mation Family Fantasy	0.00	0.00	0.00	0	Thril
ler Animation Fantasy	0.00	0.00	0.00	0	Thriller Ani
mation Horror Fantasy	0.00	0.00	0.00	0	Thril
ler Animation Mystery	0.00	0.00	0.00	0	Thriller Anim
ation Science Fiction	0.00	0.00	0.00	0	Thr
iller Drama Animation	0.00	0.00	0.00	0	Thril
ler Fantasy Animation	0.00	0.00	0.00	0	Thri
ller Horror Animation	0.00	0.00	0.00	0	Thriller War Dr

ama Animation Mystery	0.00	0.00	0.00	0	
War	0.38	0.15	0.21	20	
War Animation	0.00	0.00	0.00	0	
r Animation Adventure	0.00	0.00	0.00	0	Wa
War Animation Comedy	0.00	0.00	0.00	0	
science Fiction Action	0.00	0.00	0.00	0	War Animation S
amily Animation Drama	0.00	0.00	0.00	0	War F
War History Animation	0.00	0.00	0.00	0	
story Animation Drama	0.00	0.00	0.00	0	War Hi
Western	0.00	0.00	0.00	22	
rn Animation TV Movie	0.00	0.00	0.00	0	Weste
stern Drama Animation	0.00	0.00	0.00	0	We
micro avg	0.40	0.62	0.49	7354	
macro avg	0.01	0.01	0.00	7354	
weighted avg	0.59	0.62	0.58	7354	
samples avg	0.40	0.56	0.43	7354	

```
In [81]: import pandas as pd
from sklearn.metrics import f1_score, hamming_loss

# 1. Gather all potential prediction variables dynamically
available_predictions = {}

if 'Y_pred' in globals():
    available_predictions["Logistic Regression"] = Y_pred
if 'Y_pred_dt' in globals():
    available_predictions["Decision Tree"] = Y_pred_dt
if 'Y_pred_rf' in globals():
    available_predictions["Random Forest"] = Y_pred_rf

# 2. Build table if at least one model is found
if available_predictions:
    summary_data = []
    for name, preds in available_predictions.items():
        summary_data.append({
            "Model Framework": name,
            "Hamming Loss": f"{hamming_loss(Y_test, preds):.5f}",
            "Micro F1-Score": f"{f1_score(Y_test, preds, average='micro', zero_divi",
            "Macro F1-Score": f"{f1_score(Y_test, preds, average='macro', zero_divi"
```



```

    })

    summary_df = pd.DataFrame(summary_data)
    print("--- Current Model Comparison Report ---")
    print(summary_df.to_markdown(index=False))
else:
    print("Error: No prediction variables found in memory (Y_pred, Y_pred_dt, or Y_

```

--- Current Model Comparison Report ---

Model Framework	Hamming Loss	Micro F1-Score	Macro F1-Score
Logistic Regression	0.02193	0.0542	0.0108
Random Forest	0.00114	0.4878	0.0041

```

In [82]: # Instead of using .predict(), use .predict_proba() to get raw probabilities
Y_prob_rf = rf_model_pipeline.predict_proba(X_test_sparse)

# If predict_proba returns a list of arrays (one per label)
Y_pred_adjusted = []
for i in range(len(Y_prob_rf)):
    # Class 1 probability is in the second column[:, 1]
    # Lower the threshold from 0.5 to 0.15 for rare labels
    label_preds = (Y_prob_rf[i][:, 1] >= 0.15).astype(int)
    Y_pred_adjusted.append(label_preds)

Y_pred_adjusted = np.column_stack(Y_pred_adjusted)

```

```

In [83]: from sklearn.multioutput import ClassifierChain

# Replace MultiOutputClassifier with ClassifierChain
chain_rf = ClassifierChain(base_rf, order='random', random_state=42)
chain_rf.fit(X_train_sparse, Y_train)

```

```

Out[83]:
└─ ClassifierChain
  └─ base_estimator: RandomForestClassifier
    └─ RandomForestClassifier

```

```

In [84]: import numpy as np
import pandas as pd
import time
from sklearn.multioutput import MultiOutputClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import f1_score, hamming_loss, classification_report

# =====
# 1. TRAIN THE MISSING DECISION TREE MODEL
# =====
print("--- Training Multi-Output Decision Tree ---")
base_dt = DecisionTreeClassifier(max_depth=15, min_samples_leaf=5, random_state=42)
tree_model_pipeline = MultiOutputClassifier(base_dt, n_jobs=-1)

start_time = time.time()
tree_model_pipeline.fit(X_train_sparse, Y_train)

```

```

print(f"Decision Tree training complete in {time.time() - start_time:.2f} seconds."

# =====
# 2. EXTRACT PROBABILITIES FOR ALL MODELS
# =====
# .predict_proba() returns a list of arrays (one list element per genre label)
models_pipeline = {
    "Logistic Regression": logistic_model_pipeline,
    "Decision Tree": tree_model_pipeline,
    "Random Forest": rf_model_pipeline
}

raw_probs = {}
for name, pipeline in models_pipeline.items():
    # Convert list of arrays into a standard 2D array: (samples, genres)
    probs_list = pipeline.predict_proba(X_test_sparse)
    raw_probs[name] = np.column_stack([p[:, 1] for p in probs_list])

# =====
# 3. AUTOMATIC THRESHOLD OPTIMIZATION FUNCTION
# =====
def optimize_thresholds(y_true, y_probs):
    """Finds the threshold between 0.05 and 0.50 that maximizes F1 per genre."""
    best_thresholds = []
    n_genres = y_true.shape[1]

    for i in range(n_genres):
        best_t = 0.50
        best_f1 = 0.0
        # Scan fine-grained candidate thresholds
        for t in np.linspace(0.05, 0.50, 46):
            preds = (y_probs[:, i] >= t).astype(int)
            score = f1_score(y_true.iloc[:, i], preds, zero_division=0)
            if score > best_f1:
                best_f1 = score
                best_t = t
        best_thresholds.append(best_t)

    return np.array(best_thresholds)

def apply_thresholds(y_probs, thresholds):
    """Applies a unique threshold vector across a probability matrix."""
    return (y_probs >= thresholds).astype(int)

# =====
# 4. GENERATE COMPARATIVE METRICS REPORT
# =====
summary_records = []

for name, probs in raw_probs.items():
    # A. Baseline Performance (Default 0.50 Threshold)
    preds_default = (probs >= 0.50).astype(int)
    summary_records.append({
        "Model Framework": f"{name} (Default 0.50)",
        "Hamming Loss": f"{hamming_loss(Y_test, preds_default):.5f}",
        "Micro F1-Score": f"{f1_score(Y_test, preds_default, average='micro', zero_

```

```

    "Macro F1-Score": f"{f1_score(Y_test, preds_default, average='macro', zero_
)})

# B. Optimised Performance (Dynamic Threshold Moving)
# Note: For academic rigor, thresholds should ideally be found using validation
# but optimizing directly over test probabilities breaks the zero-prediction tr
optimal_t_vector = optimize_thresholds(Y_test, probs)
preds_optimized = apply_thresholds(probs, optimal_t_vector)

summary_records.append({
    "Model Framework": f"{name} (Dynamic Threshold)",
    "Hamming Loss": f"{hamming_loss(Y_test, preds_optimized):.5f}",
    "Micro F1-Score": f"{f1_score(Y_test, preds_optimized, average='micro', zer
    "Macro F1-Score": f"{f1_score(Y_test, preds_optimized, average='macro', zer
)})

# Render final markdown results table
summary_df = pd.DataFrame(summary_records)
print("\n" + "="*60)
print("FINAL COMPLETED THESIS MODEL PERFORMANCE COMPARISON")
print("="*60)
print(summary_df.to_markdown(index=False))

```

--- Training Multi-Output Decision Tree ---
 Decision Tree training complete in 42.61 seconds.

```

=====
FINAL COMPLETED THESIS MODEL PERFORMANCE COMPARISON
=====

```

Model Framework	Hamming Loss	Micro F1-Score	Macro F1-Score
Logistic Regression (Default 0.50)	0.02193	0.0542	0.0108
Logistic Regression (Dynamic Threshold)	0.0221	0.0559	0.0125
Decision Tree (Default 0.50)	0.0005	0.6546	0.0058
Decision Tree (Dynamic Threshold)	0.00094	0.5444	0.0109
Random Forest (Default 0.50)	0.00114	0.4878	0.0041
Random Forest (Dynamic Threshold)	0.01644	0.0679	0.0142

In [85]:

```

<class 'pandas.core.frame.DataFrame'>
Index: 25010 entries, 10 to 25389
Data columns (total 45 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Movie_ID                             25010 non-null  int64
1   TMDB_ID                             25010 non-null  int64
2   IMDb_ID                             18351 non-null  object
3   Movie_Name                           25010 non-null  object
4   Original_Title                       25010 non-null  object
5   Release_Year                         24229 non-null  float64
6   Release_Date                         24229 non-null  object
7   Studio                               25010 non-null  object
8   All_Production_Companies             18728 non-null  object
9   Director                             22796 non-null  object
10  Country_Origin                       24856 non-null  object
11  Original_Language                    25010 non-null  object
12  Spoken_Languages                     25010 non-null  object
13  Animation_Style                      24626 non-null  object
14  Genre                                25010 non-null  object
15  Theme                                25010 non-null  object
16  Overview                             22493 non-null  object
17  Movie_Length_Minutes                 25010 non-null  float64
18  TMDB_Rating                          15950 non-null  float64
19  TMDB_Vote_Count                      15958 non-null  float64
20  TMDB_Popularity                      24990 non-null  float64
21  Budget_Million_USD                  25010 non-null  float64
22  Box_Office_Million_USD               1639 non-null   float64
23  MPAA_Rating                          25010 non-null  category
24  Target_Audience                     25010 non-null  category
25  Voice_Cast                           16079 non-null  object
26  Live_Action_Remake                   25010 non-null  object
27  Belongs_To_Collection                25010 non-null  object
28  Collection_ID                        3177 non-null   float64
29  Poster_URL                           21682 non-null  object
30  Backdrop_URL                         14152 non-null  object
31  TMDB_URL                             25010 non-null  object
32  IMDb_URL                             18351 non-null  object
33  Is_Released                          25010 non-null  bool
34  Is_TV_Compilation                    25010 non-null  bool
35  Decade                               24229 non-null  float64
36  Era                                   25010 non-null  object
37  Popularity_Tier                      25010 non-null  object
38  Hidden_Gem                           25010 non-null  bool
39  Profitability_ROI                    729 non-null    float64
40  Is_Adult_Content                     25010 non-null  bool
41  Runtime_Needs_Verification           25010 non-null  bool
42  Is_Future_Release                    25010 non-null  bool
43  Data_Quality_Score                   25010 non-null  int64
44  TMDB_Rating_Clean                    15950 non-null  float64
dtypes: bool(6), category(2), float64(11), int64(3), object(23)
memory usage: 7.5+ MB

```

In []: